WBS MRes: Learning From Data

Professor Nick Lee

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Chapter 1

Introduction and Orientation

This is an online course book, designed for students on the Warwick Business School MRes Programme, who are taking the module 'Learning from Data (and) Science'. The module consists of a set of in-person lectures, linked with these examples and additional expositions.

You can see to the left a list of the contents of the book. However, please note that these 'chapter numbers' do not refer to the specific lectures. Not all of the lectures use quantitative examples.

It is important to note that this module is not designed to teach you how to use R. The objective of the module is to help you learn how to use and interpret the results of quantitative social science research, understand the core principles of excellent quantitative social science, and also how to apply key principles of quantitative and scientific thinking into your own life and (where relevant) work.

This book is written in **Markdown**, using R and the **bookdown** package, and published online in HTML format. A full archive of all the R Code and data for the individual chapters from the book is available from Professor Lee on request, once the book is completed for the year.

This book is essentially a 'live' document, meaning that it is finalised for each year just before the module runs, but then modified for future runs.

The current year is 2026, and the book is currently *in progress for 2026.*

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1.1 How this book works

1.1.1 What this book is for

This book goes with the Warwick Business School *Learning From Data (and) Science* module, and its aim is the same as the module's. It isn't trying to turn you into a statistician. It's trying to help you understand what data can and can't tell us, and what good quantitative research looks like.

It's not impossible that some of you will never run a regression in your own research, and that's fine. But whether or not you do, you'll for sure read hundreds of papers that do. If all you can do with a quantitative paper is read the introduction, the hypotheses and the conclusions, you have no real way of knowing whether to trust those conclusions. I want you to finish this module able to form a genuinely informed opinion about quantitative work. And if you arrived afraid of numbers, I want you to leave not afraid of them. A little bit suspicious of them is fine. Healthy, even.

1.1.2 How it fits with the lectures

The book is a companion to the lectures, not a replacement for them. The chapters broadly follow the order of the sessions, and at several points a chapter will hand you back to the slides, because some ideas work better as a conversation in the room than on the page. The book, in turn, does things the lectures can't. It slows down, works through examples step by step, and goes further in places where an hour in a lecture theatre isn't enough.

The chapters move from describing the world with data, and the many ways that description can mislead, through causation and relationships, to uncertainty and estimation, then to comparing groups and testing hypotheses. They finish with what goes wrong in quantitative research, and how we might do better.

1.1.3 How the chapters work

A few principles run through the whole book, and it helps to know them in advance.

Why before how. Wherever there's a decision to make, such as which test to use or whether a hypothesis should be one- or two-tailed, I explain the reasoning *before* the calculation, not in the middle of it. The arithmetic is rarely the hard part. Knowing what you're doing, and why, is.

Simple methods, on purpose. Each chapter teaches its core idea through the simplest method that will do the job. That's not because the simple methods are what you'll meet in the literature (often they aren't), but because the ideas are hard enough without complicated machinery getting in the way. The same handful of ideas sits underneath almost everything more advanced.

Simulation. Many chapters build a small artificial world in which we know the truth, and then show what a method does in it. That's how, later in the book, I'll show you a relationship being manufactured out of nothing, and a "significant" effect being found where none exists. When you know the right answer in advance, you can see exactly what a method gets right and what it gets wrong.

Real analysis, out of sight. Every analysis in the book is real, and runs live in R each time the book is built. But I've kept the code out of view. This is not a programming course, and what matters here is what the results mean, not how to produce them.

Honesty about the tools. You'll notice I'm often critical of how statistics gets used in practice: significance stars, treating $p < .05$ as a finish line, "marginally significant" results. That's deliberate. Many of the conventions in quantitative research are habits rather than laws, and you'll be a better researcher, and a better reader of research, if you can tell the difference.

1.1.4 A note on the numbers

Because the simulations run afresh every time the book is built, the numbers in the book won't always match the ones on the lecture slides. That isn't a mistake. The same process, run again, gives a slightly different answer, and understanding why is one of the most important ideas in the whole module.

1.1.5 Reading the literature

Several chapters end with a section called **Reading the literature**, and more will be added as the book develops. These sections take the chapter's core idea and show you where it turns up in published research, including in methods well beyond anything this book covers. The aim is that when you meet an odds ratio, a control function or a pre-registration in a paper, you'll know what it's claiming, which core idea it rests on, and what questions to ask of it. You won't be able to run these methods after reading these sections, and that isn't the point. You'll be able to read them with some confidence, which for most of you is what will matter.

1.1.6 Further reading and references

Chapters often end with a few suggestions for further reading, chosen for people who want to understand the ideas rather than become specialists. One book has had a particularly big influence on how I teach much of this material: David Spiegelhalter's *The Art of Statistics* (Spiegelhalter, 2019). If you read one book alongside this one, that I didn't write myself (N. Lee & Lings, 2008), make it that. Each chapter also lists the references it cites at its end, so you can follow up anything that interests you.

Please enjoy the book, and learn from it.

Feel free to email me with any questions or comments: nick.lee@wbs.ac.uk

Chapter 2

Describing the World With Data

This Chapter is designed to work with Lecture 2.

Here, I will demonstrate a number of important concepts about data displays and descriptions.

First, let's find some data to play with. In the following examples I will use some data from the Johns Hopkins Coronavirus Resource Center (n.d.) as at April 26 2022, which contains data by country about cases, deaths, and mortality for COVID.

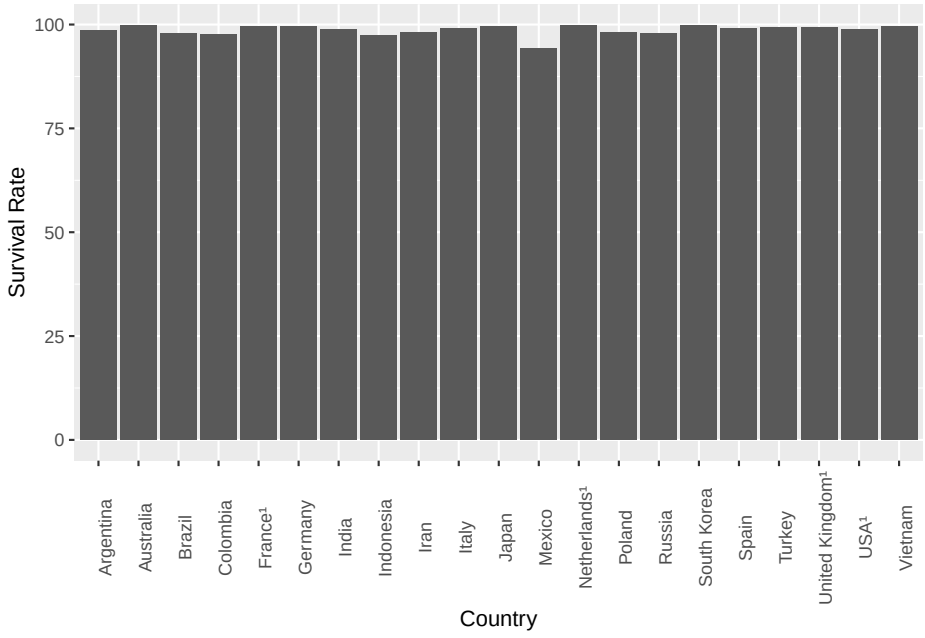
However, as of 2023, the JHCRC ceased collecting data, and their archival data is available on their GitHub (Johns Hopkins University Center for Systems Science and Engineering, 2023).

However, I do think it would be quite hard to recreate the exact snapshot in time of the data set we are using here.

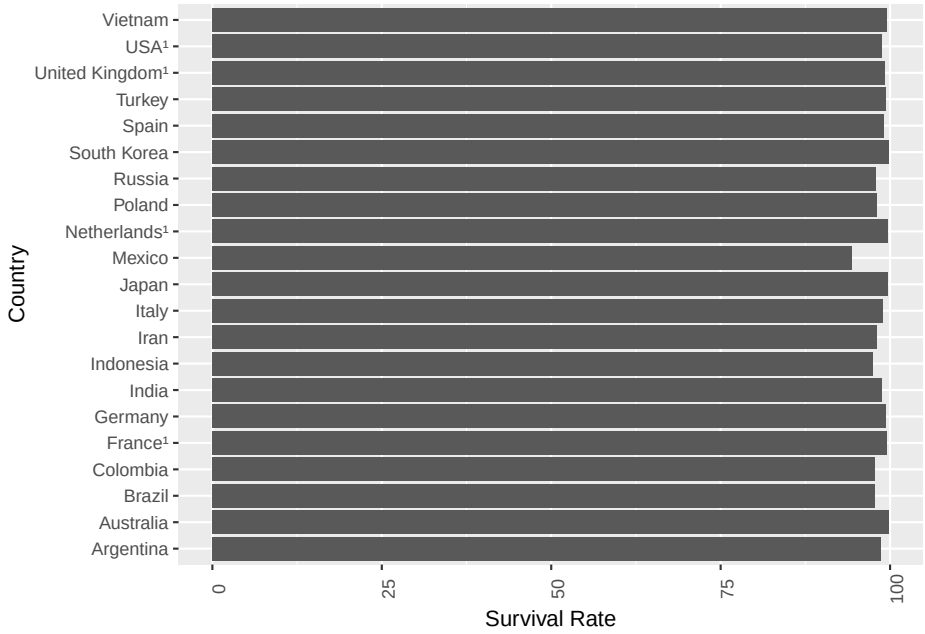
I'm going to use the top 20 countries in terms of raw number of cases

To demonstrate how we can easily change people's views of a data set, let's run some bar charts:

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One thing we could do is change the orientation of the bars…

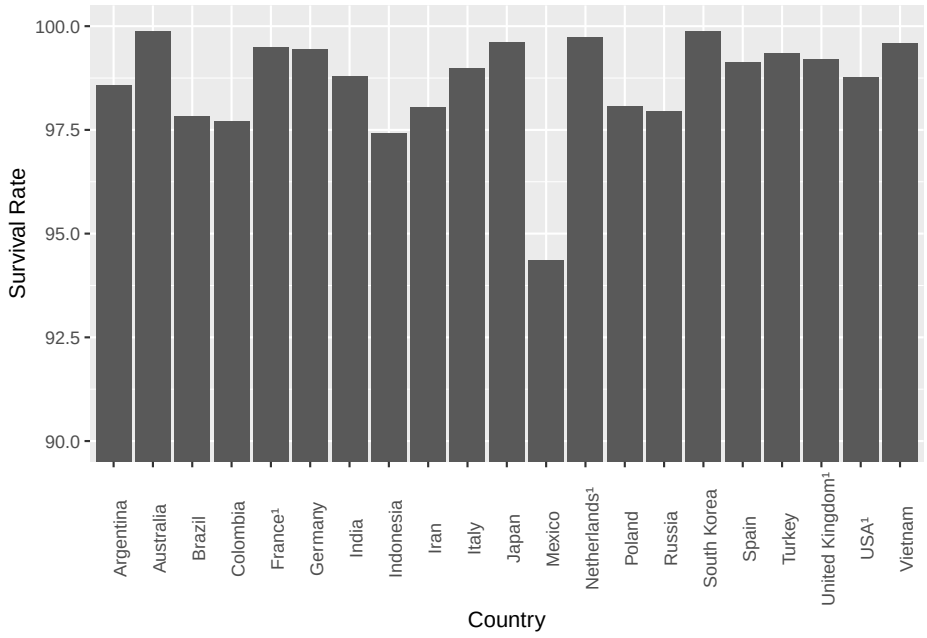


That's nice. The basic conclusion we can draw is everyone is pretty much the same in terms of survival rate, although Mexico is a bit down.

Interestingly, it seems to me that the variation is more pronounced in this

presentation where you look 'down' a line of results, than above where you look across. I'm not aware of any theory behind why, or even if anyone has ever noted that before, but it's definitely something I noticed here.

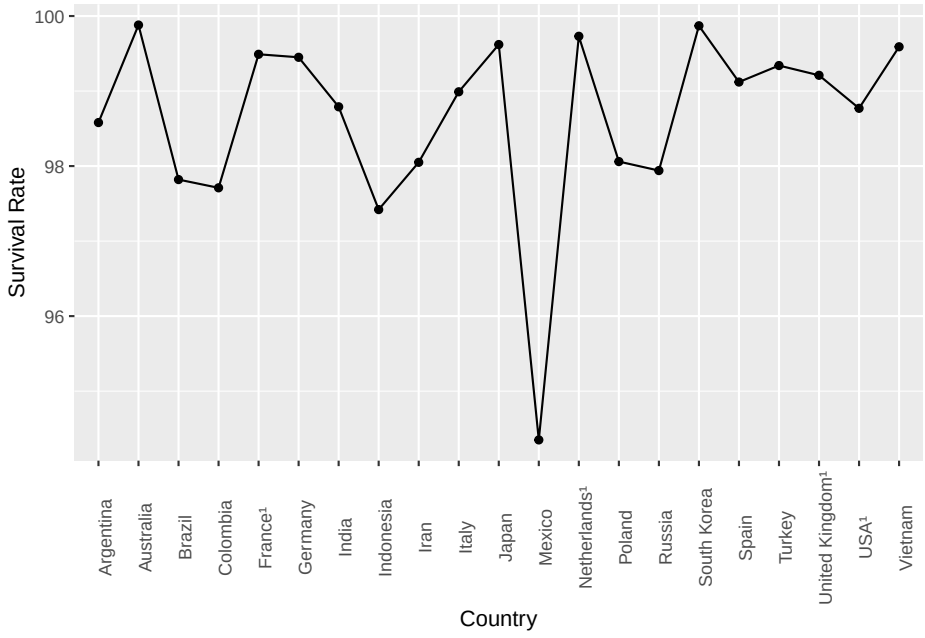
Either way, if I wanted to make everyone think that COVID survival rates were really different, there's a simple trick I could use:



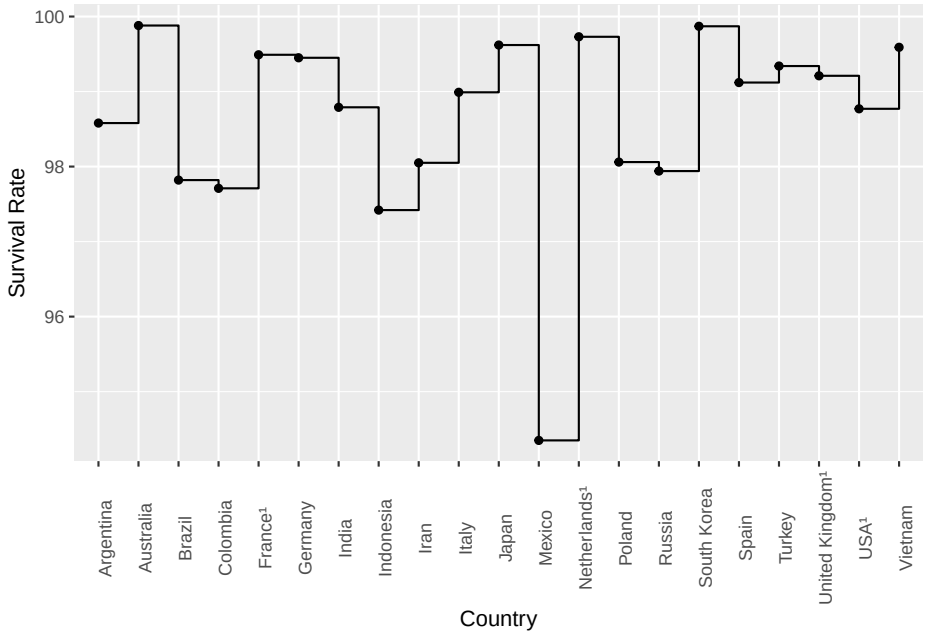
Wow, how do you like that! What's different?

Changing the scale of the y-axis is an old trick, and when used to manipulate opinion it's a bad thing. However, there is a case to say we need to know what we want to say, before we decide how to say it. When does 'making sure we see the right message' move into 'outright manipulation / misrepresentation?'.

Here's another common way this type of data is presented in media and non-scientific reports:



or a variant:



Technically, it's wrong to present this data in a line chart, because it is discrete values and does not represent a trend, but it is surprisingly common - probably because it 'looks more sciency'…

Ironically, there is some logic behind using line charts with this sort of data, depending on what you are trying to get across - let's move to excel to demonstrate…

While we're at it, we'll also look at pie charts…

Quick transition to MS Excel, then back to the slides…

2.1 When the numbers mislead: relative and absolute risk

Everything so far in this chapter has been about pictures that mislead: axes that start in the wrong place, baselines chosen to flatter. The same trick works with numbers alone, and it's arguably more dangerous, because a number feels more objective than a picture. The example I use in the lecture is one I have a personal stake in.

A few years ago a set of headlines did the rounds suggesting that bacon, ham and sausages belonged in the same bracket as cigarettes. The research behind the most recent wave was a good study, in a good journal, using the UK Biobank, a huge and genuinely excellent dataset (Bradbury et al., 2020). Its headline finding was that each extra 25 grams of processed meat a day was associated with a 19 percent higher risk of bowel (colorectal) cancer. I ate a ham sandwich pretty much every day for years, and as a parent I wanted to know whether I should care what my kids ate. So I did what I think anyone reading research should do, and worked through it step by step.

First, is the amount realistic? The authors helpfully translate 25 grams into about one rasher of bacon or one slice of ham. That's not an absurd dose like the ones behind some food scares. Plenty of people eat that much, so this is worth taking seriously.

Second, is it causal? This is a correlational study, and Chapter 4 will give you a whole toolkit for worrying about that. Maybe heavy processed-meat eaters also drink more, or move less, or eat fewer vegetables. The authors adjusted for a long list of such factors, and the association survived. More importantly, it sits within a much larger body of evidence that led the International Agency for Research on Cancer to classify processed meat as a cause of bowel cancer (Bouvard et al., 2015). So let's accept that the link is probably real.

Third, 19 percent of what? This is the question the headlines skipped. A 19 percent increase is a *relative* risk: it tells you how much your existing risk goes up, not what your risk is. To turn it into something you can use, you need the *base rate*. In the UK, roughly 1 in 17 men and 1 in 20 women will be diagnosed with bowel cancer in their lifetime (Cancer Research UK, 2026), so call it about 6 in every 100 people. Add 19 percent to 6 and you get a little over 7. In *absolute* terms, then, the extra daily slice moves you from about 6 in

100 to about 7 in 100, as Figure 2.1 shows.

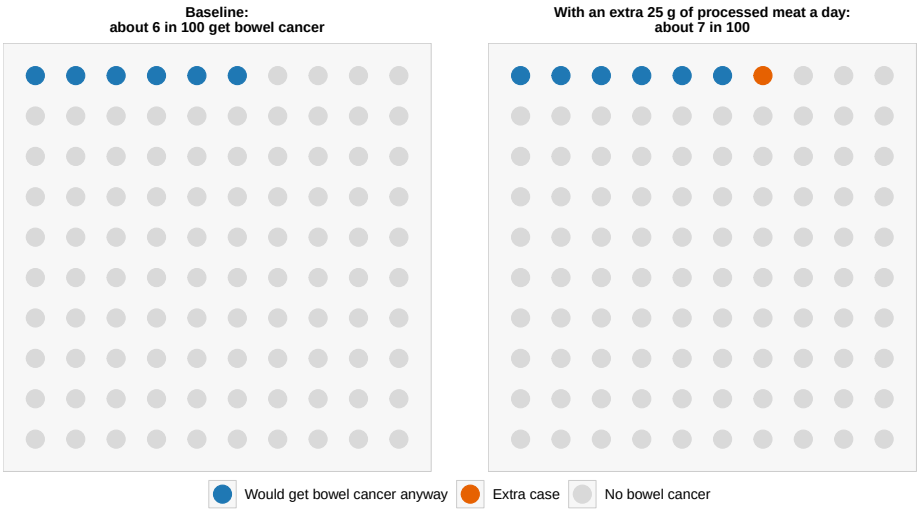


Figure 2.1: The same result in absolute terms. Each dot is one person. A 19 percent increase in relative risk moves bowel cancer from about 6 in 100 people to about 7 in 100: one extra case (orange). Figures are rounded; see text.

Notice the language trap here. The risk went up by 19 *percent*, but by only about one *percentage point*. Both statements are true. Mixing them up is one of the most common errors in the whole of quantitative reporting, and you'll see it made by journalists, managers, and, I'm sorry to say, academics.

Another way to say the same thing is to ask how many people would need to eat that extra slice for one extra case to appear. You might expect the answer to be 100, since we went from about 6 to about 7 in every 100. But the 7 was rounded. Nineteen percent of 6 is about 1.1, so the true figure is nearer 7.1 in 100, which means a little more than one extra case per 100 people. Turn that around, and it takes only about 90 people, each eating an extra slice of ham every day for the rest of their lives, for us to expect roughly one extra case of bowel cancer among them. (Epidemiologists call this the "number needed to harm.") That's a rough translation. It stretches a result from about six years of follow-up across a lifetime, and it treats the population's average risk as everyone's starting point. But it is the right order of magnitude, and it tells you far more than "19 percent" does.

Fourth, why the comparison with cigarettes? Processed meat and tobacco sit in the same IARC category. But that category describes how *strong the evidence* is that something causes cancer, not how *large* the risk is (International Agency for Research on Cancer, 2015). For smoking, the numbers are in a different universe. Roughly 1 in 100 lifelong non-smokers in the UK will get lung cancer, compared with more than 20 in 100 people who smoke a pack

a day, as David Spiegelhalter pointed out at the time (Science Media Centre, 2015). That's a relative risk of around 20, which, as you'll see in Chapter 4, is the kind of number that is very hard to explain away by confounding. Nineteen percent is a relative risk of 1.19.

Finally, whose risk is it? Here it's worth holding two ideas at once. For me, one extra case per 90 or so lifelong ham-eaters is a small risk, and I'm entitled to weigh it against how much I like a ham sandwich. The numbers can't tell me how to make that trade. But across a whole country, small individual risks add up: Cancer Research UK estimates that around 13 percent of UK bowel cancer cases are caused by eating processed meat (Cancer Research UK, 2026). "Trivial for me" and "important for public health" can both be true, and a lot of confused arguments come from people talking about one while the other person is thinking about the other.

It's also worth knowing that an average can hide a lot. In the Biobank study, the association with processed meat showed up in men but not in women. A single headline number describes a population, not a person, and certainly not you.

2.2 Summarising Data with Numbers

Above, we simply plotted a single variable of interest, and showed that some very basic changes to how it was displayed had the potential to 'lead' the reader to different conclusions. Even that simple tweaking is a powerful way to change opinion, and used in the wrong way can be very misleading.

Now, let's think about the idea of an *even simpler* way to describe data - by summarising it with just some numbers.

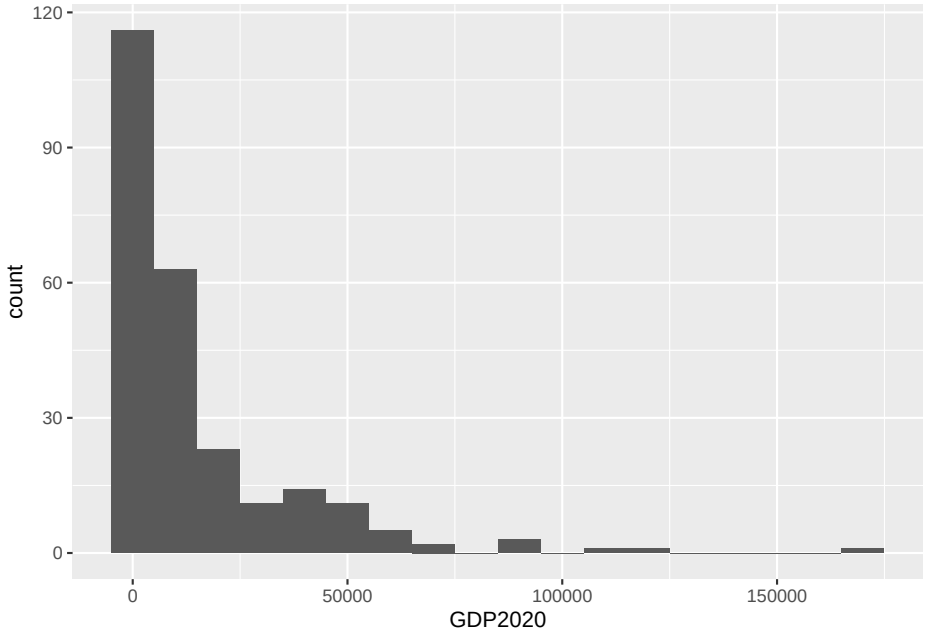
For this example, I am going to use data on GDP per head in 2020 for each nation, from the World Bank.

Before I do anything else, I'm going to graph it again. This time I will use a histogram, which is similar to a bar chart, but not the same thing.

Specifically, a bar chart is used when we are displaying discrete categories. So, above, we had 20 countries, and a single bar for each country.

A histogram is instead when we divide our data into 'groups' of similar values, which are called 'bins'. Thus, each 'bar' on a histogram represents the number of elements (in this case, countries) which fall in that bin. So, in this case, we can divide the GDP variable into bins.

As an example, we can divide GDP per head into 'bins' of US$10000, and plot a histogram



As you can see, there are nearly 120 countries in the first bin, which you can also see runs from -5000 to 5000 dollars. Of course, we know there are no countries with negative GDP per head, but this is just an artefact of the binning process.

This sort of chart gives us something very different to a bar chart. Instead, it is really a **frequency distribution**, and it charts the frequency of values of a given variable - in this case GDP per head.

So, here we have some idea of the **distribution** of the variable. It's clearly skewed quite heavily, with some outliers at the high end.

Let's move on to summarise this variable with some numbers:

```
##
##                    GDP2020
## --------------- -----------
##            Mean    15162.61
##         Std.Dev    22146.53
##             Min      233.84
##          Median     6117.49
##             Max   173688.19
##         N.Valid      251.00
##               N      266.00
##       Pct.Valid       94.36
```

What we have here are the classic 'summary statistics' for 2020 GDP per head.

One thing missing - which admittedly is not very useful in this case, is the Mode,

or the 'most frequent' value.

```
## [1] 1875.441
```

There you have it, the mode is 1875.441.

One important thing to note here is that the word **average** is badly misused when talking about data. Specifically, the *mean*, *median*, and *mode* can all at one time or the other be referred to as the **average** value of a given variable. But, as you can see, they are all different numbers! The word 'average' is a very vague word to use in this context, so do be careful about when you use it. You should always default to using the specific term you are talking about.

A few other summary statistics are calculated for 2020 GDPph below

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.     NA's
##    233.8   2127.5   6117.5  15162.6  18072.4 173688.2       15
```

```
## [1] 15944.94
```

```
## [1] 173454.4
```

What we have here now includes the range (173454.4) and the interquartile range (15944.94), which are both useful measures of how 'spread out' the data is.

Each of the summary numbers above has strengths and weaknesses when describing different types of data. Here, we will mainly talk about the sensitivity to outliers.

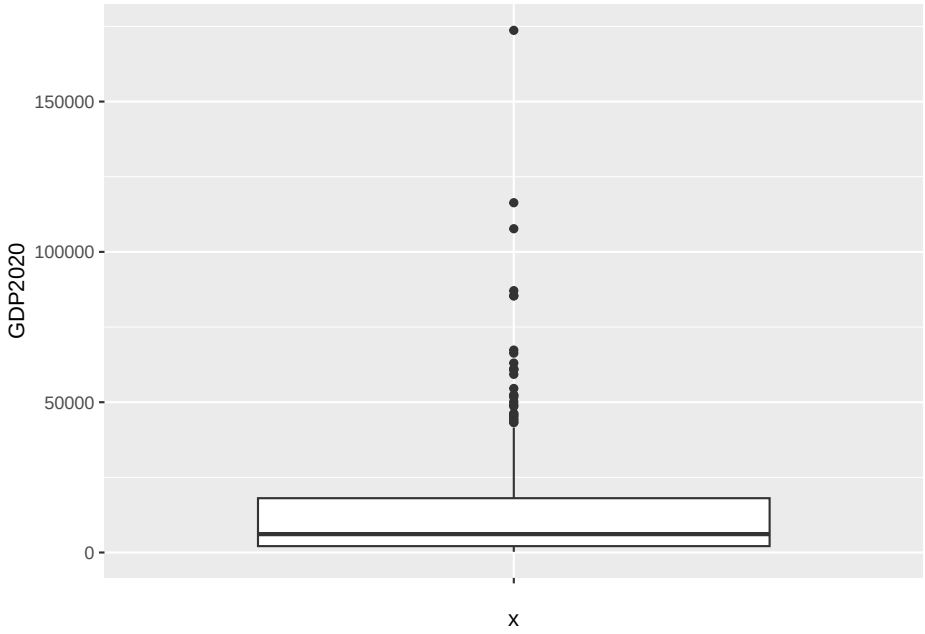
Specifically, the **median** and **interquartile range** are less sensitive to outliers as measures of central tendency and range respectively than their counterparts the **mean** and the **range**. So, that's why you commonly see the median used so frequently in these sorts of situations.

But, what do we mean by outliers? Let's look at this in the context of another graphical display, the **box plot**.

2.3 Box Plots

The box plot (and the other displays below) are criminally underused in my opinion. Most of us calculate a box plot at school, and then never again. But, it's a really useful way to understand a data set.

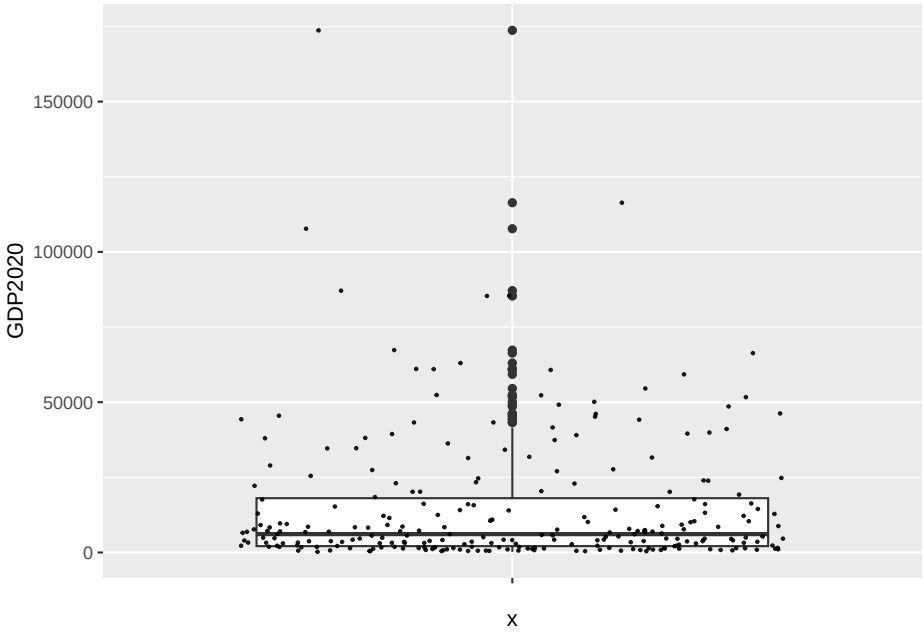
Let's do a boxplot for the 2020 GDPph data:



A box plot presents a lot of information very efficiently. The **median** is the thick horizontal, the white rectangle is bounded by the **interquartile range**, and outliers are represented as dots.

We can see here that the data is very skewed, and there are some very significant outliers at the high end (no prizes for guessing which countries they are).

A weakness of a box plot is that it doesn't actually show the distribution of the data. One way to do this is to add what is called *jitter*, which actually plots the values of each case, with a 'jitter' to avoid overlapping identical values.



So, we can see here that most of the data is clustered towards the bottom of the range - i.e. very low GDPph. This is of course the same conclusion as the histogram allowed us to draw. But, the box plot also gives us plenty of *other* information about the variable.

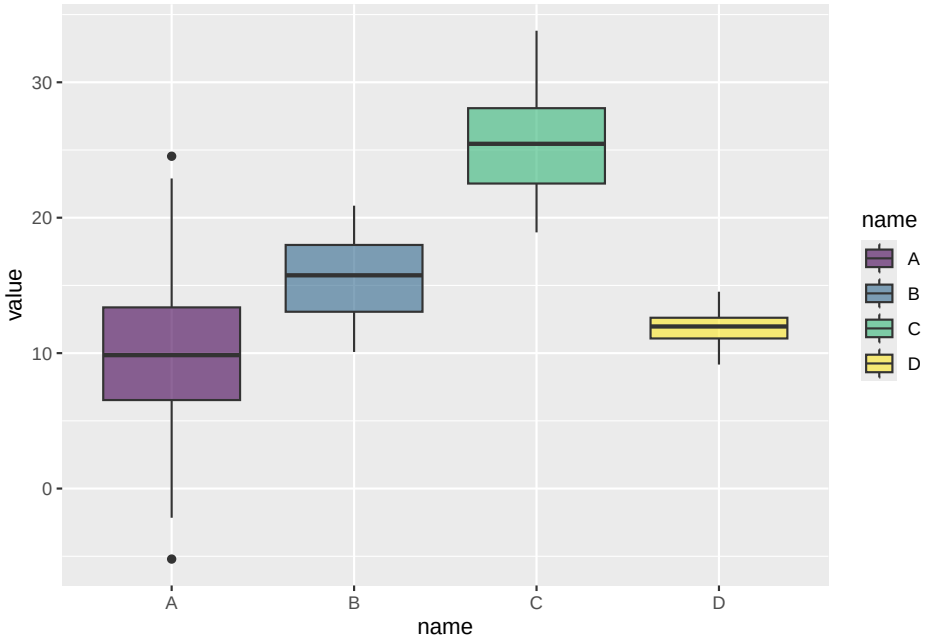
Even so, while box plots are nice, they can also obscure important facts about the distribution. Adding the jitter helps, but there are other ways to display this type of data that make the distribution more obvious.

2.4 Violin Plots and Ridge Plots

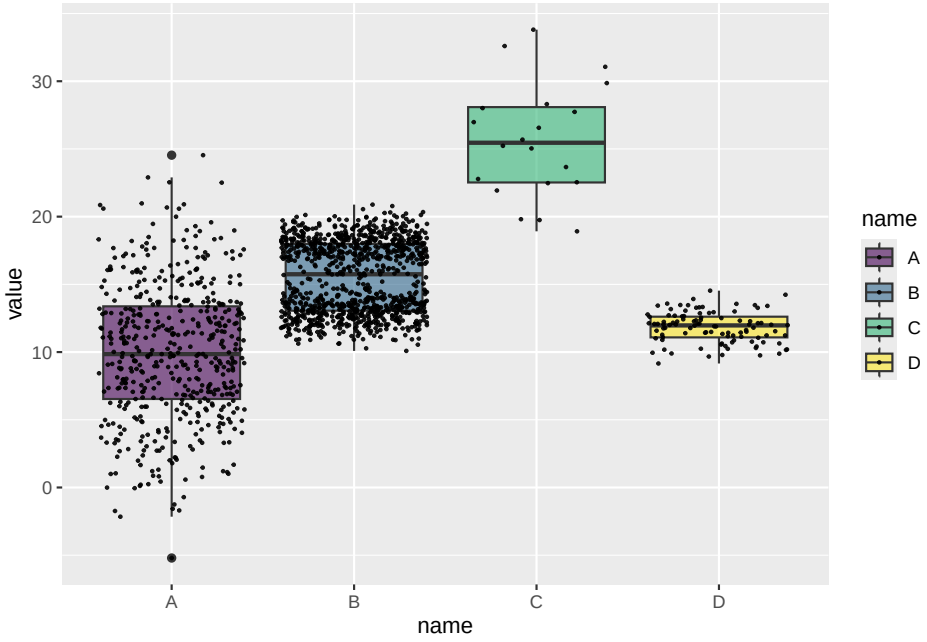
Here, I'm going to give you a set of *even more underused* data displays, which can show some powerful things about variables.

In order to do so, I am going to simulate some data, to make sure it has exactly the features I want to show.

First, let's create the data and use box plots to display it, such as you are already experienced with above.

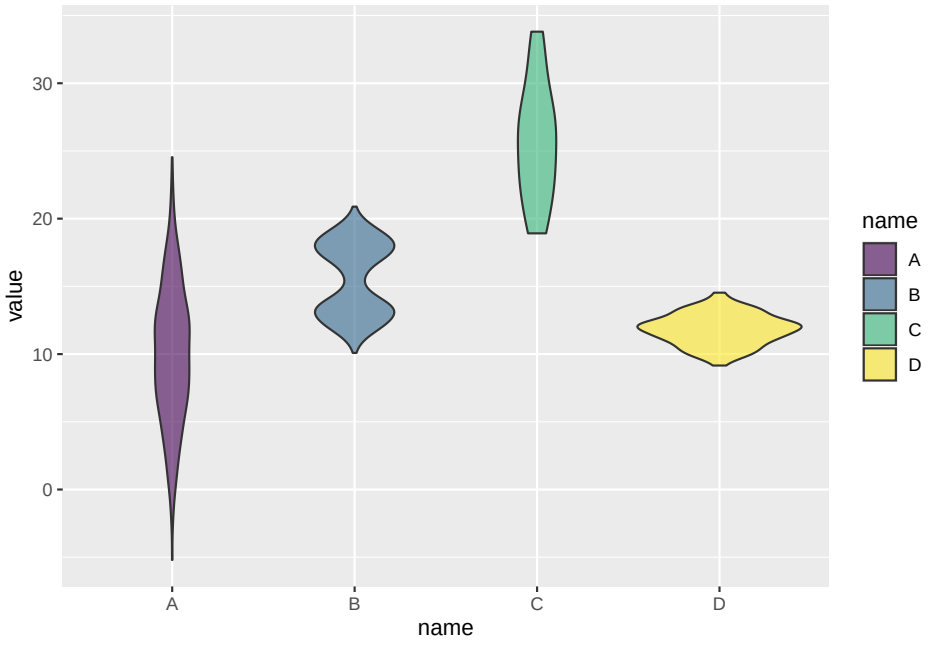


That's all very well, but as we know, this can obscure the distribution. So, let's add jitter:



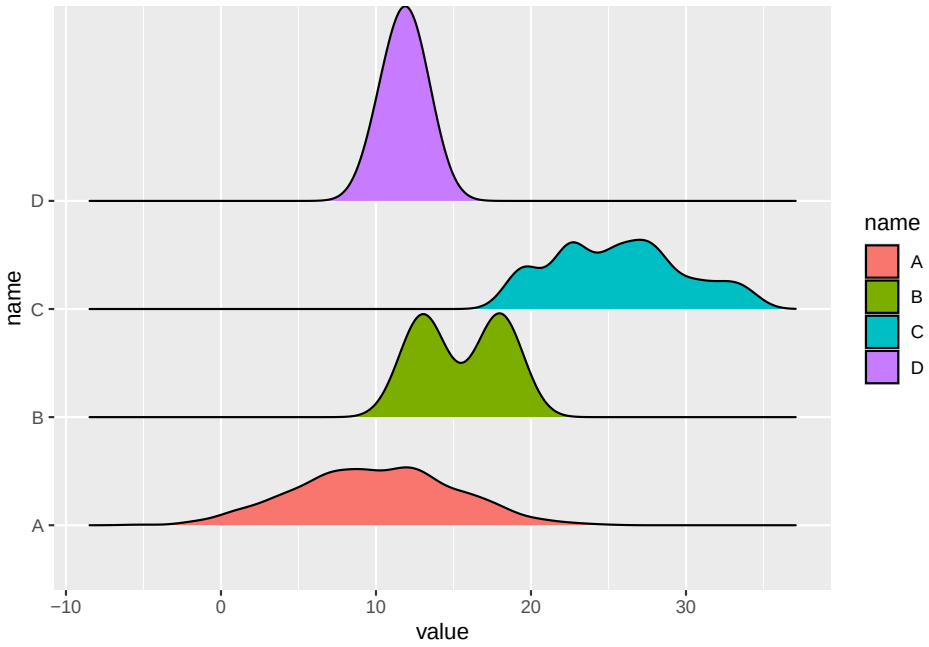
Well, look at that. B has what is called a *bimodal* distribution.

Let's look at what is called a **violin plot** to emphasize this:



Neat! You can see how the violin plot really emphasizes the differences in the distributions, which are totally obscured by the box plots, and only partially shown by adding the jitter.

Another way to visualise this is with **ridges**, which basically compare the distributions of variables together:

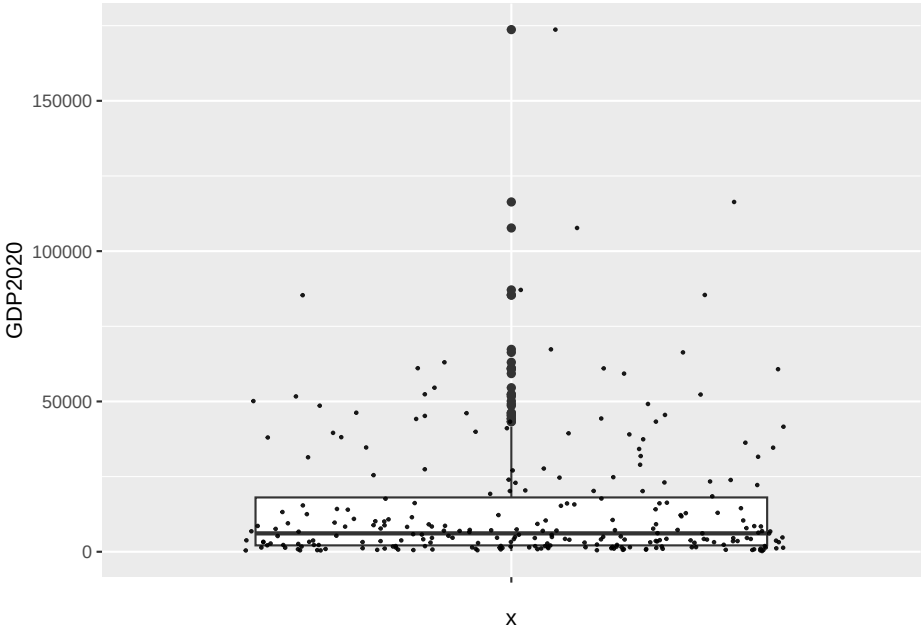


These are really cool in my view. And, if you are asking me, my preference in this type of situation would be to combine jittered box plots, which show the basics, and emphasize the outliers and summary numbers, with ridge plots, which show a really nice visualisation of the distributions of each variable. Taken together, these plots allow us to understand a surprisingly large amount of important information about variables.

OK, let's move back to the slides to discuss another crucially important concept related to data distributions: **transformations**.

Back to slide deck

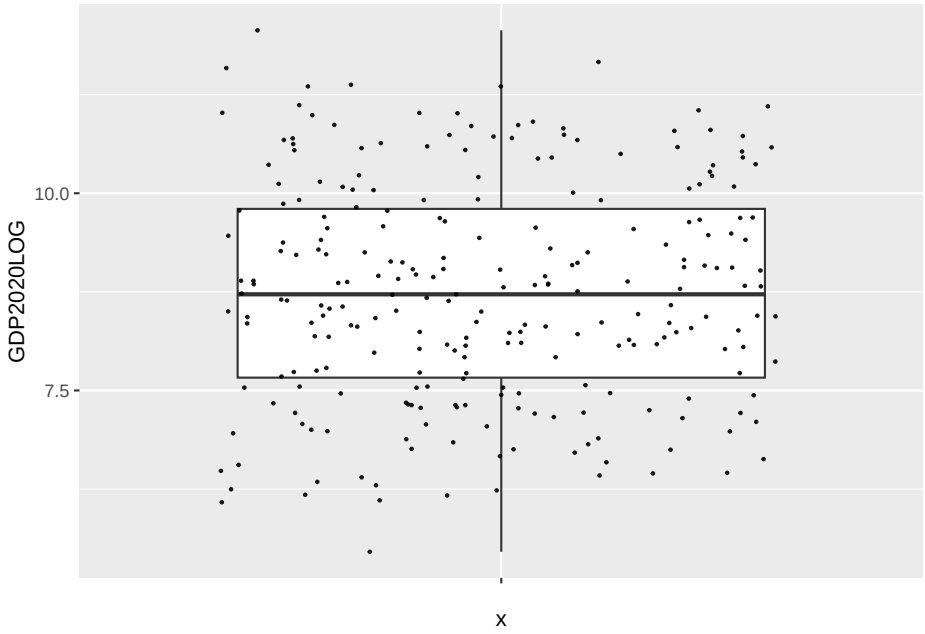
2.5 Transformations



As you can see, the data is heavily skewed, or 'squished' towards the bottom. This makes it hard to see any patterns by eye.

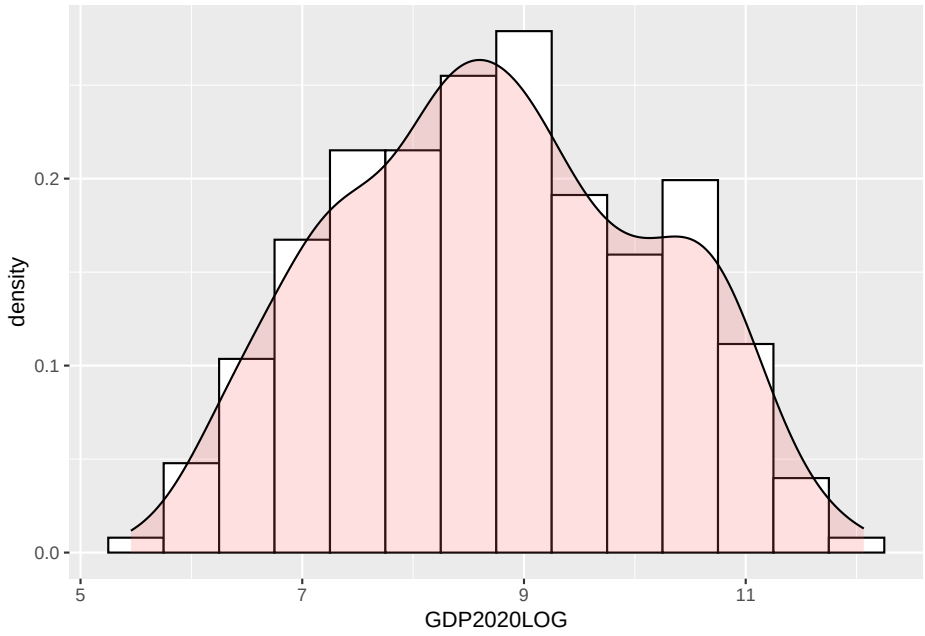
We can make it easier to interpret by *transforming the variable*. Here, using a logarithmic transform.

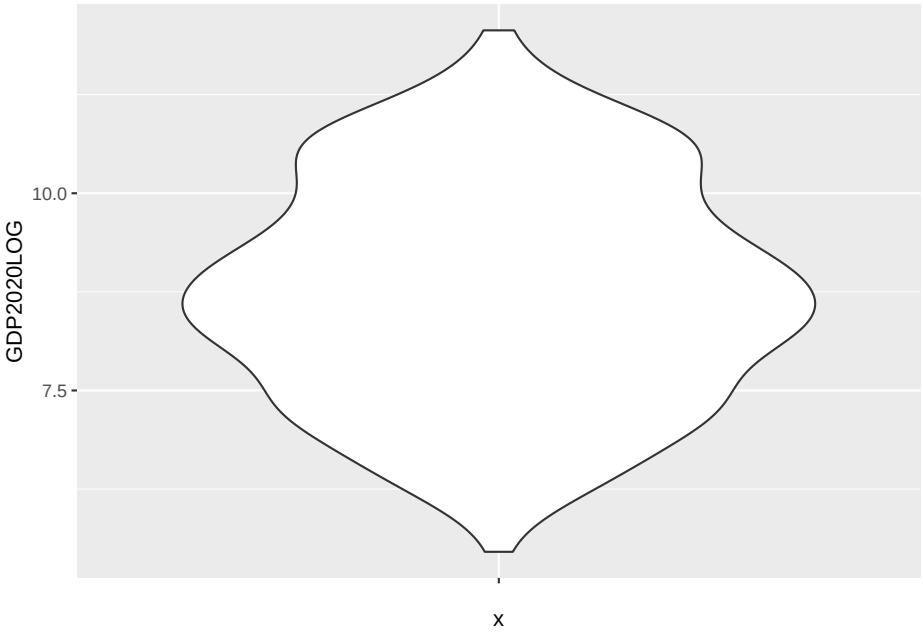
Let's run the box again:



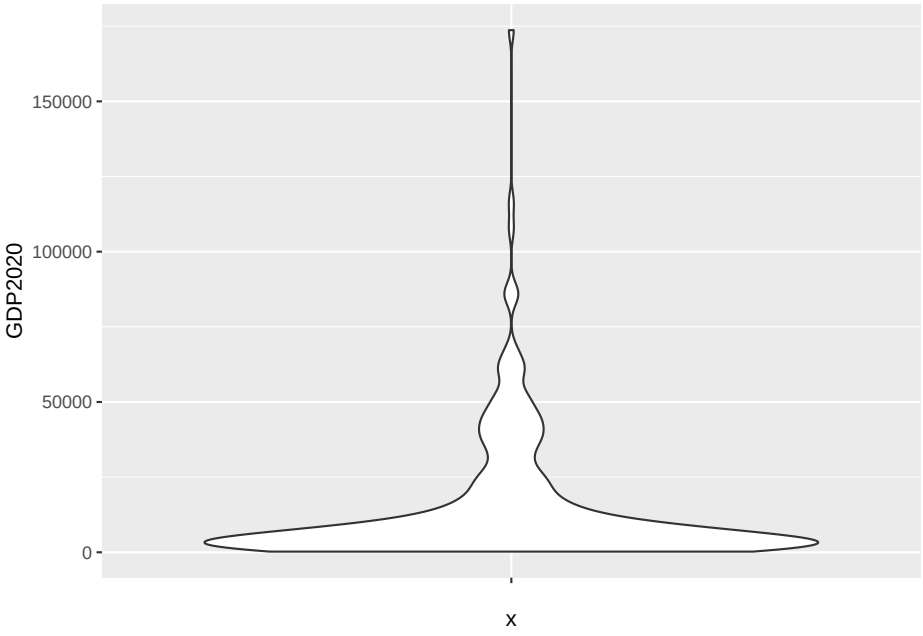
Nice, see how the spread is easier to interpret here? Log transforms are essentially 'inverse exponentials', so what they do is bring down extreme high outliers.

Let's see this in some more plots:





Compare the violin of the log transformed variable with the original:



In fact, we can use the log transform to revisit some of our COVID data back in EXCEL…

Quick transition to MS Excel, then back to slides…

2.6 Reading the literature

The bacon example comes from medicine, but the trap it illustrates is everywhere in management research. Almost every effect you'll read about is reported *relative to something*, and the something is often left out, buried in a table, or never mentioned at all.

2.6.1 Where you'll meet this

Ratios. Three ratios turn up again and again in papers, usually from models predicting whether something happens: an employee leaves, a firm adopts a practice, a customer churns. They're easy to confuse, so it's worth separating them.

- A *relative risk* (or *risk ratio*) compares probabilities. If 50 percent of one group leave and 40 percent of another, the relative risk is $50 \div 40 = 1.25$. The first group is a quarter more likely to leave.
- An *odds ratio* compares odds instead. The odds of something are the chance it happens divided by the chance it doesn't. A 40 percent chance of leaving is odds of 40 to 60, or 0.67, and a 50 percent chance is odds of 50 to 50, or 1. The odds ratio is $1 \div 0.67 = 1.5$. Odds ratios are what logistic regression, the most common model for yes/no outcomes, produces naturally, which is why you'll see so many of them.
- A *hazard ratio* compares how quickly something happens. It comes from models that follow people or firms over time, and it asks: among those it hasn't happened to yet, how much faster does it happen in one group than the other, at any given moment? The bacon figure earlier in this chapter was actually a hazard ratio.

All three are relative numbers, so all three need a base rate before they mean anything. A hazard ratio of 1.5 for employee turnover means one thing if the baseline is 30 percent a year and something much smaller if it's 1 percent.

Odds ratios need one extra warning. When the outcome is rare, an odds ratio and a relative risk are almost the same, which is why people swap them freely. When the outcome is common, which many management outcomes are, the odds ratio exaggerates. You've just seen it happen: the same turnover figures give a relative risk of 1.25 but an odds ratio of 1.5. The number that sounds bigger is usually the one that gets reported (Zhang & Yu, 1998).

Percent versus percentage points. Conversion rates, response rates, market shares and turnover rates are all percentages already, so changes in them can be described either way. A conversion rate going from 2 to 3 percent is a one-point increase and a 50 percent increase. Both are true, and a paper (or a consultant's slide deck) will usually pick whichever sounds better for the argument being made.

Percentage effects from models. When a model uses the logarithm of an

outcome such as sales or salary, its coefficients are usually read as percentage changes. They're relative by construction, so the same question applies: a percentage of what? A 10 percent sales uplift means something very different for a corner shop than for a supermarket chain.

Standardised effect sizes. Cohen's d and standardised regression coefficients express an effect relative to the spread of the data rather than in its natural units. They have real uses, and you'll meet them properly in Chapter 8. But they share the weakness of all relative numbers: a "medium" standardised effect can be large or trivial in pounds, days or customers. Good papers translate back into natural units. It's worth noticing when they don't.

Pictures in papers. Everything earlier in this chapter applies to journal figures too. Watch especially for bar charts of means on a 1-to-7 rating scale whose axis starts at 4, and for interaction plots where two lines diverge dramatically across a range of barely a scale point.

2.6.2 The connection back

A truncated axis and a relative risk are the same trick. Each shows you a comparison with its anchor removed. The axis hides where zero is, and the relative risk hides the base rate. The defence is identical in both cases: go and find the anchor before you decide how impressed to be.

2.6.3 Questions worth asking

1. **Is this a relative or an absolute number?** If it's relative, where is the base rate?
2. **Percent or percentage points?** Which one did the authors choose to put in the abstract, and why?
3. **Is it an odds ratio for a common outcome?** If so, the effect is probably smaller than it sounds.
4. **What does it mean in natural units?** Pounds, days, people, customers. If the authors never say, can you work it out from their descriptives table?
5. **Whose risk is this?** An average across a population can hide groups where the effect is much larger, and groups where it's absent. And a small effect on each person can still add up to a large effect across a population, or the other way round.

2.6.4 One red flag

The abstract that reports only the relative effect: "increased the likelihood of turnover by 50 percent," with the base rate nowhere in sight. It's not always an attempt to mislead, since journal word limits are real. But authors know which framing sounds more impressive, and a paper whose headline number only works without its anchor is telling you something. Go to the descriptives, find

the base rate, and do the arithmetic from this chapter yourself. It rarely takes more than a minute.

2.6.5 If you want to read more

David Spiegelhalter's *The Art of Statistics* (Spiegelhalter, 2019) is the best general book I know on reading numbers sensibly, and it was Spiegelhalter's explanation of the bacon story that first got me thinking about it properly. Gerd Gigerenzer's *Reckoning with Risk* (Gigerenzer, 2002) is older but still the classic on why relative risks mislead, and on how much clearer things get once you switch to natural frequencies like "7 in 100."

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 3

The Normal Distribution

This chapter supplements the content in Lecture 3.

3.1 A Simple Demonstration of the Normal Distribution

First, let's consider the 'empirical distribution', which is the pattern your data set makes.

Let's consider births in England and Wales in 2020 for our example, since they're a subject close to my heart (my first child was born in 2020).

Looking online, one can pull down the distribution of birthweights in England and Wales in 2020, available from the ONS website, and which are reported in 500g 'bins' (Office for National Statistics, 2021).

We can use these numbers to create a chart

(please note, the code I use for the demonstrations in this lesson is mostly modified from David Spiegelhalter's originals, available on his GitHub page, with some of my own bits and bobs and new data…)

Spiegelhalter's GitHub can be found here: https://github.com/dspiegel29/ArtofStatistics

The first thing I want to do is create a data set that accurately reflects the ONS distribution of birthweights in England and Wales in 2020. To do that, I use the following code, which in this instance I will display below, with annotations.

```
weights=c(1500, 2000, 2500, 3000, 3500, 4000, 4500,5000,5500) #modified
#categories from 2020 UK Data
mids=weights-250 # Note the DS original is +250, but this does not work for UK
#data as the numbered is the UPPER bound of the weight category in the ONS stats.
```


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```
#So the babies in the '2000' bin weigh 1500-1999, so the midpoint is 1750, not
#2250
n=c(5015, 7103, 27554, 99220, 218442, 179088, 53530,6952,635) # numbers in each
#bin, UK2020
N=sum(n)  # total number of babies
area=N*500  # number * binwidth = total area of histogram

#I think lbw should be sum of groups 1-3 not 1-2 as in the book code. Using 1-3
#gives a result for the UK of 6.6% which tallies with the official stat given
#that I remove the 'no recorded weight' and 'implausible' categories'. I imagine
#that the DS book data has categories which may be the lower bound not the upper
#as in the UK data table.

lbw   = sum(n[1:3])   # number with low birth weight (less than 2500)
lbw.percent=100*lbw/N  # % low birth weight
# 6.6% which tallies with Nuffield stat for 2020.
```

Remember though, the above just sets up the numbers required to recreate the empirical distribution, it isn't analysis itself.

Next, I want to summarise the distribution I just created and a few important numbers regarding Low Birthweight that we will need to use for our later discussion.

Displayed below are the total N for the distribution, the number of low birthweight (LBW) babies born, the percentage of LBW babies, the mean, the standard deviation, the skewness, and kurtosis.

```
## [1] 597539
```

```
## [1] 39672
```

```
## [1] 6.639232
```

```
## [1] 3345.263
```

```
## [1] 577.7594
```

```
## [1] -0.4955778
```

```
## [1] 4.068451
```

Two of those numbers need a word of explanation, because you'll meet them in the descriptives of almost every quantitative paper. *Skewness* measures how lopsided a distribution is. A perfectly symmetrical distribution, like the normal, has a skewness of zero. Birthweights come out at -0.5. That's negative, meaning the distribution has a longer tail on the left, the lighter side, than on the right. That makes sense: babies born early can be very much lighter than average, while babies much heavier than average are rare. *Kurtosis* is about the tails, meaning how much of the distribution sits a long way from the mean compared

with a normal distribution. The normal distribution has a kurtosis of 3, and ours is 4.07, so birthweights have somewhat heavier tails than a normal distribution would. (Some software reports *excess* kurtosis instead, which subtracts the 3 so that a normal distribution scores zero, so always check which you're looking at.) Neither number is dramatic, and as you'll see in a moment, the normal distribution still does a respectable job of describing these data. But it isn't a perfect description, and the skewness tells you which way it departs.

Next, I will calculate the estimated proportion of low birthweight babies using a normal distribution approximation, and see if it tallies OK with the figure that is taken from the distribution of 6.6%

```
## [1] 7.173338
```

I will also calculate the middle birthweight proportion as a sense check - should be very close to 50%!

```
## [1] 49.98182
```

For our reference, we will calculate the 25th, 50th, 75th percentiles

```
## [1] 2955.57
```

```
## [1] 3345.263
```

```
## [1] 3734.956
```

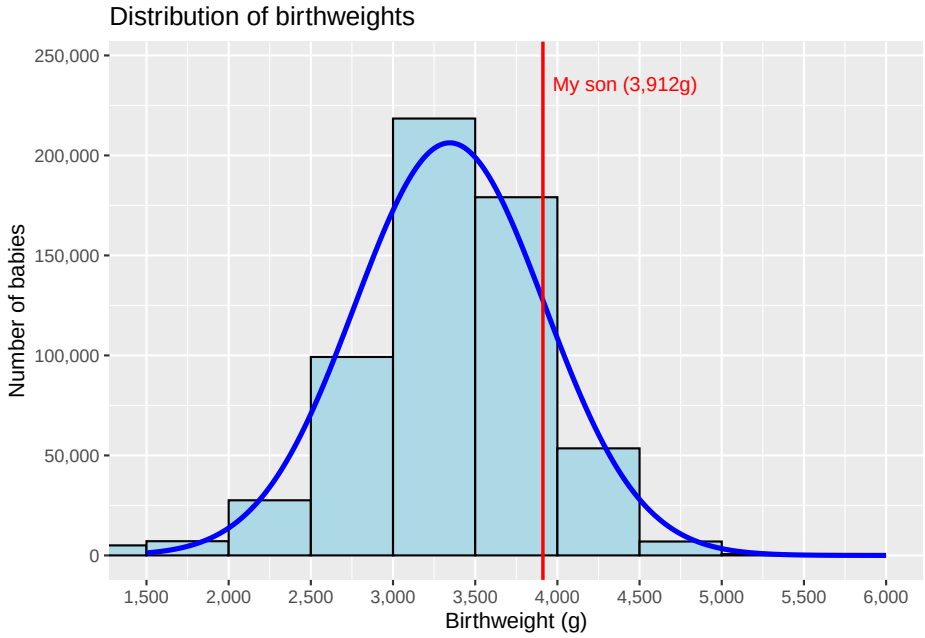
Next I want to calculate the birthweight percentile of my son.

```
## [1] 0.8366843
```

Now, at last, we can do some plots!

First let's set up the histogram and overlay a normal distribution on to it.

My son's weight is shown as a red line.



3.2 Measuring distance in standard deviations

So, was my son's weight unusual? He weighed 3,912g, which is 567g above the mean. On its own, that number doesn't tell us much. Whether 567g is a lot depends entirely on how spread out birthweights are in general. If almost every baby were born within a couple of hundred grams of the mean, he'd stand out. If babies routinely varied by a kilo or more, he'd be pretty ordinary.

That's exactly what the standard deviation measures. You saw it in the list of summary numbers at the start of this chapter, and roughly speaking, it's how far a typical value sits from the mean. In Chapter 2 we met the range and the interquartile range as ways of describing spread. The standard deviation is another, and it's the one the normal distribution is built on. And here's the key thing about the normal distribution: it's completely described by just two numbers. The mean tells you where the centre of the curve is, and the standard deviation tells you how wide it is. Give me those two numbers and I can draw the whole curve.

So from here on I'm going to set the bars (the actual counts) aside and work with the smooth curve, treating it as a model of the population of birthweights. And instead of measuring distance from the mean in grams, I'll measure it in standard deviations.

The figure below shows why that matters: the same 567g gap, under two different spreads.

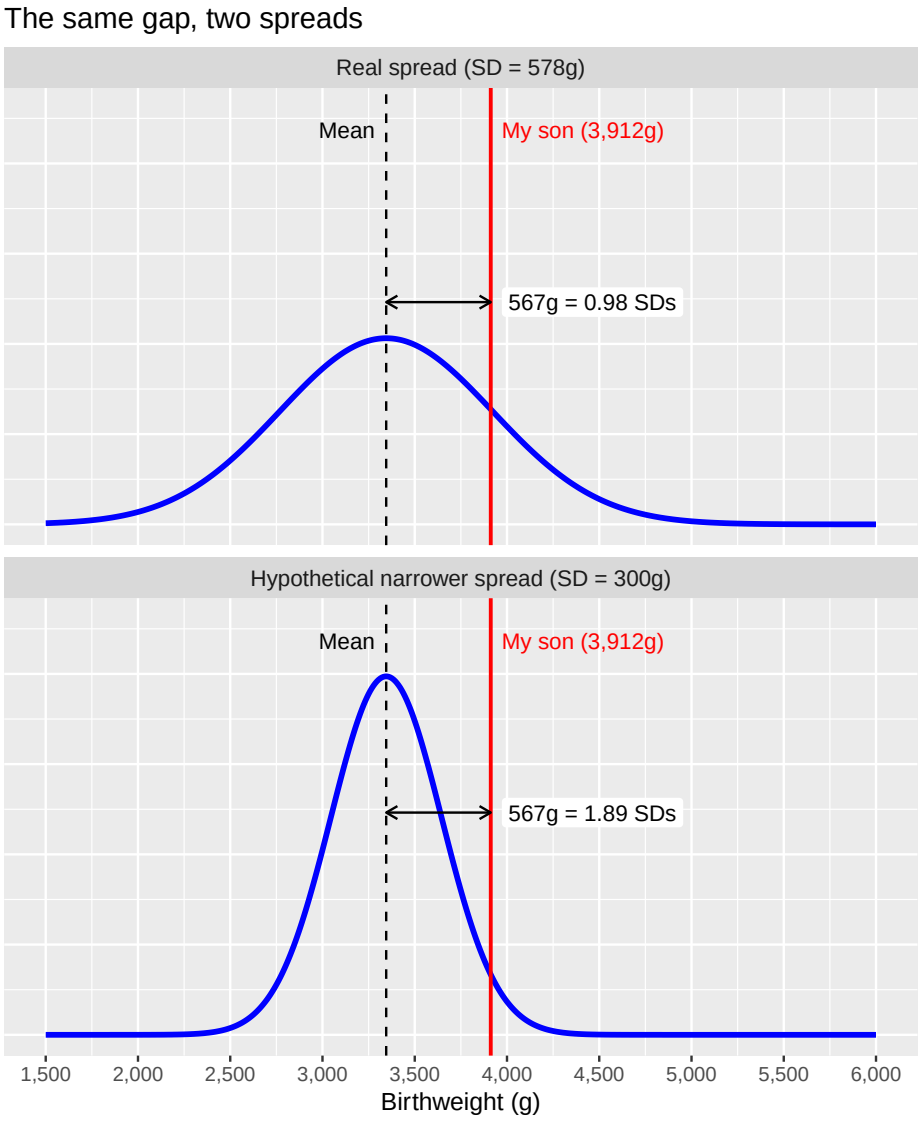


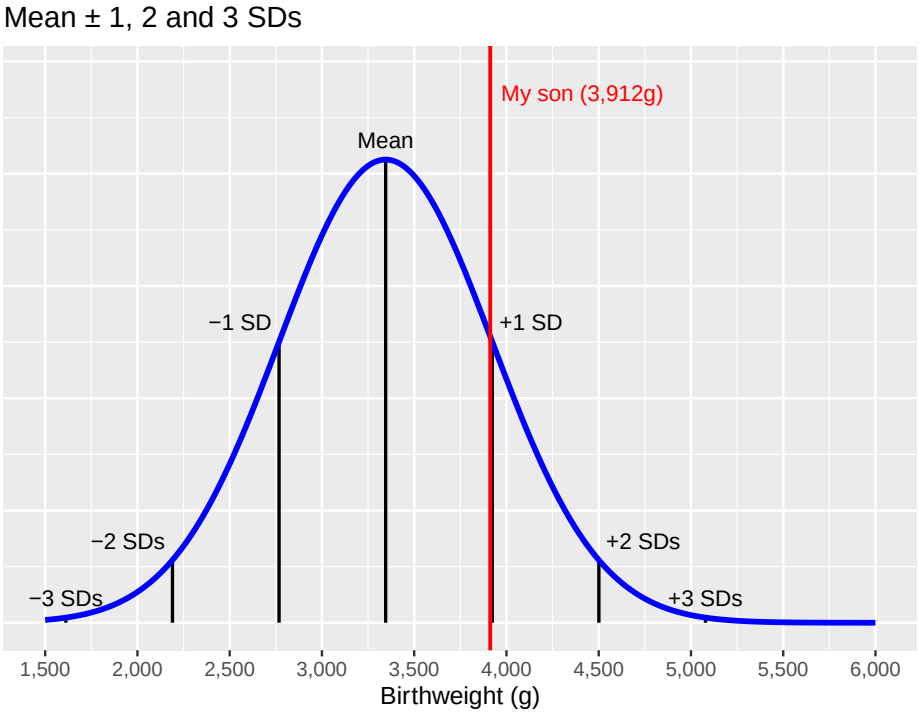
Figure 3.1: The same 567g gap above the mean, under two spreads. Top: the real birthweight distribution. Bottom: a hypothetical, narrower distribution with the same mean and an SD of 300g, drawn to the same scale. The narrower the spread, the more standard deviations the same gap covers.

Mathematically, about 68% of the area under the curve lies between −1 and +1 SDs, about 95% between −2 and +2 SDs, and 99.7% between −3 and +3 SDs.

Now, another way to think about this is that the area under the normal curve represents 100% of the probability of a given value (here, weight) occurring. So, you could say that it is around 95% probable that any given birthweight (randomly sampled) will occur between −2 and +2 SDs of the mean.

We can calculate this easily, given that the mean is 3,345g, and the SD is 578g

So, it is 95% probable that a random baby born in England and Wales in 2020 will weigh between 2,190 and 4,501g



Now we can answer the question properly. To find out how many standard deviations my son's weight was above the mean, we calculate what's called a *z-score*: take his weight, subtract the mean, and divide by the standard deviation.

$$z = \frac{X - \mu}{SD} = \frac{3912 - 3345}{578} = 0.98$$

So he was almost exactly one standard deviation above the mean. That puts him just inside the band containing the middle 68% of babies: on the heavy side, but not unusual. We'll go through this calculation on the slides as well.

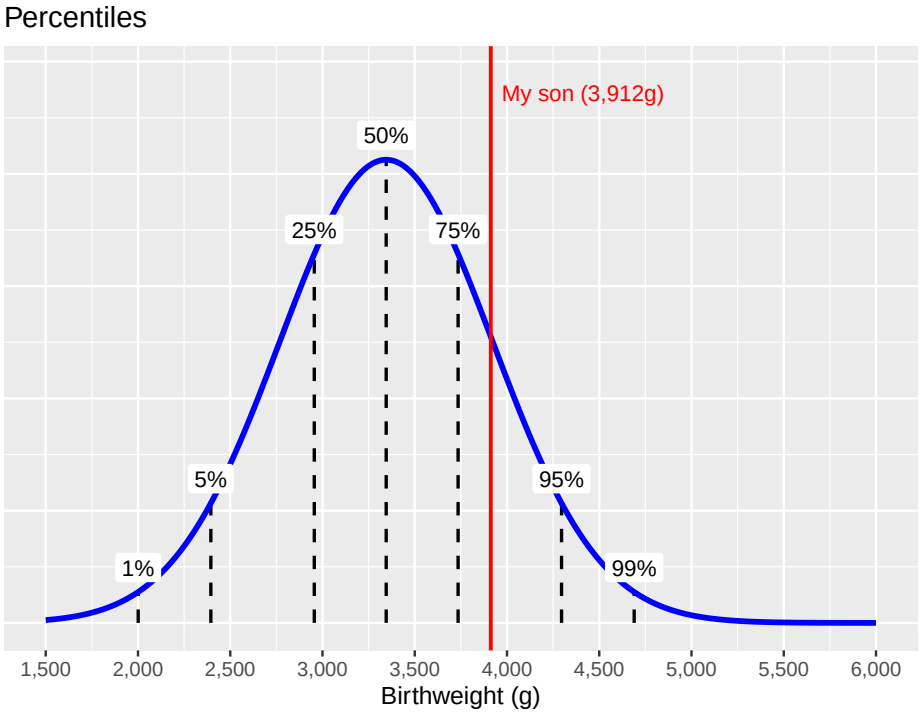
A z-score tells you where someone sits in standard deviations; a percentile tells

you the same thing as a share of the population. We can overlay the 1st, 5th, 25th, 50th, 75th, 95th and 99th percentiles on the chart, which are also useful ways to describe a distribution like the Normal.

50th is of course the median, and the 25th and 75th percentiles are the quartiles.

Here, the same things we have previously discussed in relation to samples are applied to populations

We can also see that my son's weight lies on the 83rd percentile, that is 83% of babies born in 2020 weighed less. This tallies with what we were told at the time, and then things are monitored quite closely from that time on in terms of percentiles.



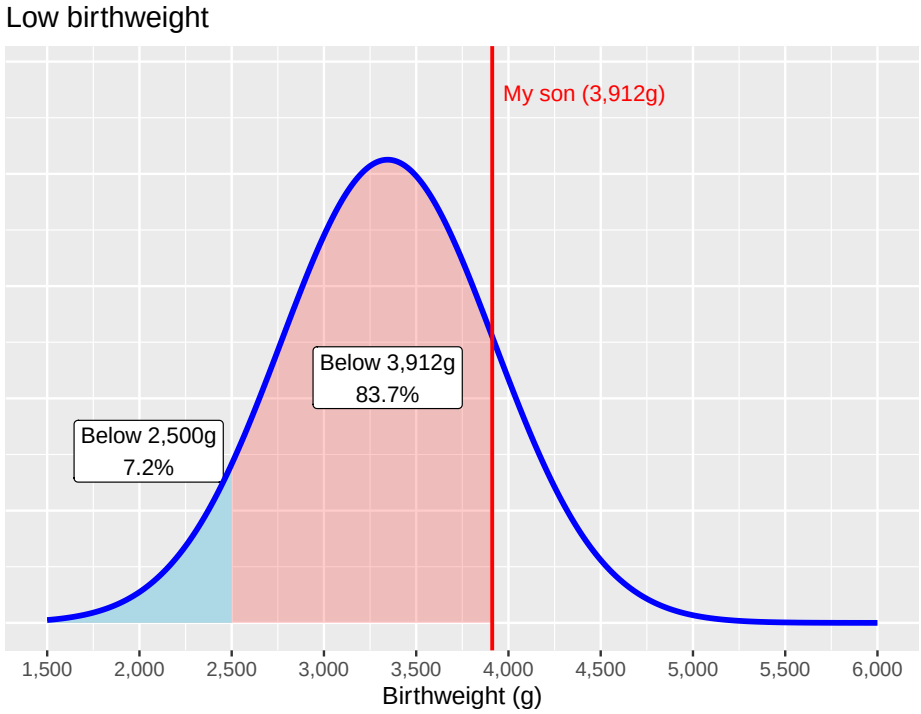
Moving to a discussion of low birthweight. It is medically defined as low if below 2500g. Nuffield reports that 6.6% of babies in 2020 were born at low birthweight, which is fairly good.

This tallies fairly well with the predicted proportion from the normal distribution of 7.2%

What we gain from the light blue shaded area is the proportion of the population which is expected to be born at low birthweight, but also an indication of the probability that a randomly-chosen baby born in 2020 would be of low weight.

This concept can easily be transferred to other weights in this population (and

by extension to other variables and populations distributed like this)



3.3 Reading the literature

Most quantitative papers include a table of descriptive statistics: a mean and a standard deviation for every variable, sometimes with minimums, maximums, medians or skewness. It usually sits alongside the correlations in Table 1, and it's usually skipped. The normal distribution in this chapter is the benchmark that makes that table readable. A mean and a standard deviation summarise a normal distribution almost perfectly. The further a variable is from normal, the less those two numbers tell you, and the more carefully you need to read them.

3.3.1 Where you'll meet this

Survey scales. Many management variables are averages of several Likert items, such as "strongly disagree" (1) to "strongly agree" (7). There's a long-running argument about whether it's legitimate to take means of these at all, since the gaps between the response options needn't be equal. In practice, averages of several items usually behave well enough for the standard methods (Norman, 2010), but it's worth reading the numbers in terms of the scale. A mean of 4 on a 1–7 scale could mean that everyone chose the neutral midpoint, or that half the sample strongly agreed and half strongly disagreed, and only the

standard deviation will tell you which. Watch for means close to either end of the scale. A mean of 6.3 on a 1–7 scale, with a small standard deviation, means almost everyone gave nearly the same answer. There's then very little variation for any other variable to explain, and correlations with that variable will be weaker than the underlying relationship because the range is so restricted.

Skewed business variables. Plenty of the variables in management research are nothing like normal. Firm size, sales, executive pay, patents and citations are typically *right-skewed*: most firms are small, and a few are enormous. For variables like these the mean is dragged upwards by the largest cases and can be a poor guide to a typical firm. Where a paper reports both the mean and the median, compare them; a mean far above the median is the signature of right skew.

Logged variables. Because of that skew, papers often analyse the logarithm of a variable, as you'll see Chapter 5 do with GDP. The log pulls in the long right tail, and it changes what a coefficient means. When the predictor is logged, its coefficient describes the effect of a *proportional* change: roughly, a 1% increase in X goes with a change of the coefficient divided by 100 in Y. When the outcome is logged too, the coefficient becomes an *elasticity*: a 1% increase in X goes with a change of roughly the coefficient, in per cent, in Y. So check the variable descriptions for "ln" or "log" before interpreting any coefficient. Also check what was done about zeros, which can't be logged. Many papers add 1 to every value first, and with count variables full of zeros, that choice can matter.

Normality checks. Papers sometimes report that variables were tested for normality and found wanting, or not. Two things are worth knowing. First, what ordinary regression actually assumes is roughly normal *errors* (residuals), not normally distributed variables. A skewed predictor isn't a problem in itself. Second, in large samples formal tests of normality will flag departures far too small to matter. Skewness and kurtosis figures, or a plot, usually tell you more than a significance test does. Chapter 4 shows one place where the shape of a distribution matters a great deal: some endogeneity corrections only work if the variable is clearly non-normal.

3.3.2 The connection back

The normal distribution in this chapter isn't something real data have to follow. It's a reference point. Birthweights sit close to it, which is why the mean, standard deviation and normal curve described them so well and predicted the low-birthweight proportion reasonably well. Much business data sits far from it. The descriptives table is where you find out which kind of variable you're dealing with, and therefore how much to trust its mean and standard deviation.

3.3.3 Questions worth asking

1. **Is the mean a sensible summary of this variable?** Compare it with the median, the minimum and maximum, and the scale's limits.
2. **Is there enough variation for the analysis to find anything?** A scale where everyone answered much the same can't correlate strongly with anything.
3. **Which variables were logged?** How does that change what their coefficients mean?
4. **When normality is discussed, is it the right thing that's being checked?**

3.3.4 One red flag

Check the descriptives for a standard deviation that's bigger than the mean, on a variable that can't be negative, such as revenue, employees or pay. A mean of 450 and a standard deviation of 1,200 means the distribution is extremely right-skewed: a handful of very large cases are pulling both numbers upwards. If the paper then puts that variable into a regression untransformed, those few cases may be driving the results. That's not necessarily wrong, but the paper should say so, and ideally show what happens without them.

3.3.5 If you want to read more

Geoff Norman's "Likert scales, levels of measurement and the 'laws' of statistics" (Norman, 2010) is a short, punchy defence of using ordinary statistical methods on survey data. It takes the arguments against head on, so you can judge for yourself. It'll equip you for the next time a reviewer, or a supervisor, tells you that you can't take the mean of a Likert scale.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 4

Causation

This chapter supplements the content in Lecture 4.

Every chapter so far has been building toward a question this one finally asks directly: when two things seem to appear together, or somehow to be related or associated, what would it actually take for us to say that one *causes* the other?

Start with a claim you'll meet constantly once you leave this course: a company runs a sales-training programme, and afterwards, salespeople who scored higher on the training assessment also post higher sales numbers. Training works — right?

Maybe. But notice what that claim is doing. It's drawing an arrow: training score → sales performance. One thing causing another. And the honest first question about any arrow like that isn't "is the correlation strong enough?" — it's "what's actually at the other end of it, and is anything else pointing in?"

Here's a trick worth keeping, because you'll use it constantly from here on: a variable is *endogenous* if something else in your explanation points into it — if it's being explained. In other words, it appears at the *end* of an arrow in a diagram. It's *exogenous* if nothing is — if it's just a starting point you're taking as given. The easiest way to remember which is which: the end of an *arrow* points *into* the *endo*genous variable. I genuinely came up with that myself, and no one else has ever found it as satisfying as I do, but it's yours now too.

In the training example, sales performance is what we're trying to explain — arrow pointing in, endogenous. Training score is what we're using to explain it. But is training score really where the story starts? Or is something else pointing into *that* too — something that also has its own arrow running straight to sales performance, independent of training altogether?

That's an example in microcosm of the whole problem this chapter is about. The question is "have I actually found the arrow I think I've found, or am I

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mistaken? For example, have I mistaken two effects of the same cause for one thing causing the other?" Sometimes the honest answer is genuinely hard to reach. Occasionally, someone reaches it anyway, against a famous opponent, with nothing but the strength of their argument. That's where we'll start.

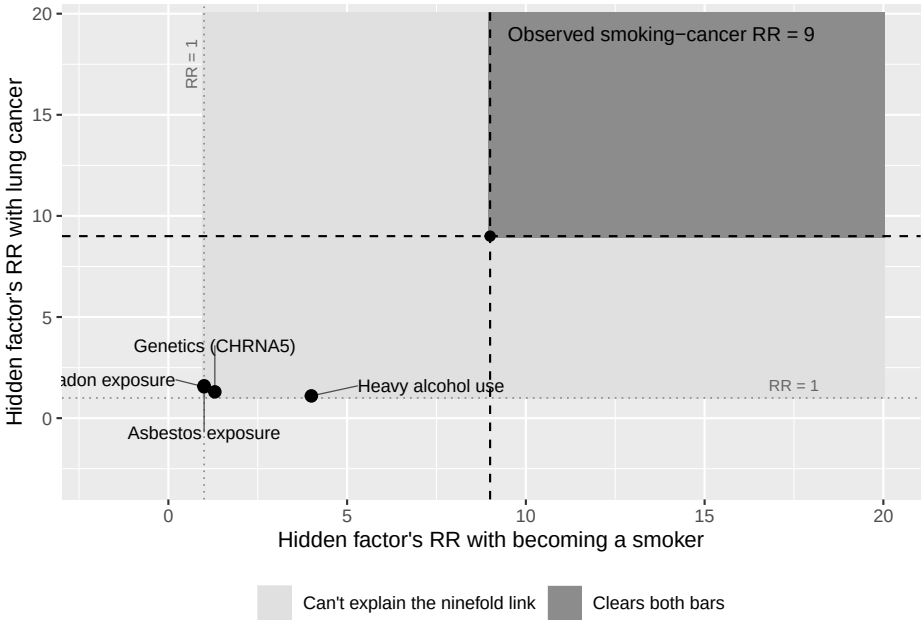
4.1 Confounding

4.1.1 The Cornfield Test

By the late 1950s, the evidence linking smoking to lung cancer was overwhelming in one specific sense: smokers were getting lung cancer at roughly nine times the rate of non-smokers. Not a modest bump — ninefold.

And yet Ronald Fisher — one of the most important statisticians who ever lived, a genuine giant of the field — spent his final years arguing, in print, that this proved nothing about causation. His alternative: some unmeasured (e.g. genetic) factor might make a person both more likely to smoke *and* more likely to develop lung cancer. That would mean no causal arrow between smoking and cancer at all — just two effects of the same hidden cause. Fisher wasn't a crank. He was applying a rule this whole chapter is going to keep insisting on: correlation isn't causation, and a third variable can produce exactly this pattern. He was right that it *could*.

Jerome Cornfield's answer wasn't to argue Fisher's logic was wrong, because it wasn't. But to ask a sharper question: *how big would this hidden factor have to be?* For a confounder to fully explain away a ninefold relationship, it isn't enough for it to be loosely associated with smoking and loosely associated with cancer. It has to be at least as strongly tied to smoking as smoking is to cancer, *and* at least as strongly tied to cancer as smoking is to cancer (Cornfield et al., 1959). Both bars, not one. Let's look at this visually.



Below and to the left of both dashed lines, a candidate confounder simply can't do the job — it isn't strongly enough tied to *both* smoking and cancer to produce a ninefold relationship on its own. Above and to the right of both, it isn't mathematically ruled out — but now you have to actually name a real factor that's this strongly linked to becoming a smoker *and* this strongly linked to cancer, independently of smoking itself.

Here's what happens when you actually go looking. The closest thing to Fisher's hypothesised factor that modern genetics has found is a variant near the *CHRNA5* nicotine-receptor gene: three independent genome-wide studies linked it to roughly a 30% increase in lung cancer risk (RR $\approx$ 1.3). A related genetic marker had separately been linked to nicotine dependence (Saccone et al., 2007), but Amos et al. (2008) found only weak, inconsistent evidence that their own variant affects smoking behaviour directly — and, in their UK sample, no elevated cancer risk at all among people who'd never smoked, hinting the gene's effect may run specifically through smoking rather than sit independently upstream of it, in the way a genuine confounder would need to. Even on the most generous reading then, it's real and replicated, but still nowhere near either bar.

Radon gas is a genuine lung cancer risk factor — high residential exposure raises risk by roughly 50–60% (RR $\approx$ 1.5–1.6, see Heinzl et al., 2024) — but has essentially no relationship to who smokes (RR $\approx$ 1). Occupational asbestos exposure lands in almost the same place: RR $\approx$ 1.6 for lung cancer (Goodman et al., 1999), no meaningful link to smoking status. Heavy drinking is the mirror image — heavy drinkers are roughly four times more likely to smoke than light or

non-drinkers (RR ≈ 4, see Garnett et al., 2022) — but alcohol's own independent effect on lung cancer, once smoking is separated out, is essentially null (RR ≈ 1.1, not statistically significant, see Bagnardi et al., 2011). Four real, well-studied factors, marked on the plot above, four different ways of falling short. Not one of them gets remotely close to doing the job Fisher's argument needed. The point isn't that clearing both bars is impossible. It's that nobody, in over sixty years, has ever produced a real candidate that gets close.

Nobody has ever found that factor. Fisher never named one. The argument didn't kill the possibility of confounding — it just showed exactly what confounding would need to look like, and let the silence answer the rest.

This kind of reasoning got a name and a formula decades after Cornfield: the E-value (VanderWeele & Ding, 2017), the minimum strength a confounder would need with both the exposure and the outcome to fully explain an observed relationship away, on its own — one number instead of two bars to check. It's simple enough to work out yourself: $E = RR + \sqrt{RR(RR-1)}$. Run the smoking figure through it: $E = 9 + \sqrt{9 \times 8} \approx 17.5$. Notice that's higher than 9 — the precise version of Cornfield's argument sets the bar even further out of reach than his own simplified statement did.

Management research rarely reports risk ratios — you'll see correlations and standardised regression coefficients instead. The same test still works: a correlation (or a standardised coefficient) plugs directly into $RR \approx e^{0.91r}$ (Mathur & VanderWeele, 2020; VanderWeele, 2017), and from there, the same formula as above. Take r = 0.3 — a correlation many reviewers would treat as a genuinely strong finding. RR ≈ 1.31, E-value ≈ 1.96. A confounder barely stronger than the effect itself would fully explain it away. Compare that to smoking's 17.5, and you get a real sense of how differently "solid" two published findings can actually be, even when both get called significant.

It's worth keeping this formula, because you'll meet plenty of causal claims that don't hold up nearly this well once you check. Next time a headline reports a relative risk — or a paper reports a correlation — this is the one-line test for how seriously to take it.

4.1.2 The Everyday Version

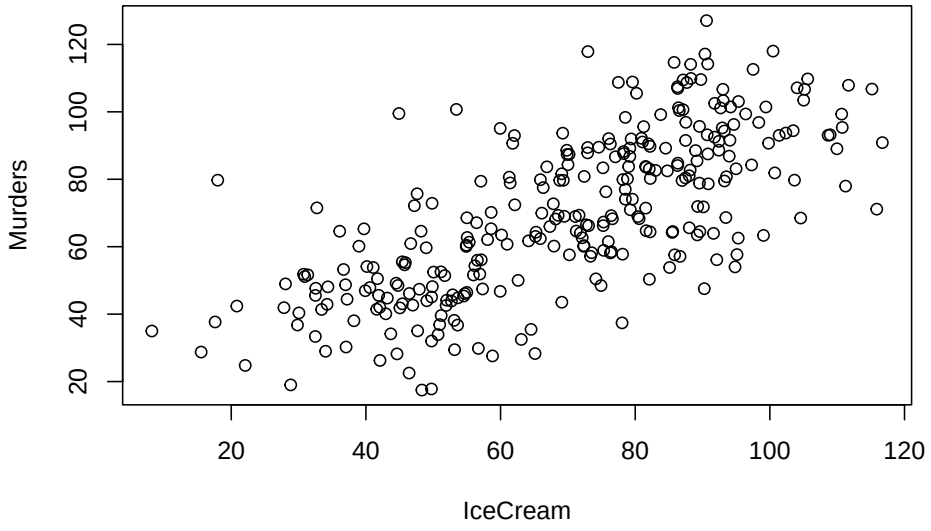
You won't often get to run an argument as tight as Cornfield's yourself — it took decades, a real historical dataset, and the simple fact that nobody ever found the missing factor. But the everyday version of watching a confound do its work doesn't need any of that. It shows up anywhere you have the data and don't think to check.

You've probably met the classic case already, possibly at a dinner party where someone was showing off. Across US cities, ice cream sales and murder rates rise and fall together — and not weakly, either. Plot one against the other and you'll get a relationship strong enough that if you'd found it in your own data,

you'd be drafting the abstract. This isn't a made-up example either — Tyler Vigen's *Spurious Correlations* project puts the real figure at r = 0.715 across US data from 1990–2021 (Vigen, 2021).

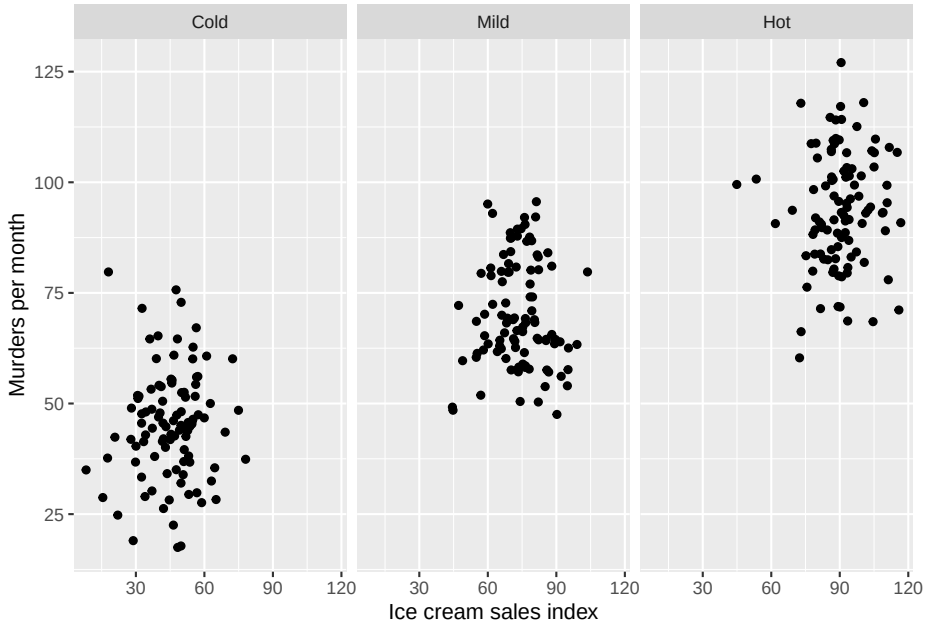
Nobody thinks ice cream causes murder. The obvious culprit is temperature: hot weather means more people buying ice cream, and — for reasons criminologists actually take seriously — more violent crime too. Ice cream and murder never talk to each other. They're each just reacting to the same third thing.

That's a confounder: a variable that causes both of the things you're looking at, with no arrow running between them at all. And here's what makes it worth more than a paragraph — we can simulate exactly this, watch the correlation appear, and then watch it disappear.




```
## [1] 0.7137528
```

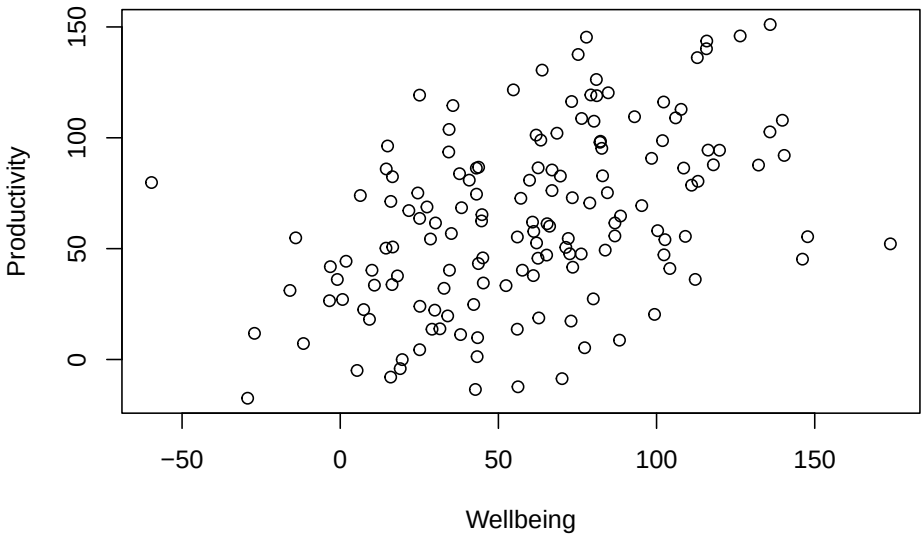
There's the strong relationship, and it's entirely manufactured — neither variable had any effect on the other anywhere in that code. Both were generated purely from `Temp`, city-months' average monthly temperature. Now let's split the sample into cold, mild, and hot months, and look again within each band.




```
##         Cold         Mild          Hot
##  0.053207223 -0.009546515  0.067120654
```

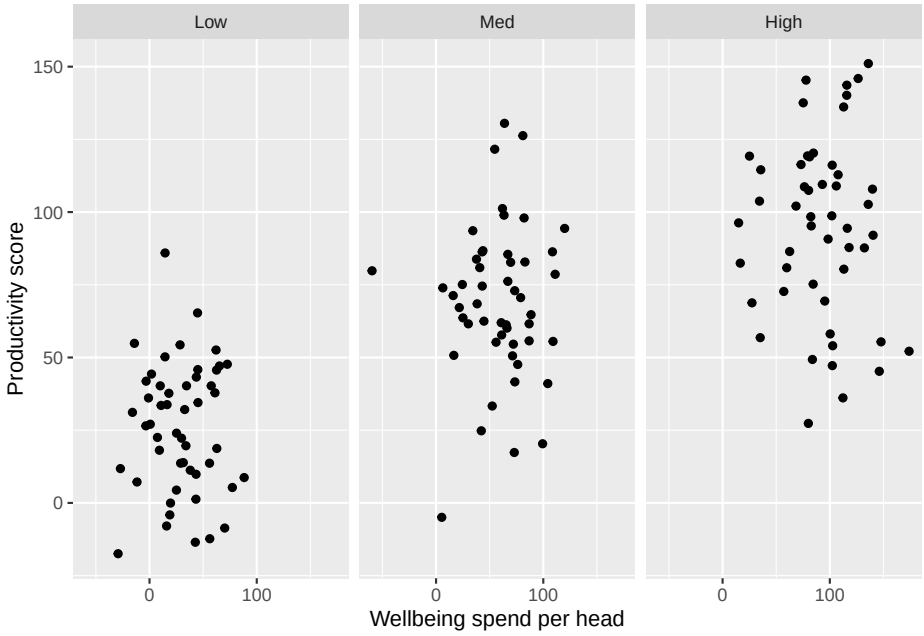
Not weakened. Gone. Within each band the correlation is close enough to zero to be noise. Nothing about ice cream or murder changed; we just stopped letting temperature do the confounding.

Suppose a well-meaning HR department rolls out a wellbeing programme, and a year later, teams that spent more per head on it are visibly more productive. It's tempting to write the business case right there.




```
## [1] 0.4463021
```

But run the same check first. Departments with bigger budgets can afford *both* better wellbeing programmes *and* better everything else: staffing, tools, time.




```
##          Low          Med         High
## -0.006996962  0.046068957 -0.031799717
```

Split by department resourcing, and the wellbeing–productivity relationship

folds up exactly the way ice cream and murder did. A lot of strategy concepts are likely to fall prey to this type of confounding. Companies with more resources can spend money on 'cool strategy idea x' as well as all the stuff we already know makes firms successful.

Confounding is a story about a third variable doing its work invisibly behind two others. Selection bias is stranger still — because sometimes there isn't a third variable at all. The relationship gets built entirely by how the sample was assembled.

4.2 Selection Bias

4.2.1 All the Data Isn't All the Data

Here's the plainest version, and can happen right in your classroom. If I ask everyone in my PhD Quant Methods class a question — say, how satisfied are you with the course so far — and I use their answers to draw a conclusion about the whole cohort, I've made an error before I've even started analysing anything. The students didn't choose to be in the sample by accident. They came to this specific class, at this specific time, for a reason, which makes them systematically different from the people who didn't. Maybe more conscientious. Maybe just better at reading a timetable. Either way, whatever I conclude from the room doesn't automatically generalise to everyone I'm claiming to describe.

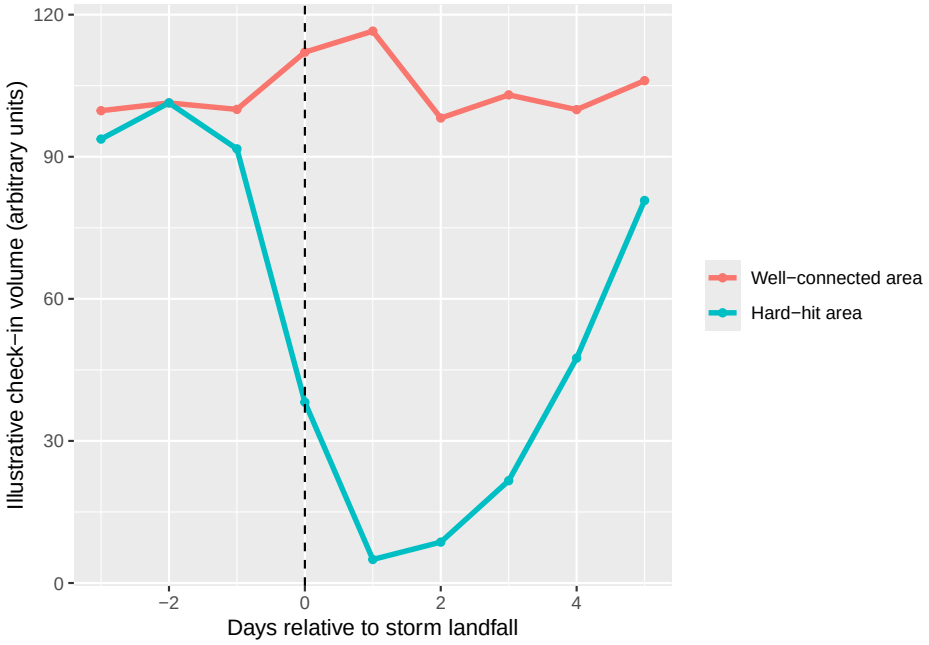
That much is intuitive once it's pointed out — nobody's surprised that a self-selected sample is biased. What catches people out is the belief that this problem goes away once you stop sampling and start using "all the data." A lot of big-data enthusiasm rests on exactly that idea: no sample, no sampling bias, done.

Here's why I don't buy it. A 2013 study (Grinberg et al., 2013) analysed millions of geotagged Twitter and Foursquare posts to track real-world activity through Hurricane Sandy — genuinely all the check-in data available, not a survey, not a sample. Their headline finding: a spike in nightlife check-ins right after the storm passed, which they read as people celebrating having made it through. Taken at face value, that's a nice, tidy story, and the paper reports it as evidence their method was picking up something real.

Read it a little more suspiciously, though, and a different story is sitting right there in the same dataset. Check-in activity — on both platforms — comes overwhelmingly from wherever population density and phone connectivity are highest. During Sandy, that wasn't evenly distributed across New York at all: swathes of Lower Manhattan lost power outright, but Midtown and Upper Manhattan largely didn't, while the outer boroughs and the Long Island shoreline — hit hardest by flooding and the longest power outages — simply went dark and stopped generating data almost entirely. "All the data" quietly became "all the data from the people whose phones still worked," and the people whose phones still worked were disproportionately the ones for whom Sandy had been the

least disruptive. A "celebration" spike and "the sample skewed toward people having a comparatively normal night" would look identical in a dataset built this way — and only one of those explanations was on the table.

The chart below isn't a reproduction of their actual figures, but an illustration of the same pattern, built from numbers I've made up for the purpose. For the real data and figures, see Grinberg et al. (2013).



To be clear, the original researchers don't raise the geography question themselves, and their finding is entirely honest as reported. It's what happens when you take a real published result and ask, on your own, who actually generated this data and whether that's the population the conclusion is really about. That's a habit worth having with everything you read, not just this one paper.

That's selection bias without anyone choosing anything, in the ordinary sense — nobody self-selected into being a well-connected New Yorker with power that night. The selection happened in how the data was collected, not in who volunteered. It's the same underlying problem wearing a different disguise, and it's the reason "we have all the data" is a claim worth being suspicious of rather than reassured by.

Sometimes, though, selection bias is more deliberate than either of those cases, and what it does to a relationship is stranger still, because it can manufacture one that was never there at all.

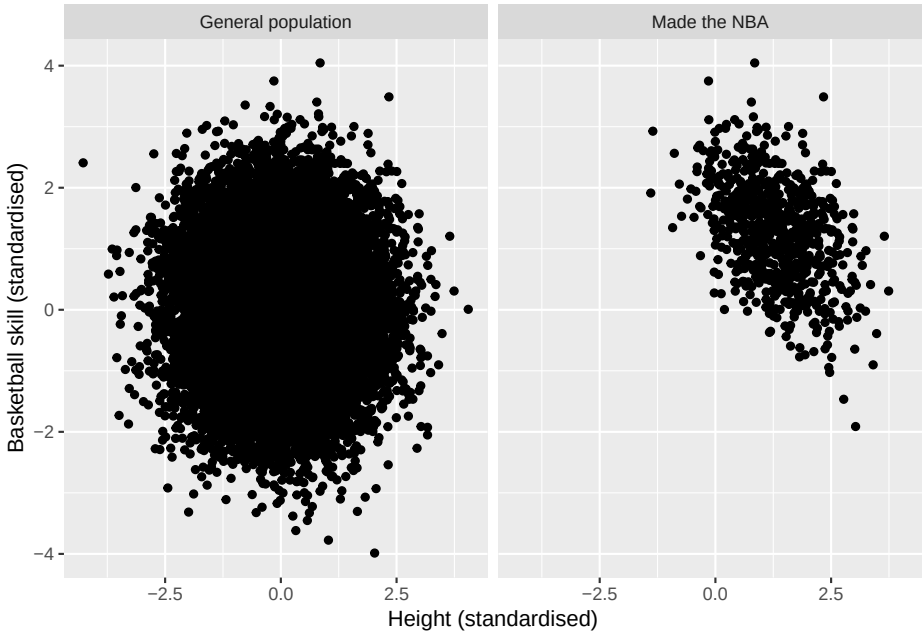
4.2.2 Manufacturing a Relationship from Nothing

Here's one that should feel more surprising than the last, because there's no obvious third variable to blame. The surprise is that a relationship shows up *at all*.

Take height and basketball skill. Across the general population, these two things are essentially unrelated — being tall doesn't make you more skilled with a basketball, and being skilled doesn't make you tall. But look only at NBA players, and you'll find something odd: among people who've actually made the league, height and skill are *negatively* correlated. The shorter players are, on average, the more skilled ones. Michael Jordan wasn't especially tall for an NBA player. Elite players like Allen Iverson and John Stockton were shorter still. This isn't a fluke of famous names — it holds up in the data, and it's been documented both informally (Johnson, 2017) and in peer-reviewed work on NBA draft sampling bias (Economou et al., 2021).

```
## [1] 0.00365453
```

```
## [1] -0.4462717
```



In the pool of 20,000 draft-eligible prospects, height and skill are unrelated, exactly as we built them to be. Select down to the players good enough or tall enough (or both) to clear the bar, though, and a clear negative relationship appears out of nothing.

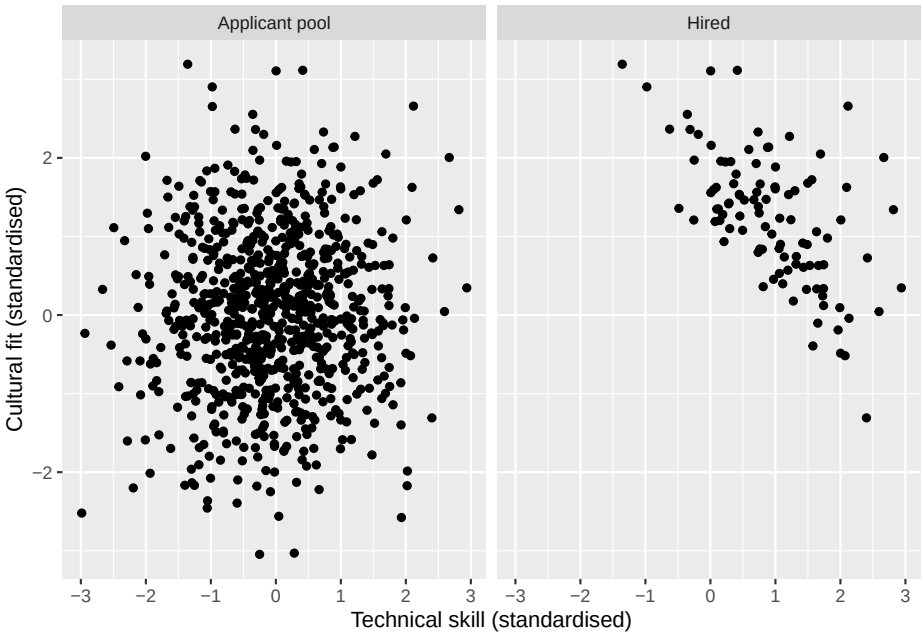
Nothing about height or skill changed between "general population" and "NBA players." What changed is how the sample was built. The NBA doesn't select

on height, and it doesn't select on skill — it selects on *some combination of the two clearing a bar*. You can make a roster by being exceptionally tall, or by being exceptionally skilled, but rarely need to be both. Once you only look at the players who cleared that bar, you're looking at a sample where — among the tall ones, the skill requirement was more negotiable, and among the short ones, it wasn't. Select on a sum, and you manufacture a relationship between the parts that was never there in the population you drew from.

Now the version with more relevance. A company hiring for a role almost always screens (typically implicitly) on some blend of technical skill and cultural fit — nobody demands maximum on both, but the combination has to clear a bar. In the applicant pool at large, skill and fit have nothing to do with each other.

```
## [1] 0.04425896
```

```
## [1] -0.6027258
```



Among the people who get *hired*, though, they look negatively related — the highly skilled hires will tend to be the ones who scraped by on fit, and vice versa. If a manager later notices "our most technically brilliant hires are always the awkward ones" and starts wondering whether skill and personality genuinely trade off, they're not wrong about the pattern they are seeing, they're wrong about where it came from.

4.2.3 Why selection feels different

Most people find confounding intuitive once it's pointed out. There's a culprit, a third variable you can name, like temperature, and the fix is to hold it steady. Selection feels different, and the feeling is right. In the NBA and hiring examples there is no culprit at all. Nothing in the population links height and skill, or skill and fit. The relationship was manufactured entirely by the filter that decided who we got to see.

Here's the way I find most helpful to think about it. Put the two kinds of demonstration side by side, which is what Figure 4.1 does, with freshly simulated versions of each. In the ice cream example, looking *within* one band of temperature made a spurious relationship disappear. In the NBA example, looking *within* the group of people who made the league made a spurious relationship appear. Both times we did essentially the same thing, which was to look at a subset. The opposite results come from where the subsetting variable sits. Temperature is a common *cause* of ice cream sales and murders, and holding a common cause steady removes a fake link. Making the NBA is a common *consequence* of height and skill, and holding a common consequence steady creates one. If you read further, you'll find this called a "collider," and medics know one version of it as Berkson's paradox.

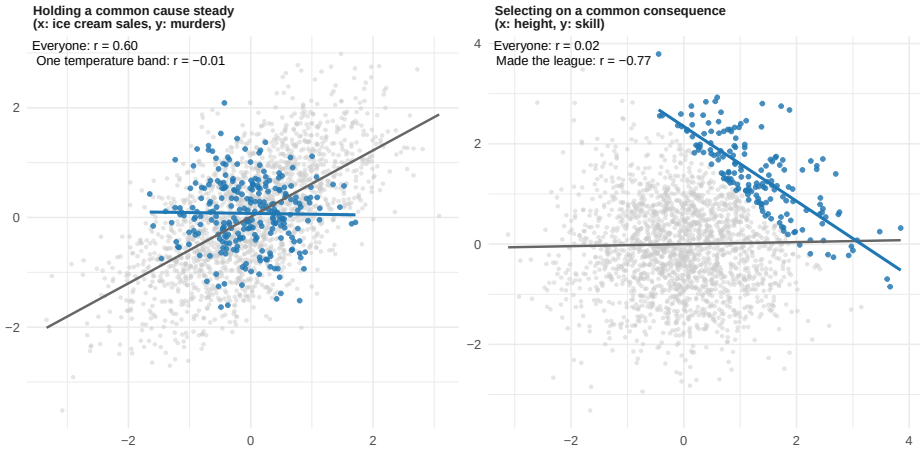


Figure 4.1: The same move, opposite results. Left: looking within one band of a common cause (temperature) makes a spurious relationship disappear. Right: looking only at cases that passed a filter depending on both variables (making the league) makes a relationship appear from nothing. Grey: everyone. Blue: the subset.

One uncomfortable implication follows directly. Holding a variable steady by *selecting* on it and holding it steady *statistically* are, for this purpose, the same move. So if a researcher "controls for" something that is itself a consequence of both X and Y, they manufacture a relationship in exactly the way the NBA

filter did. More control variables are not automatically safer.

It's also worth knowing that "selection" gets used for two quite different problems. **Selection into treatment** is about who *chooses* the thing being studied. Suppose we're back with the sales training from the start of this chapter, and the most motivated salespeople sign up, and motivated salespeople would have sold more anyway. That is really a confounding problem wearing a selection label, because motivation is the culprit. **Selection into the sample** is about who ends up in the data at all, and that's the NBA problem, the one that genuinely behaves differently. Measuring more things doesn't fix it, and neither does a bigger sample, because the distortion is built into who you're looking at.

Two clues give selection away. The first is a restricted range. If a variable's spread in your data is noticeably narrower than it is in the population, something has filtered it, which is exactly the compressed cloud in the NBA plots. The second is a sample defined by an *outcome*: players who made the league, candidates who were hired. Whenever inclusion depends on how things turned out, it's worth asking what the missing cases would look like, and whether adding them back would change the relationship.

4.3 Endogeneity

Here's a useful way to see all three of these threats side by side, because they don't compete with each other — the same study can be hit by more than one at once, and they often travel together in the same variable pair.

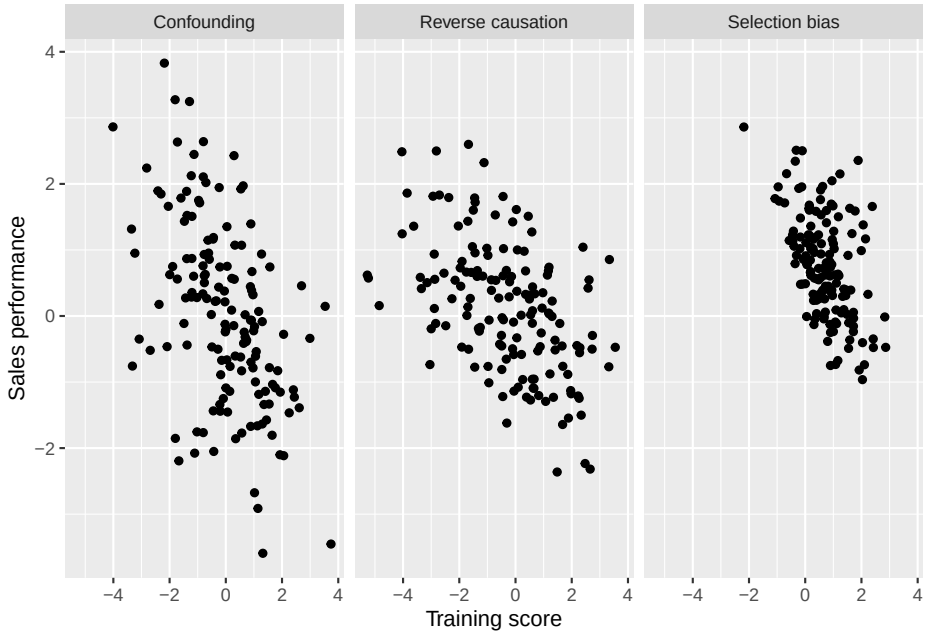
Go back to the sales-training question this chapter opened with: does a training course actually improve sales performance? And, let's switch it up a bit to the idea that we might observe a *negative* relationship in the data. How?

Confounding, you've already watched at work. Here it's tenure: more experienced salespeople tend to coast through a mandatory training assessment — they've heard it all before, so they don't try especially hard, and their scores show it — but that same experience is exactly what makes them better at actually selling (or, you could say that salespeople get to have a long tenure in the job *because* they keep making their numbers…). Leave tenure unmeasured, and part of what looks like "training hurts performance" is really "people who don't need the training are also the people who are already good at the job". That's an arrow pointing into both variables from somewhere you never accounted for.

Reverse causation is a different failure with the same symptom. Maybe training doesn't cause sales performance at all — maybe sales performance causes how seriously someone takes training. A salesperson who's already succeeding has no urgency to engage with a mandatory course; someone who's struggling might throw themselves into it, hoping it turns things around. If that's what's happening, the arrow runs backwards from what the study assumed. Your performance level causes your engagement with the training.

Selection bias adds a third route in. If the training course is optional, who actually chooses to do it? Often the people who feel they need it — not the stars. If untrained salespeople outperform the trained group on average, an easy but wrong conclusion is "training makes people worse." The real explanation is that the people who opted in started from a lower baseline; training might have helped every one of them, and you'd never see it, because the comparison was never fair to begin with.

Here's the same idea made visual: three simulated datasets, one per mechanism above, each built to produce a similar-looking negative relationship between training score and sales performance — similar strength, similar slope. The spread is a different story, and deliberately so.



Confounding and reverse causation really are that hard to tell apart — same strength, same slope, same scatter. Selection bias gives itself away a little: the third panel is visibly tighter, because restricting the sample to whoever clears a bar doesn't just distort the relationship, it shrinks the range of data you ever get to see in the first place. That's a signature of selection effects, though spotting it still depends on already suspecting where to look. Three entirely different mechanisms — an omitted variable driving both sides, an arrow running the wrong direction, a systematically unrepresentative sample — and all three produce the identical statistical symptom: your independent variable ends up tangled up with the very thing you're trying to explain, the error term.

That symptom has a name — *endogeneity* — and it's the word you'll meet most often if you go on to read (good) empirical papers seriously. It doesn't tell you

which of the three is happening. It just tells you that something is, and that the confident arrow you drew earlier might not be pointing the way you think.

4.4 The Power of Randomisation

So confounding, reverse causation, and selection bias share something important: all three can manufacture a relationship — or hide one — without you doing anything wrong with your statistics. The correlation you calculate is correct. It's the story you tell about why it's there that goes wrong.

There's one design choice that neutralises confounding before you ever reach the analysis stage: randomisation. If who receives a treatment is decided by something like a coin flip, rather than by any pre-existing characteristic, then every confounder — measured or not, thought of or not — ends up, on average, equally represented on both sides of the comparison. Randomisation doesn't remove confounders. It makes them irrelevant, because they can no longer line up systematically with which group you're in.

Hold onto that idea. A few chapters from now, you'll meet a two-study design built around exactly this problem — and you'll be equipped to see why the second study's fix actually works before the book explains it.

4.5 Reading the literature

This chapter used deliberately simple tools: splitting data into bands, simulating populations, drawing scatter plots side by side. That was on purpose, because the ideas are hard enough without machinery getting in the way. But when you open a management journal, nobody is banding ice cream sales by temperature. What you'll find instead, usually near the end of the method section, is a paragraph headed something like "Addressing endogeneity," followed by a list of techniques you've probably never run and may never run. In the lecture I rattle off a few of these names partly to sound clever. Here I want to do something more useful than that.

The aim is for you to read that paragraph with some confidence, even if you couldn't do any of it yourself. There's a trick to it. Almost every technique in there is an attempt to solve one of the three problems in this chapter: confounding, reverse causation, or selection. And every one of them works the same basic way. It swaps the assumption "nothing else is going on" for a different assumption that is hopefully easier to believe. So your job as a reader isn't to check the maths. It's to ask two questions: which problem is this aimed at, and what is it asking me to believe instead?

4.5.1 Where you'll meet this

It helps to sort the techniques into four families, because each family has one characteristic assumption, and that assumption is what you're actually evaluating.

Family one: measure the confounder.

Control variables. "Controlling for firm size, age and industry…" This is the stratification from Section 4.1.2, done inside an equation instead of by splitting into bands. It's the same logic, and it has the same limit. It only deals with confounders that somebody thought of and then measured. The assumption is that nothing important is missing from the list, and the list is usually in a table you can check.

Matching and propensity scores. These build a comparison group that looks like the "treated" group on the variables that were measured. Papers often show a "balance table" to demonstrate how similar the groups are after matching. Done well, matching is a genuinely solid method, and it has one real advantage over simply adding control variables: it makes the researcher show you how comparable the groups actually are. What it can't do is make them comparable on anything that wasn't measured. So read it for what it is: a careful way of dealing with the confounders somebody thought of, with the same limit as control variables.

Family two: use structure in the data.

Fixed effects. Here each firm, person or family is compared with itself. That removes everything stable about the unit, whether it was measured or not, which is genuinely powerful. A firm's culture, a manager's personality, a family's genes: if it doesn't change during the study, it's gone. But fixed effects do nothing about things that *do* change over time, and those are often exactly the things you'd worry about.

Difference-in-differences. This compares how a treated group changed over time with how an untreated group changed. It relies on the two groups having been on parallel paths before anything happened. Good papers show you the pre-treatment trends so you can judge for yourself. If they don't show you, ask why not.

Family three: find something that moves X from outside.

Instrumental variables (often "two-stage least squares" or 2SLS). The idea is to find something that pushes X around but affects Y *only* through X. Confusingly, IV here means "instrumental variable," not the "independent variable" I've been abbreviating the same way.

Two things matter. First, the "only through X" part (the *exclusion restriction*) can't be tested with the data. It has to be argued. So read the argument, and try hard to think of another route from the instrument to Y. Second, the

instrument has to actually move X. Papers report a "first-stage F" to show this, usually against a rule of thumb of 10. It's a threshold, and as Chapter 11 will argue, thresholds are heuristics, not laws. More recent work suggests that for the usual significance test to behave as advertised, the F may need to be closer to 100 (D. S. Lee et al., 2020). You don't need to settle that argument. Just notice when a paper treats "F > 10" as the end of the conversation.

Control functions. These are increasingly common, especially in marketing, and they're easier to read than they sound (Petrin & Train, 2010). The researcher uses an instrument to predict X. They then take the part of X the instrument *couldn't* predict (the residual) and add it to the main model as an extra variable. That leftover part is where the contamination lives. So a control function is, quite literally, a way of turning an unmeasured confounder into a control variable, which puts you right back on familiar ground from family one.

In simple linear models it gives the same answer as 2SLS. Its appeal is that it still works in models where 2SLS gets awkward, such as models of which brand a customer chooses. The key point for you as a reader is that it is not an escape from the instrument. It rests on exactly the same exclusion restriction, so the same questions apply.

Gaussian copulas. These go one step further and need no instrument at all (Park & Gupta, 2012). In practice most applications use a control-function version, adding a "copula term" to the model as an extra variable. What replaces the instrument is an assumption about the *shape* of X's distribution. The method only works if X is clearly non-normal, and simulation work suggests X needs to be quite strongly skewed for the method to perform well in the sample sizes typical of management research (Becker et al., 2022).

This is where Chapter 3 pays off. Many survey constructs are an average of several Likert items, and such averages will often be much closer to symmetrical than that. So check the descriptives table. If the variable being "corrected" looks roughly normal, the correction is standing on thin ice.

Family four: model who got into the sample.

This family deals with the second kind of selection from Section 4.2.3, the filter problem. It's worth knowing where that problem hides in management research, because papers rarely announce it. It shows up in who answers surveys, in which firms survived long enough to appear in a database, in which start-ups got through an investor's gate, in which employees stayed at a firm rather than leaving, and in which studies got published. That last one is big enough to get its own treatment in Chapter 13.

One caution before going further, because this is exactly where research myths start. A low response rate is not automatically a problem. What matters is whether the people who didn't respond differ *systematically* from those who did, in ways connected to what's being studied. A survey with a modest response rate, where non-response is essentially unrelated to the variables of interest, can

tell you more than one with a high response rate, where the people who didn't reply are precisely the ones the study is about. It's the same logic as the rest of this chapter: a filter only distorts a relationship if it's connected to the variables in that relationship.

You'll meet a handful of responses to it, and it helps to know one piece of vocabulary first. Papers distinguish **selection on observables**, where inclusion depends on things the researcher measured, from **selection on unobservables**, where it depends on things they didn't. The first is the easier problem, and the methods for it carry the same assumption as family one. The second needs something stronger.

Non-response checks. In survey research, especially in marketing, the most common move you'll see is to compare early and late respondents, on the theory that late ones resemble the people who never replied, and then to conclude from a non-significant difference that non-response bias isn't a problem. This is almost always cited to Armstrong and Overton (Armstrong & Overton, 1977). But it isn't really what they recommended. Their paper was about *estimating and adjusting for* non-response bias, not testing for its absence. When Armstrong later checked how the paper had been used, he and his co-author found that 49 of the 50 studies they examined had reported its findings improperly (Wright & Armstrong, 2008). The most common error was running a significance test, finding nothing, and concluding that no bias was present, a move you'll learn to distrust in Chapter 11. So treat the early-versus-late comparison as something that has commonly been done, to questionable effect, not as a solution. A more informative check compares respondents with known figures for the whole population they were drawn from, such as firm size or sector from a database, though it can only check the variables that happen to be available for everyone.

Weighting. This reweights the sample so that it matches the population on known characteristics, counting under-represented cases more heavily. It's selection on observables, so it fixes only what it can see.

Heckman models. These are the ones you'll see most often in strategy and management journals (Heckman, 1979). They work in two stages. The first stage predicts *who gets into the sample*. From that it builds an extra variable, usually called the inverse Mills ratio or lambda, which is added to the main model as a correction. If that sounds familiar, it should: it's the control-function idea from family three, applied to selection instead of to X. It also comes with the same catch. For the method to be trustworthy, the first stage needs at least one variable that affects *whether* a case is selected but not the outcome itself. That's an exclusion restriction again. Without one, the correction rests almost entirely on the assumed shape of a statistical distribution, and it can do more harm than good. One review of strategy and organisation theory journals found that only about a third of papers using the method even claimed to have such a variable (Wolfolds & Siegel, 2019).

Another review of Heckman models in strategy research found them used in-

creasingly but inconsistently, and made three points worth carrying into any paper you read (Certo et al., 2016). First, selection only biases the result you care about if X itself helps predict who got selected, so look at the first stage. Second, a significant lambda does not by itself show that selection bias exists. Third, Heckman addresses bias from the sample, not other sources of endogeneity, so it isn't a general-purpose fix.

Treatment-effects and switching models. You'll sometimes see Heckman invoked to deal with who *chose* a strategy, such as which firms entered a market or adopted a practice. That's selection into treatment, which, as the chapter showed, is closer to confounding than to filtering. There are related models designed for exactly that case, usually labelled treatment-effects or endogenous switching models. When a paper says "we used a Heckman correction," a useful first question is simply: selection into what?

Bounds. Occasionally, rather than correcting the estimate, a paper will report the range of values the true effect could plausibly take, given what's missing. It's rare in management and more common in economics. It's also refreshingly honest when you do see it, because it shows you the size of what we don't know rather than hiding it.

4.5.2 The connection back

None of these is randomisation. Randomisation is the only design that breaks the link between X and *every* confounder, measured or unmeasured, without asking you to take anything else on trust. It does this on average across all the random assignments you might have got, not perfectly in any single sample. That's why randomised studies still report balance and can still get unlucky.

Everything on the list above is closer to randomisation by argument. That's not a criticism. For most interesting questions in management you can't randomise, and these methods are how serious people deal with that. But it does mean the quality of the argument is what you're evaluating, not the name of the technique.

4.5.3 Questions worth asking

1. **What verb are they using?** Results sections say "is associated with." Discussion sections have a habit of drifting into "drives," "shapes" or "leads to." Did the design earn that drift?
2. **What's the most obvious third variable?** If you can think of one in thirty seconds, is it in the model? If it isn't, did the authors say why?
3. **Which family is their fix from, and what does that family assume?** Did they argue for the assumption, or just name the method and move on?
4. **Who is in the sample, and how did they get there?** Is this selection into treatment (a confounding problem) or selection into the sample (a

filter problem)? Was inclusion decided by an outcome? What would the missing cases look like?

5. **Did the correction change anything?** If the corrected and uncorrected estimates are nearly identical, that could mean endogeneity wasn't a problem. It could equally mean the correction wasn't strong enough to find it. Which reading do the authors choose, and what's their evidence?
6. **How strong would an unmeasured confounder need to be?** You can ask the E-value question even if the authors never computed one.

4.5.4 One red flag

The endogeneity paragraph that reads like a ritual: one technique, one sentence, and "endogeneity was therefore not a concern." Its most common form is a test. The control-function residual, the copula term or the Heckman lambda is "not significant," and the authors conclude there's no endogeneity. As you'll see in Chapter 11, that doesn't follow: a non-significant result is an absence of evidence, not evidence of absence. These tests can also be weak, especially with a feeble instrument or a barely skewed variable.

My strong suspicion, and I say this as someone who has been on both ends of the review process, is that a lot of these paragraphs exist because a reviewer asked for them, not because the authors thought hard about their design. A check that the authors never expected to fail isn't really a check. When you see one, go back to question 2 and do the thinking yourself.

4.5.5 If you want to read more

Two books do this properly and are written for people who want to understand the ideas rather than become econometricians. Both are free to read online. Scott Cunningham's *Causal Inference: The Mixtape* (Cunningham, 2021) covers every technique in this section with real examples and a good sense of humour. Nick Huntington-Klein's *The Effect* (Huntington-Klein, 2025) is, if anything, gentler. Its first half builds the concepts almost without equations, and its treatment of colliders and selection is the clearest I know. If you'd rather start with something you can read on a train, Judea Pearl and Dana Mackenzie's *The Book of Why* (Pearl & Mackenzie, 2018) is a popular account of the same ideas, written by one of the people who developed them. It's also where much of my own thinking about Cornfield and the smoking debate, earlier in this chapter, came from. Their chapter on it is superb.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 5

Associations and Relationships

This chapter supplements the content in Lecture 5.

5.1 Correlations and Associations

First, let's look at the basic concepts of correlation, using some data from the World Happiness Report, and GDP from the World Bank, all put together by the team at Our World in Data (n.d.).

Their site is well worth exploring in its own right.

I can just double check the data by describing it. `describe()` is what I'll reach for throughout the book whenever I'm looking at a data set for the first time — it gives a fuller statistical snapshot than just eyeballing the first few rows, including how much data is missing, which turns out to matter here.

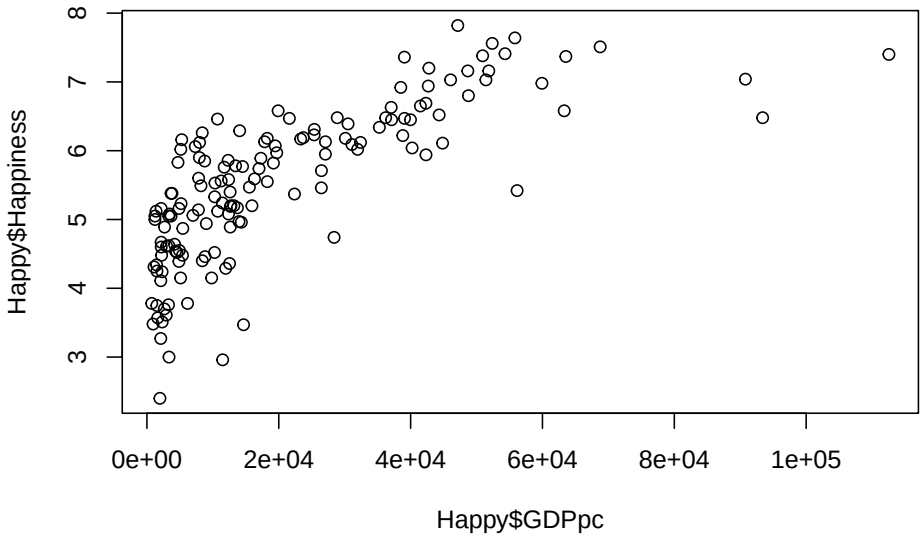
```
##           vars   n        mean           sd     median     trimmed        mad
## Country*     1 249      125.00        72.02     125.00      125.00      91.92
## Happiness    2 153        5.49         1.12       5.53        5.52       1.16
## GDPpc        3 197    20463.88     20717.34   12655.00    17037.01   13338.95
## Pop          4 242 59178643.60 331869505.09 5596196.00 12318073.38 8185922.38
##              min          max        range  skew kurtosis          se
## Country*     1.0 2.490000e+02 2.480000e+02  0.00    -1.21        4.56
## Happiness    2.4 7.820000e+00 5.420000e+00 -0.26    -0.38        0.09
## GDPpc      731.0 1.125570e+05 1.118260e+05  1.58     2.55     1476.05
## Pop        809.0 4.663087e+09 4.663086e+09 11.65   152.44 21333379.77
```

We've got 249 countries, with population, GDP per capita, and Happiness. There's a fair bit of missing data, which you can see by looking at the num-

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bers in the n column. That's ok for now, as long as we are aware of it.

Let's plot this data using a very basic scatterplot.



Interpretation: there's seemingly an association here. As one variable increases, the other does too.

It's worth noting that I have chosen to put GDPpc on the x-axis, which *implies* that the driver of Happiness is GDP. This replicates the way that most of the media articles that are regularly written on this kind of topic do it.

Nevertheless, this is just a scatterplot, so there is no real requirement to put any particular one of the variables on the x-axis — it's just a correlation, and Chapter 4 is where we deal properly with why that's *not* the same thing as causation.

Either way, all we have here is a basic data display. While, as I showed previously, simply displaying data can be very powerful, it is still only essentially an 'eyeball' analysis, and different people might draw different conclusions here (we'll discuss that later on).

What we want to do, is have some sort of objective **metric** for the relationship, and for this, we have the **correlation** statistic. We met the idea of correlations in the last chapter, but here we can dig into them properly.

```
##
##  Pearson's product-moment correlation
##
## data:  Happy$Happiness and Happy$GDPpc
## t = 13.502, df = 146, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
```

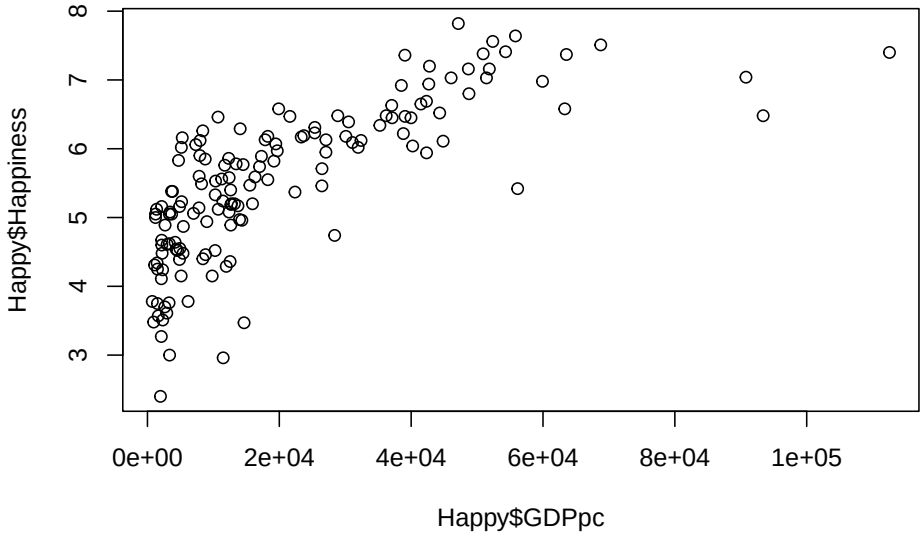
```
##  0.6636288 0.8092331
## sample estimates:
##      cor
## 0.745184
```

Here, we can see that the correlation is 0.75

But what does that mean? We can **return to the slides for some explanation**.

5.2 Introducing Non-linear Associations

Let's return to the scatterplot of GDPpc and Happiness



As the *Economist* analysis discussed in the lectures suggests, this association also looks kind of *non-linear*.

However, unlike the *Economist*, we can also say that there are multiple ways to look at this:

First, you could take the *Economist's* view that this is probably a **non-linear relationship.** Indeed, we can check this out by *transforming* GDPpc and comparing the results between transformed and non-transformed data.

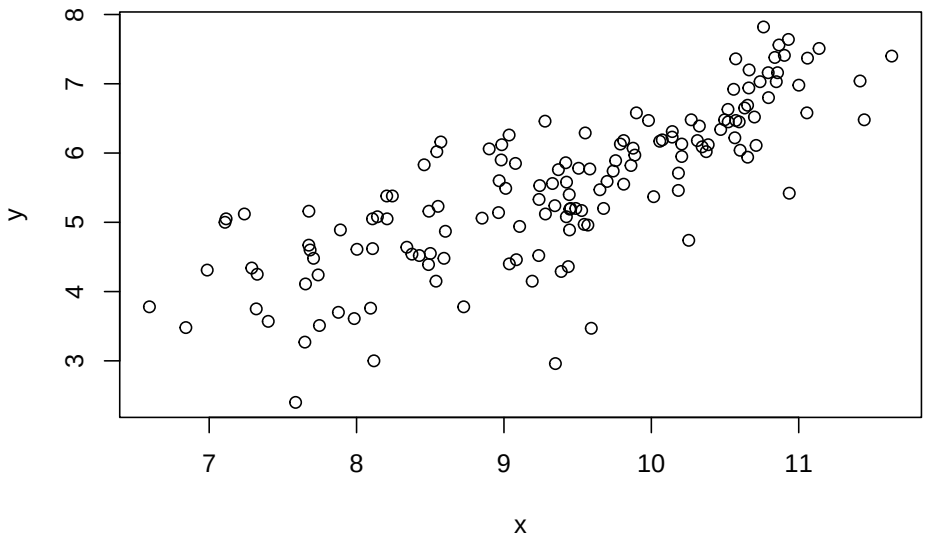
Let's have a try at that. What we will do is transform the GDPpc variable using a log transform (i.e. taking the natural logarithm of GDPpc).

```
##           vars   n        mean          sd     median     trimmed        mad
## Country*     1 249      125.00       72.02     125.00      125.00      91.92
## Happiness    2 153        5.49        1.12       5.53        5.52       1.16
## GDPpc        3 197    20463.88    20717.34   12655.00    17037.01   13338.95
```

```
## Pop          4 242 59178643.60 331869505.09 5596196.00 12318073.38 8185922.38
## GDPpc_log    5 197        9.38         1.14       9.45        9.41       1.33
##               min          max        range  skew kurtosis          se
## Country*     1.00 2.490000e+02 2.480000e+02  0.00    -1.21        4.56
## Happiness    2.40 7.820000e+00 5.420000e+00 -0.26    -0.38        0.09
## GDPpc      731.00 1.125570e+05 1.118260e+05  1.58     2.55     1476.05
## Pop        809.00 4.663087e+09 4.663086e+09 11.65   152.44 21333379.77
## GDPpc_log    6.59 1.163000e+01 5.040000e+00 -0.25    -0.79        0.08
```

We can see there is a new variable now, GDPpc_log, which is the log transformed GDPpc variable - basically the natural log of the original GDPpc variable.

Of course, there are many other possible transforms. I touched on them earlier, in Chapter 2, and I'll elaborate a little more on it below, for now let's rerun the plot using GDPpc_log



That relationship definitely looks more linear, and if that was my only criteria for deciding what to do with the data, I would definitely now run with this idea, as the *Economist* did in their article

With that in mind, let's take a quick look at the correlations for these variables. First the original GDPpc, and then the log transformed one.

```
##
##  Pearson's product-moment correlation
##
## data:  Happy$Happiness and Happy$GDPpc
## t = 13.502, df = 146, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
```

```
##  0.6636288 0.8092331
## sample estimates:
##      cor
## 0.745184
```

```
##
##  Pearson's product-moment correlation
##
## data:  Happy$Happiness and Happy$GDPpc_log
## t = 15.916, df = 146, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.7287785 0.8487456
## sample estimates:
##       cor
## 0.7964703
```

We can see that the association is stronger (the correlation is higher) for the log GDPpc variable, although to be honest, it is not a very big difference here, compared to other data sets that I have seen discussed in the media. That's interesting in itself (note we are using 2020 data, so maybe the pandemic has something to do with it?)

However, **transformations** of data are kind of a complicated topic though. In essence, what we are doing is changing the distribution of one (or more) of the variables, in a systematic way, in order to convert a non-linear relationship (which may be unable to be tested with methods like correlation and regression) into a linear one.

There are many different types of transformations available to you, and the one you choose depends on the form of the non-linearity. A **log** transform converts an exponential (or similar) relation into a linear one. But, there are specific transforms for many different forms. Oftentimes, it's not always obvious what exactly to do, and people often default to transforming **any** non-linear-looking relationship, or highly skewed variable, using log or maybe the quadratic (squaring). It's not really that simple though, and I think we need to be a bit more logical and theory-driven on **why** we transform variables.

In this case, I admit that the logic makes sense, that there is still a relationship between GDPpc and happiness at the top end, but you just have to increase GDPpc by a **lot more** to get the same change in happiness, compared to the change needed at the lower end of GDPpc. And, making that transform to come to that conclusion actually makes a difference to how we think of the influence of GDPpc on happiness, and could conceivably lead to policy changes compared to the prior non-transformed analysis.

Even so, I have always had a slightly different possibility on my mind. In fact, it may be that there are **two distinct groups** of country, low and high income, and different **linear** associations within those groups. That's a similar-ish idea

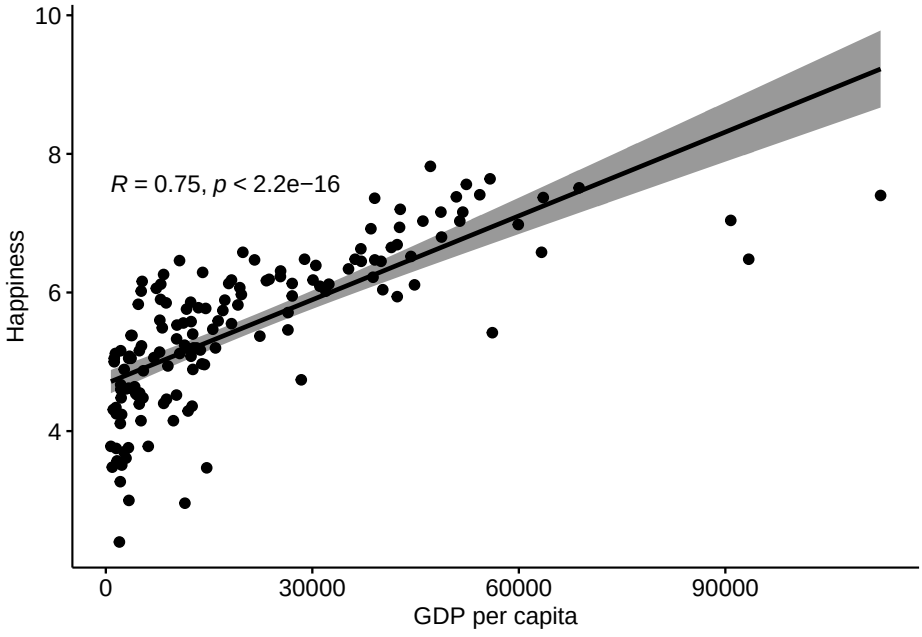
to the non-linear one, but it is a bit different, and I actually prefer that idea. A nice challenge would be to work out a way to test which of the two alternative explanations is actually better supported by the data…

Maybe I'll leave that for another day though.

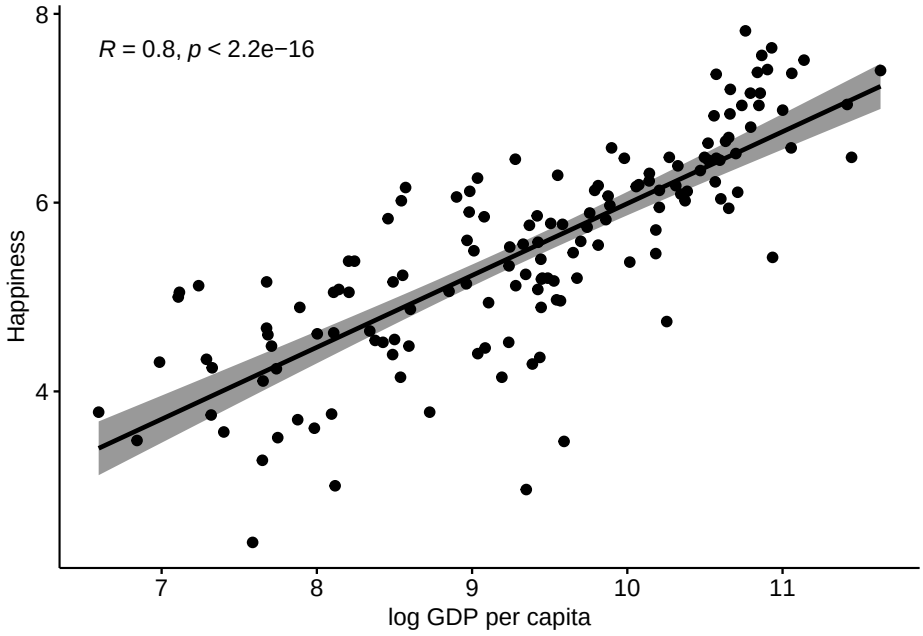
We can **return to the slides** for a bit more discussion.

5.3 Regression

We can add a **regression line** to the GDPpc / Happiness scatterplot, for some extra information over the correlation.



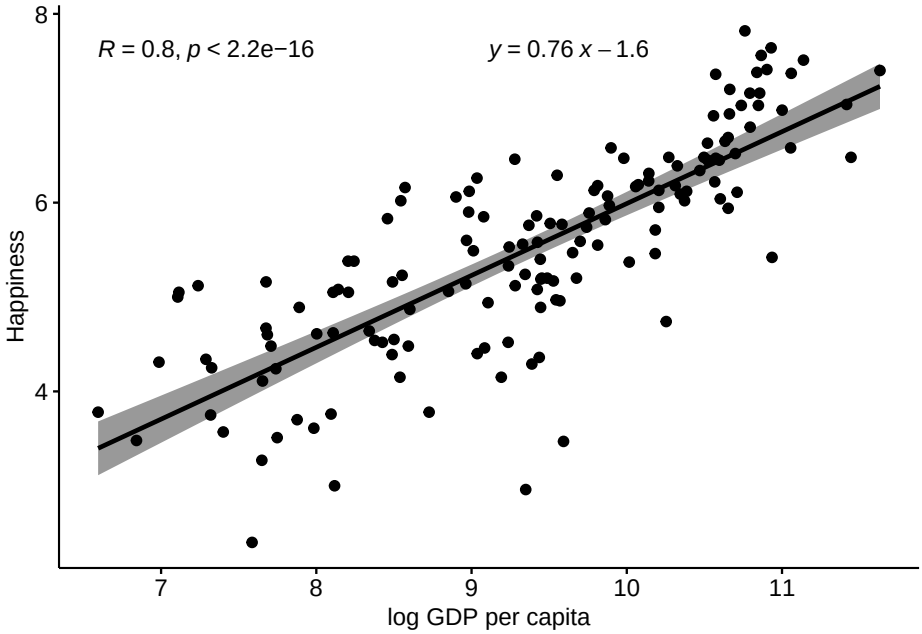
We can also do this with the log GDPpc to compare.



You can see that using the log GDPpc variable does improve things, but not by that much in my view. That said, it's clear that the line is a bit more 'precise' (the errors are lower which can be seen by the narrower shaded regions). So, if we are looking for the 'best' model, it's pretty clear that the log GDPpc model is better than the simple GDPpc model.

Either way, the regression line adds a layer of information on to the correlation, which allows us to predict y from x.

How so? Well, let's add the regression equation to the chart to see:



This adds an *intercept*, and with that plus the coefficient we have all that we need to plot a straight line, which we can *extrapolate* to higher values of GDPpc (although remember this is the logged GDP variable) and predict what the happiness scores would be.

This is obviously viable to the extent we can justify the relationship, and also within the parameters of our variables, and how confident we are that the relationship is consistent at all levels of the variables.

return to the slides

5.4 Multiple Regression

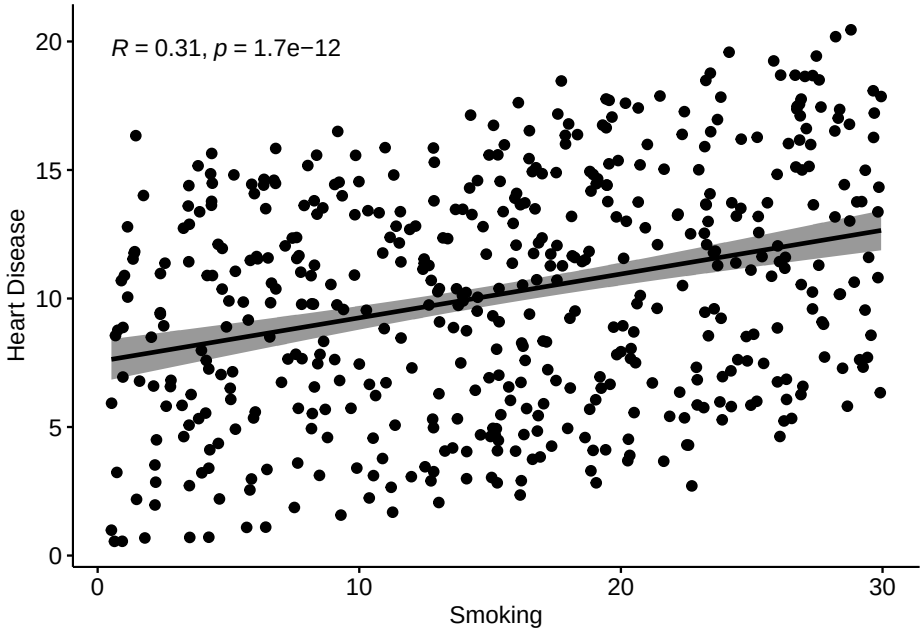
Here, we will use a simple three-variable set of simulated data, which represents rates of smoking, rates of cycling, and heart disease incidence. This data is available from Scribbr's guide to linear regression in R (Bevans, 2020).

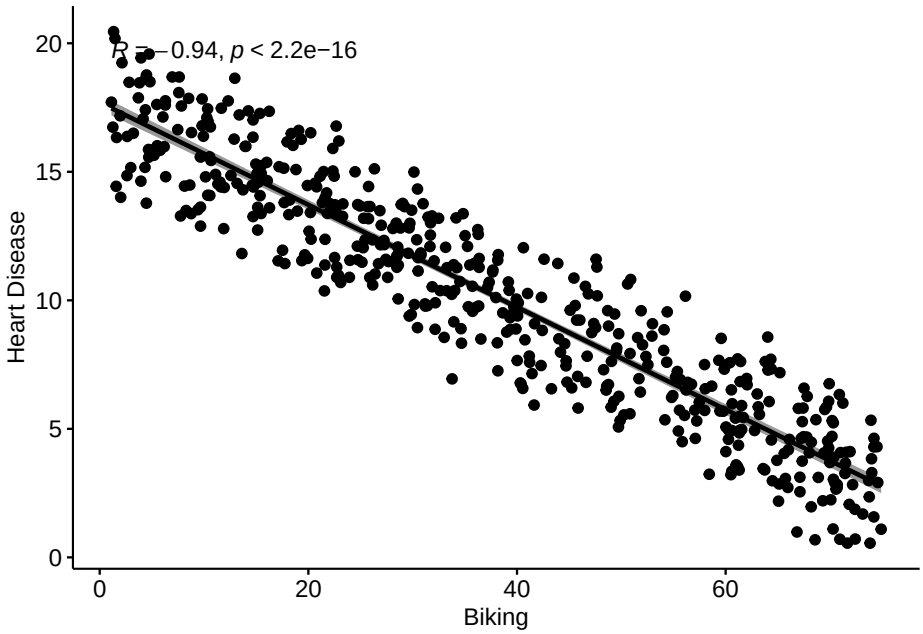
Remember, this is **simulated data** not real data.

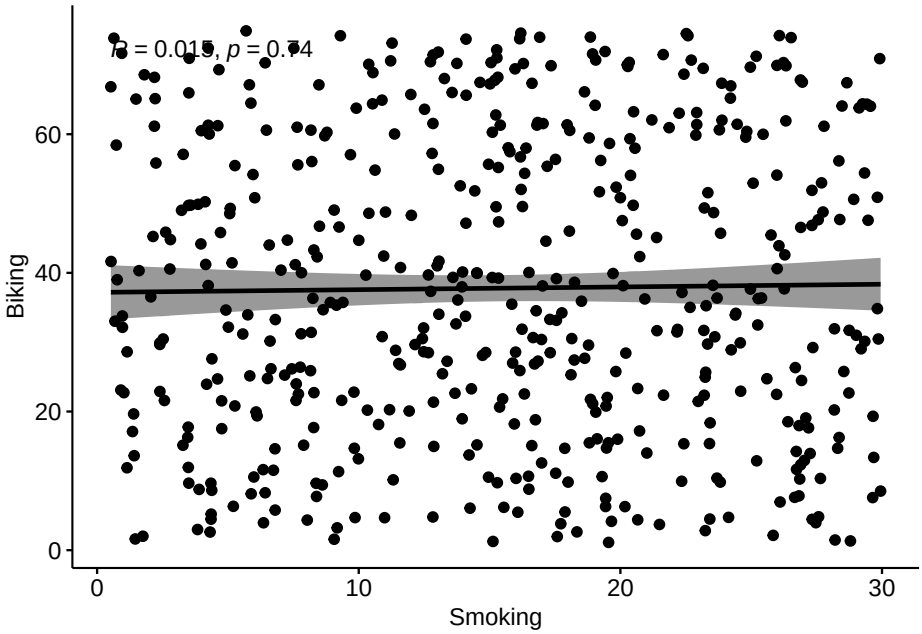
```
##               vars   n   mean     sd median trimmed    mad  min    max  range
## ...1             1 498 249.50 143.90 249.50  249.50 184.58 1.00 498.00 497.00
## biking           2 498  37.79  21.48  35.82   37.71  27.51 1.12  74.91  73.79
## smoking          3 498  15.44   8.29  15.81   15.47  10.86 0.53  29.95  29.42
## heart.disease    4 498  10.17   4.57  10.39   10.18   5.42 0.55  20.45  19.90
##                skew kurtosis   se
## ...1           0.00    -1.21 6.45
## biking         0.07    -1.22 0.96
```

```
## smoking       -0.04    -1.12 0.37
## heart.disease -0.03    -0.93 0.20
```

Next, let's run three simple regressions among the three variables







Remember, these are *simple* regression lines, with bivariate scatterplots. What I mean, is the effect of biking on heart disease does not take account of the effect of smoking on heart disease. What we need to do is run a model which takes account of *both* the predictors, as **we can explain on the slide-deck...**

Now we can run the Multiple Regression Model

```
##
## Call:
## lm(formula = heart.disease ~ biking + smoking, data = Heart)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1789 -0.4463  0.0362  0.4422  1.9331
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 14.984658   0.080137  186.99   <2e-16 ***
## biking      -0.200133   0.001366 -146.53   <2e-16 ***
## smoking      0.178334   0.003539   50.39   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.654 on 495 degrees of freedom
## Multiple R-squared:  0.9796, Adjusted R-squared:  0.9795
## F-statistic: 1.19e+04 on 2 and 495 DF,  p-value: < 2.2e-16
```

First, we look at the R2, or more accurately the Adjusted R2, which takes account of the number of variables (they are usually very similar). The R2 essentially measures how much variation in your Dependent Variable is explained by your model (i.e. your regression predictors). It ranges from 0 to 1, and higher means more variance explained. Here we see a very high AdjR2 of 0.98, almost 1. This is simulated data of course, and it is quite rare to see such high R2s in real data. For the most part, you want R2 to be high, but if it gets too high (say above 0.8 but this is not a 'rule') you might start to get worried about whether your IVs and DVs are actually distinct things. Apart from that, standards of what makes an acceptable R2 are highly dependent on the context and field you are working in, and you'll no doubt see lots of different ideas throughout your career on what is acceptable. Of course, that only scratches the surface of this issue, and I'll say more in the lecture slides.

Next, we look at the coefficients, and they can be interpreted as follows:

A 1-unit increase in biking is associated, on average, with a 0.2-unit decrease in heart disease, holding smoking constant. A 1-unit increase in smoking is associated, on average, with a 0.17-unit increase in heart disease, holding biking constant.

Notice the wording there. It's tempting to say that biking *leads to* less heart disease, and in everyday conversation most of us would. But a regression coefficient on its own describes how the variables move together in these data, after adjusting for the other predictors in the model. "Holding smoking constant" is the stratification from Section 4.1.2, done inside an equation instead of by splitting into bands, and it has the same limit: it only deals with the variables that are actually in the model. Whether biking *causes* less heart disease is a question about the design and the argument, not about the regression output. With simulated data the question is academic. With real data, it's the whole question.

Of course, what these 'units' mean depends on how the variables were measured. That's exactly why it's hard to say from these numbers alone whether either effect is large or small: we'd need to know what a unit of biking, smoking or heart disease actually represents.

However, because the *scales of the variables* are different, its not really possible to compare the sizes of the effects. So to some extent we don't know which of the two variables has the 'bigger' effect here, relatively at least.

To do that, we need to standardise the coefficients, which basically means putting all the variables on the **same scale**. Beforehand, they all were on different scales, which you can see by going back up and looking at the range, mean, and standard deviation of the original variables. You can see for yourself they are all different.

We can solve this issue by creating a new set of standardised data, and running the model on that, as follows:

```
##                vars   n mean sd median trimmed  mad   min  max range  skew
## ...1              1 498    0  1   0.00       0 1.28 -1.73 1.73  3.45  0.00
## biking            2 498    0  1  -0.09       0 1.28 -1.71 1.73  3.43  0.07
## smoking           3 498    0  1   0.05       0 1.31 -1.80 1.75  3.55 -0.04
## heart.disease     4 498    0  1   0.05       0 1.19 -2.10 2.25  4.35 -0.03
##                kurtosis   se
## ...1              -1.21 0.04
## biking            -1.22 0.04
## smoking           -1.12 0.04
## heart.disease     -0.93 0.04
```

You know these variables are standardised as they now all have a mean of 0 and a standard deviation of 1. Now, let's run our model on this standardised data set.

```
##
## Call:
## lm(formula = heart.disease ~ biking + smoking, data = std_Heart)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47658 -0.09762  0.00792  0.09671  0.42283
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.205e-16  6.410e-03    0.00        1
## biking      -9.403e-01  6.418e-03 -146.53   <2e-16 ***
## smoking      3.234e-01  6.418e-03   50.39   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1431 on 495 degrees of freedom
## Multiple R-squared:  0.9796, Adjusted R-squared:  0.9795
## F-statistic: 1.19e+04 on 2 and 495 DF,  p-value: < 2.2e-16
```

You can see the things that have changed are the 'Estimates', which are now standardised.

In decimal, they are -0.94 for biking and 0.32 for Smoking.

The correct interpretation of these standardised effects is that for every 1SD increase in biking, you expect a 0.94SD decrease in heart disease, and for every 1SD increase in smoking, you would expect a 0.32 increase in heart disease.

So, in this sample, biking has a much **larger** standardised association with heart disease than smoking does (although in the opposite direction). One caution, though. Standardised coefficients are measured in standard deviations, and those standard deviations come from this particular sample. If another sample had much more variation in smoking, say, the standardised coefficient for smok-

ing would change even if the underlying relationship didn't. So comparisons like this are specific to the sample, and are safest between samples that look alike.

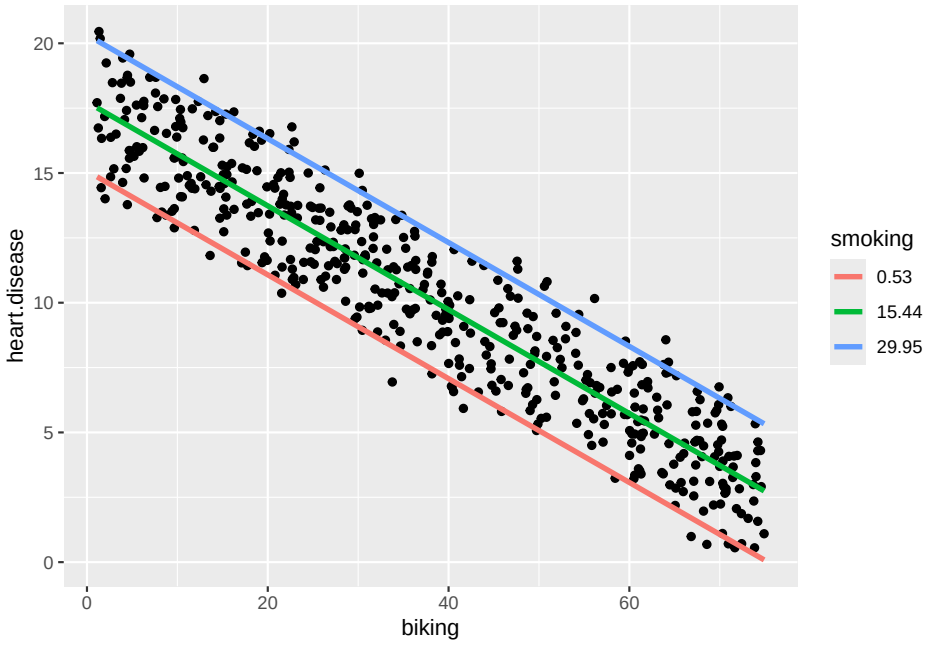
5.5 Visualising Multiple Regression

There are loads of different ways to visualise multiple regression. It's not trivial, because we have more than two variables, so we can't use the techniques we used already.

Some people like to use 3D-style plots, which look cool, but are not always easy to interpret, and take quite a lot of extra work, for what I would say is not that much pay-off (if any).

In this case, we can use a pretty simple visualisation, where we could plot the relationship of biking to heart disease at different levels of smoking.

This would be a fairly typical way to do things if we thought the relationship between biking and heart disease changed according to the level of smoking, in which case it would be a moderator. Here, it doesn't really work that way, but it's a cool visualisation regardless. It does require some data prep, but not that much, and I took the basic idea from the website where I sourced the data: https://www.scribbr.com/statistics/linear-regression-in-r/



Here, we can see that the effect of smoking is really just to raise the likelihood of heart disease, however much biking you do.

So, for a given person who bikes a given amount, if they smoke more they will have a higher risk than a person who bikes the same amount but smokes less.

But, *for a given smoker*, the more they bike, the lower their risk of heart disease, to the extent that if a heavy smoker bikes enough, their actual risk of heart disease could even be lower than a non-smoker who does not bike at all.

Don't forget, this is not real data, but the point stands. Get on your bike.

5.6 Reading the literature

Almost every quantitative paper based on survey or archival data has the same first table. It lists every variable in the study, gives its mean and standard deviation, and then fills a triangle with the correlations between every pair. It's usually Table 1, and most readers skip straight past it to the regressions. I think that's a mistake. Table 1 is the rawest view of the data you'll ever get from a paper, and almost everything that follows is built out of it. This section is about reading that table, and the regression tables that come after it, with the ideas from this chapter in hand.

5.6.1 Where you'll meet this

Table 1: the correlation matrix. I'll leave the means and standard deviations aside and focus on the correlations. Three things are worth checking. First, find the correlation between the main predictor and the outcome. If the paper's central claim is that X drives Y, you'd usually expect X and Y to be correlated. If they aren't, but X becomes significant once other variables enter the regression, something unusual is going on. It can be real, but the paper should explain it. Second, look at how strongly the predictors correlate with each other, because very high correlations there foreshadow the multicollinearity problem below. Third, look at how the control variables relate to the main ones. A control that correlates with almost nothing is doing very little work, while one that correlates strongly with both X and Y is exactly the kind of third variable Chapter 4 worried about. It's also worth knowing that, for ordinary regression, the correlations, standard deviations and sample size are enough to reproduce the main estimates. So a paper that leaves them out is withholding the table that would let you check its work.

Alpha on the diagonal. Many Table 1s put a number on the diagonal, where each variable would otherwise correlate with itself. It's usually *Cronbach's alpha*, a measure of the reliability of a multi-item scale. Alpha is built from the correlations between the items in a scale: the more consistently they correlate with each other, and the more items there are, the higher it goes. You'll nearly always see it judged against a threshold of .70. That threshold is a convention, not a law, and alpha is regularly asked to carry more than it can. A high alpha doesn't show that the items measure a single thing, still less the thing the authors say they measure, and a long enough scale can reach a respectable alpha

even when its items are only modestly related (N. Lee & Hooley, 2005). I'll admit to a wry smile at this point. That paper has been cited, more than once, as the authority *for* the .70 threshold. It's the Armstrong and Overton story from Chapter 4 again, only this time it's happening to me.

Multicollinearity. When predictors are strongly correlated with each other, a regression struggles to separate their effects, because the data contain little information about what happens when one moves and the other doesn't. Papers usually check this with *variance inflation factors* (VIFs), declaring values below 10, or sometimes 5, to be fine. Two things are worth knowing. First, in the textbook case multicollinearity doesn't bias the estimates. It makes them less precise by inflating their standard errors, so its main cost is that real effects become harder to detect. Second, the thresholds are rules of thumb, and a well-known review argued that even much higher VIFs don't, on their own, invalidate a regression (O'Brien, 2007). There is a catch, though. More recent work shows that when a model leaves out something the correlated predictors share, the correlation between them can produce false positives, whatever the VIFs say (Kalnins & Praitis Hill, 2025). So "all VIFs were below 10" is a reasonable thing to report and a weak thing to rely on.

Common method variance. If the predictor and the outcome come from the same questionnaire, completed by the same person at the same time, some of their correlation may come from the method itself: the respondent's mood, their wish to appear consistent, or a habit of agreeing with statements. This is *common method variance*, and it's probably the most frequent criticism made of survey research in management. It's a real issue, but it has gathered its own myths. It doesn't automatically inflate every relationship: depending on how it affects each measure, it can shrink a relationship as well as inflate it. And it can't manufacture an interaction or a curved effect out of nothing; if anything, it makes those harder to find (Siemsen et al., 2010). Some researchers have argued that the whole concern has been overstated (Spector, 2006). The most common defence you'll see is *Harman's single-factor test*, which checks whether one factor accounts for most of the variation across all the items. It's usually cited to a review that in fact judged it insensitive (Podsakoff et al., 2003), which is the same fate you saw Armstrong and Overton suffer in Chapter 4. The stronger defences are about design: measuring the predictor and the outcome at different times, or getting them from different sources.

R^2. You met R^2 in this chapter as the proportion of variation in the outcome that the model explains. In management papers you'll often see values that look disappointingly small, .10 or .15, sometimes less. That isn't automatically a problem. If the aim is to test whether a particular relationship exists, rather than to predict the outcome, a model can explain a small share of the total variation and still tell you something real. Most of what people and firms do has many causes. If the aim is prediction, a low R^2 matters a great deal more.

Hierarchical regression and ΔR^2. Many papers enter variables in blocks: control variables first, then the main predictors, and sometimes interactions

last. Each block is a *step*, and the paper reports how much R^2 rises at each one (ΔR^2, "delta R-squared"), usually with a test of whether the rise is significant. The logic is sensible: how much do my variables explain beyond the controls? But notice what it builds in. Any variation the controls share with the main predictors is credited to the controls, because they went in first. Change the order of entry and the ΔR^2 changes too. So a small ΔR^2 for the main predictors can say as much about the controls as about the theory. One naming trap catches almost everyone: "hierarchical regression" in this sense has nothing to do with *hierarchical linear models*, which are models for nested data such as employees within firms. Same word, completely different methods, so check which one a paper means.

5.6.2 The connection back

Almost everything in this section is correlation in a different hat. Alpha is the correlations among a scale's items. VIFs come from the correlations among the predictors. Common method variance is the worry that some correlations are partly an artefact of how they were measured. R^2 is built from the correlations between the outcome and the predictors. That's why Table 1 deserves far more attention than it usually gets.

5.6.3 Questions worth asking

1. **Do the key variables actually correlate in Table 1?** If not, how do the authors explain a significant effect in the regression?
2. **Where did each variable come from?** If the predictor and the outcome share a respondent, a questionnaire and a moment, what did the design do about it, beyond a statistical test?
3. **What do we know about each measure, apart from its alpha?**
4. **Which predictors are strongly correlated with each other?** Do the authors claim to have separated their effects, and how convincingly?
5. **In a hierarchical regression, what went in first?** Would the story change if the order changed?

5.6.4 One red flag

Look along the rows of Table 1 for two variables the paper treats as different things but which correlate at .80 or above. Suppose each has an alpha of .85. With measures that reliable, the highest correlation you could expect to observe between them is also about .85, even if they were the same thing. So a correlation of .85 means the data can't tell them apart. The paper may then be testing whether a construct predicts a slightly relabelled version of itself. It isn't rare, and it hides in plain sight in the one table most people skip.

5.6.5 If you want to read more

Paul Spector's "Method variance in organizational research: Truth or urban legend?" (Spector, 2006) is a short, sceptical and very readable look at how common method variance became a reflex rather than a question. And, at the risk of immodesty, Graham Hooley and I wrote about how coefficient alpha, and the .70 threshold that goes with it, became something close to mythology in marketing research (N. Lee & Hooley, 2005).

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 6

Beginning to Understand Uncertainty

This Chapter relates to the first part of lecture 6. Here, I'll introduce core concepts around **uncertainty** in our results. Understanding that the results of our analysis always contain some level of uncertainty is probably the most critical concept to get our heads around as quantitative social scientists. Most of our job is not really coming up with the actual *statistics*, such as the correlation coefficient, or regression beta, but is more about understanding how to interpret and use those results - i.e. **what they mean**. And, fundamental to that is understanding their uncertainty.

Again, to reiterate the message I sent in class, many of the examples in this Chapter, and later ones, involve **randomness**. This means that the results here may be slightly different numerically to the results in the slides. And, if you were to run these examples yourself, you would also get slightly different results. This is nothing to worry about, because the meaning of the results does not change.

So, to start the journey, let's grab some data.

Here, we will again use the simple three-variable set of simulated data from the demonstration of multiple regression in Chapter 5, which represents rates of smoking, rates of cycling, and heart disease incidence.

```
## # A tibble: 6 x 4
##    ...1 biking smoking heart.disease
##   <dbl>  <dbl>   <dbl>         <dbl>
## 1     1  30.8    10.9          11.8
## 2     2  65.1     2.22          2.85
## 3     3   1.96   17.6          17.2
## 4     4  44.8     2.80          6.82
```


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```
## 5     5  69.4    16.0           4.06
## 6     6  54.4    29.3           9.55
```

You'll notice I use different functions to preview data through the book, and that's deliberate rather than sloppy. `head()`, like I've done here, is just for confirming the data loaded and what variables are there. `describe()` is for a fuller statistical snapshot of a data set I haven't looked at yet — including whether anything's missing — which is why I used it in the last chapter. And a plain `summary()`, like the one below, is for a quick look at a single variable I'm about to use. Three different jobs, so three different functions.

Let's calculate some simple summary statistics from this data set to build on. For example, what is the mean and median for 'smoking'?

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.5259  8.2798 15.8146 15.4350 22.5689 29.9467
```

Now, we know this is **really** simulated data, but let's **imagine** for now that it was actually obtained by an organisation like the Office for National Statistics in the UK, using a survey. We can presume the study was done well, and thus it is based on a true random sampling method, and we assume that the study population matches whatever target population we have in mind (remember the 'inference gaps' discussed in class).

What we really want to know is, how close are these statistics (i.e. the mean and median) to the *true population values* that we would have found if we could survey the entire target population?

Let's begin to think about this by starting to build a table using these statistics, **by going back to the slide deck...**

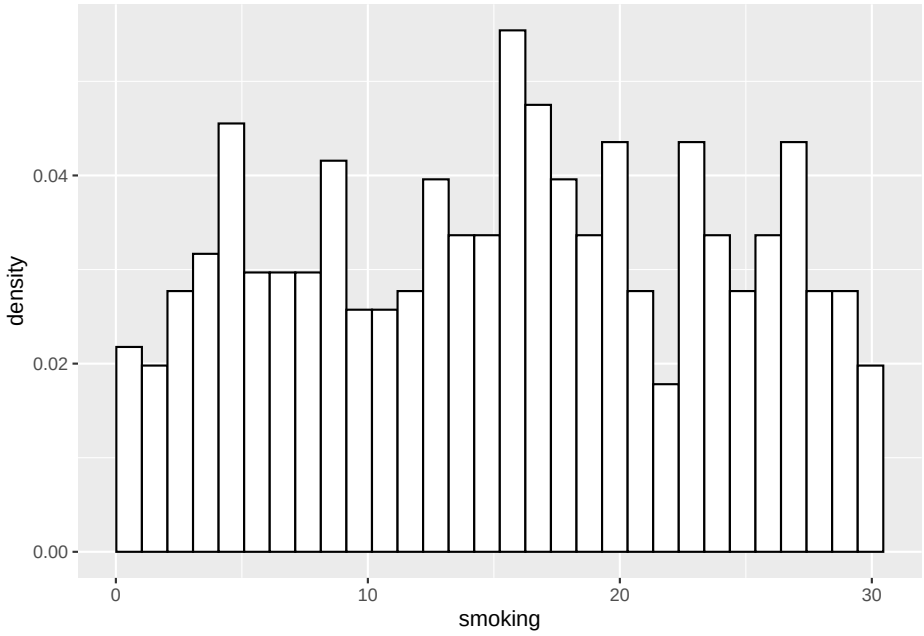
6.1 Demonstration: Sampling from a 'Known' Population

Now, let's go back **one more step**, and demonstrate the uncertainty inherent to sample statistics by way of example.

Let's *now* assume that this sample of 498 people actually *is* the population we are interested in.

What **this** means is, we can actually *draw a sample from this population of 498* and see what happens.

First, let's present the distribution for the entire 'population' of 498.



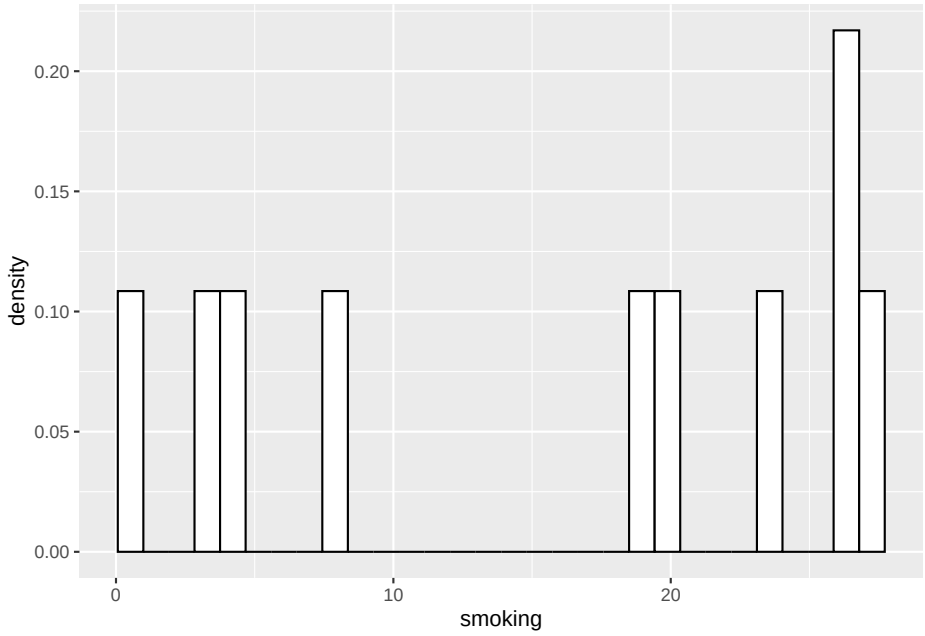
Now, let's literally take a *sample* of 10 random cases from that population of 498. Here, we are **sampling without replacement**, and are thus essentially doing exactly what a hypothetical 'researcher' would do if they drew a random sample of 10 people to complete their survey, from the population of 498.

```
## # A tibble: 10 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   334  13.9   27.3          16.0
##  2   332  51.6   23.3           8.55
##  3   230  22.7    8.29         11.3
##  4   266  66.8    0.526         0.988
##  5   273  24.7    4.60         12.1
##  6   241   6.28  20.2          17.6
##  7   473  74.2   26.1           4.63
##  8   223  49.7    3.49          5.07
##  9   333  74.0   18.9           3.30
## 10   318  43.9   26.1          11.4
```

Next, let's look at the relevant statistics (median and then mean) and distribution of this sample of 10:

```
## [1] 19.51484
```

```
## [1] 15.86817
```

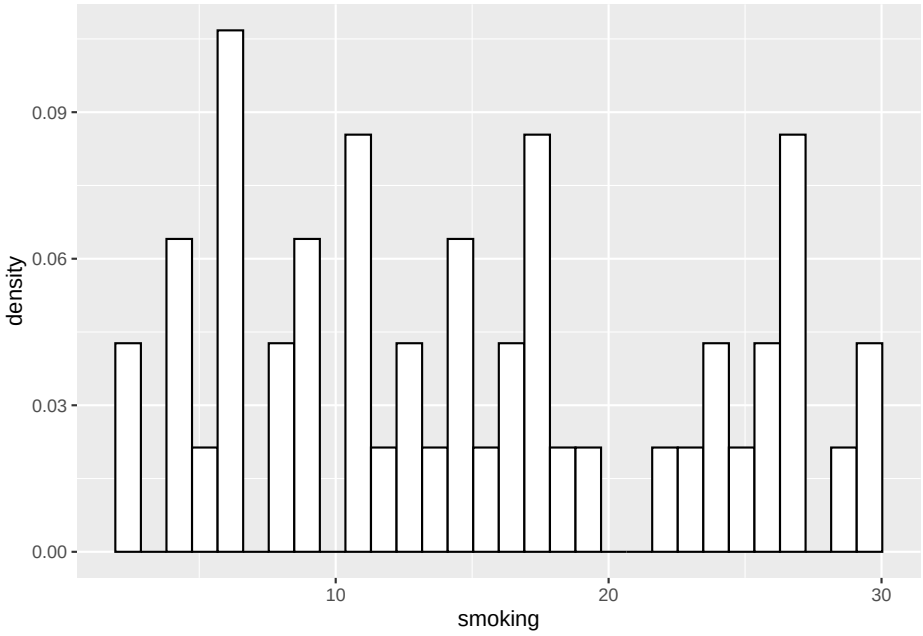


We can do the same for successively larger samples, say 50, and 200:

```
## # A tibble: 50 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   281  73.9    26.5           5.33
##  2   373  74.6    16.2           2.91
##  3   422  25.9    16.2          13.6
##  4    31  44.2     3.97          7.98
##  5   267  70.3     6.42          1.10
##  6     3   1.96   17.6          17.2
##  7    20  63.1    22.9           5.86
##  8   170  29.7     2.40          9.39
##  9   310  60.3     8.79          4.59
## 10   280  28.5    17.3          11.7
## # i 40 more rows
```

```
## [1] 14.3452
```

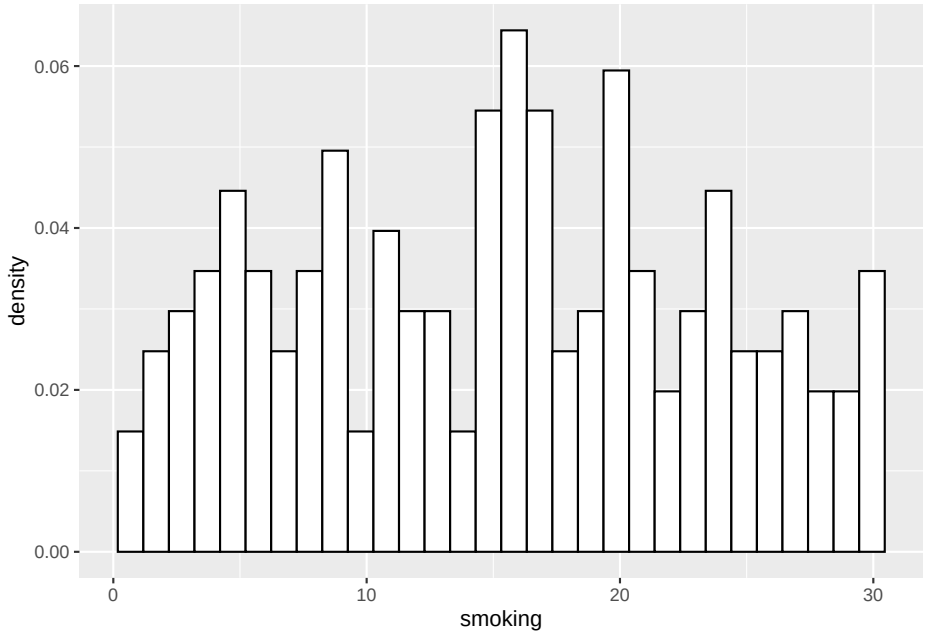
```
## [1] 15.07077
```




```
## # A tibble: 200 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   250  47.5    20.1           8.94
##  2    39  14.0    21.0          16.0
##  3    26  53.0    27.7           9.09
##  4   338   8.13    5.89         14.4
##  5   269  42.6    26.3          11.6
##  6   274  55.4    17.2           7.23
##  7   219  71.0    15.3           4.08
##  8   209   7.83   26.8          17.6
##  9   164  70.9    29.9           6.34
## 10   213  15.5    18.8          14.9
## # i 190 more rows
```

```
## [1] 15.74633
```

```
## [1] 15.74633
```



As you can see, the distributions of the smaller samples are more peaky and bumpy, because they are very sensitive to individual data points. As the sample gets larger, it starts to look more like the population right?

We can complete our table now in the slides of the sample statistics (median and mean) showing that in general, as we get closer to the population size, the statistics generally get closer too. To do so, let's go **back to the slides...**

Chapter 7

Introduction to Bootstrapping

This Chapter relates to the second part of Lecture 6. In this chapter, you will learn the core concepts of bootstrapping. That is, creating synthetic sampling distributions through multiple resampling (with replacement) of a single sample.

The basic process is fairly simple, once you have your original sample, and has the following characteristics:

1. A bootstrap sample has an equal probability of randomly drawing any of the original sample elements (data points).
2. Each element can be selected more than once - because the sample is done **with replacement**.
3. Each resampled data set (the new sample) is the same size as the original one.

First, I will demonstrate the basic principle.

Recall from the last chapter that there was a simulated data set of 498 people, with variables representing smoking, biking, and heart disease.

We treated this as **the population** and then sampled from it to demonstrate uncertainty at different sample sizes.

So, let's take the sample of **50** that we drew from that 'population' of 498, and imagine that we got this sample by (for example) doing a survey of the population (of 498), and this is our data set for analysis.

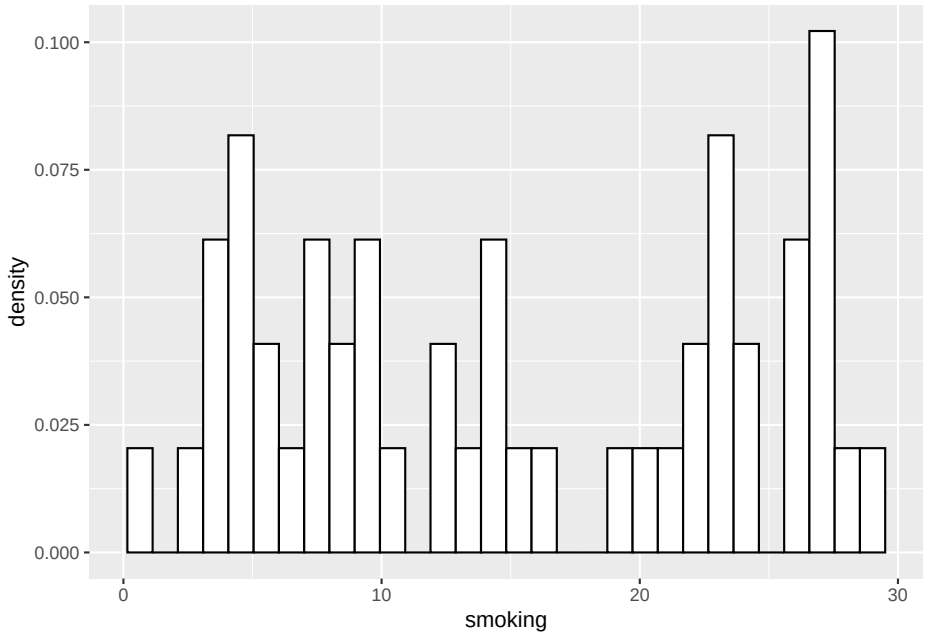
Because this chapter now draws its own sample rather than reusing the one still in memory from Chapter 6, the exact numbers below may land somewhere slightly different from what we saw there — same unseeded variation as always, not a new inconsistency.

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First we will remind ourselves of the properties and distribution of our sample, the median, mean, and distribution:

```
## [1] 14.03561
```

```
## [1] 15.15627
```



Ok, there we go. Remember, this sample is our 'data set' for analysis. Remember, it represents a **sample of 50 cases** from a **population of 498**.

We know the median and mean, but we have no real indication of the uncertainty in those estimates. What I mean is, without information about the population mean and median (which of course we would not have in a real-life sampling situation), we cannot know how close the sample median and mean are to the population mean and median.

In order to estimate that information, we will eventually use the bootstrap method. For now, we are just going to demonstrate the basic idea.

So, what we do now is draw *another random sample of 50*. However, we don't draw it from the population, we instead draw the new sample of 50 **from our existing sample of 50**.

That sounds weird to many people, and at first they often say 'but how can it be different from your original 50??'. The trick is that each time we draw a data point, we *replace* it back, so we are always drawing our sample from the full 50. This is called *sampling with replacement*.

In this way, the new sample can **only contain values which were in the**

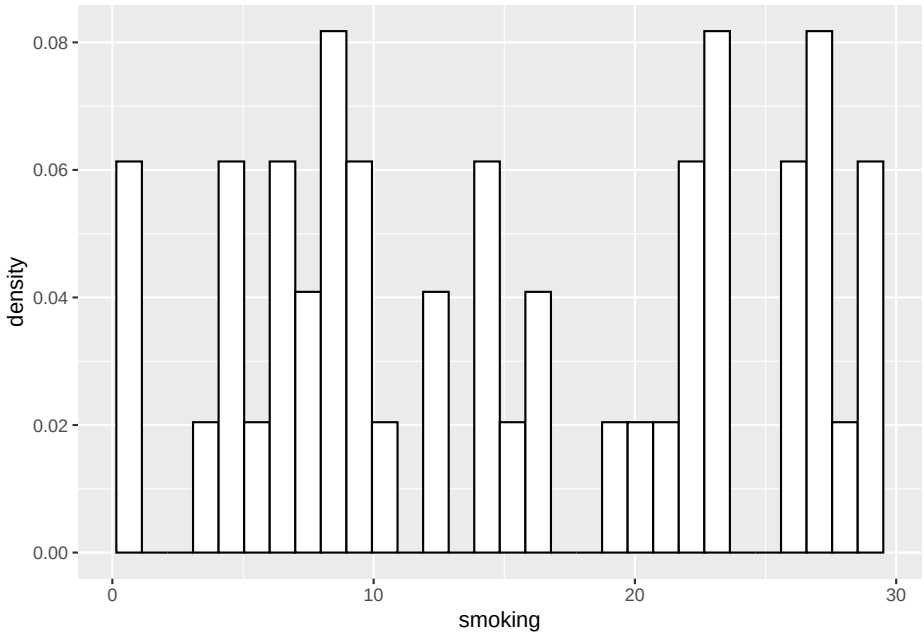
original sample, but can contain different frequencies of those values. Or in other words, each value can occur many different times, and that number of times may be different to the original sample. So the *distribution of values* in this new sample will be different to the original sample, and the statistics will therefore also be different.

Let's draw this new sample, take the median and mean of the sample, and plot it:

```
## # A tibble: 50 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   464   73.8   0.642         0.552
##  2   213   15.5  18.8          14.9
##  3   275   32.2   5.02          9.90
##  4   434   36.3   8.25          9.78
##  5    49   28.6  12.5          11.4
##  6   429   35.7   9.40          9.57
##  7   232   10.4  16.0          14.1
##  8    87   31.0  29.0          13.8
##  9   403   38.2   4.25          7.26
## 10   269   42.6  26.3          11.6
## # i 40 more rows
```

```
## [1] 14.10165
```

```
## [1] 15.37759
```

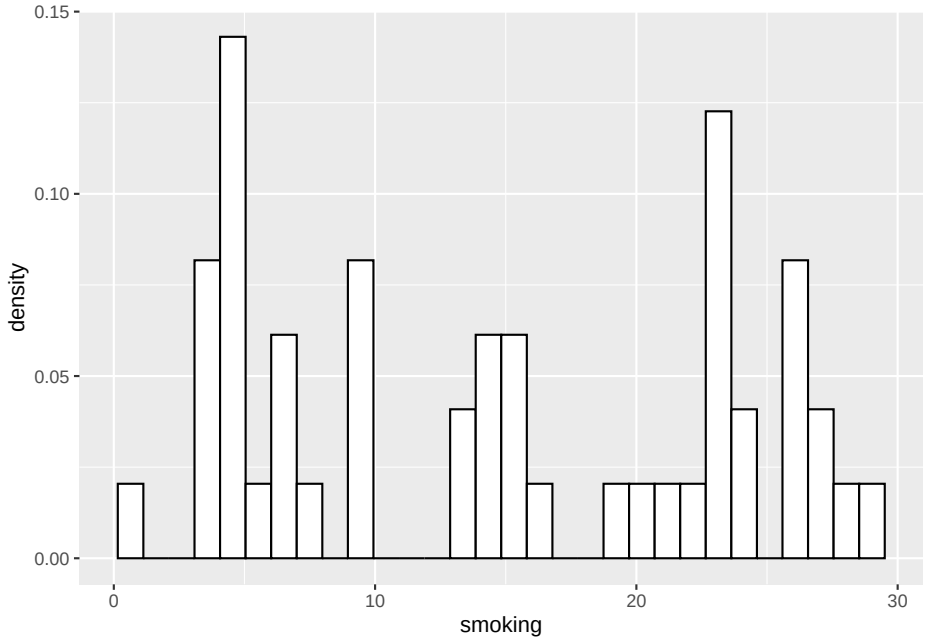


Marvellous! Now, for the purpose of example, let's draw two more of these resamples from the original 50, take their median and mean, and plot the distributions...

```
## # A tibble: 50 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   372   15.3   22.4          17.3
##  2   260   37.7   26.3          11.2
##  3   158   54.1   20.4           8.05
##  4   232   10.4   16.0          14.1
##  5   405   12.9   27.0          18.6
##  6   240   21.6    9.35         14.0
##  7   309   60.0    4.29          4.11
##  8   429   35.7    9.40          9.57
##  9   362   49.9    3.86          5.33
## 10    58   27.6    4.39         10.9
## # i 40 more rows
```

```
## [1] 13.96957
```

```
## [1] 14.70367
```

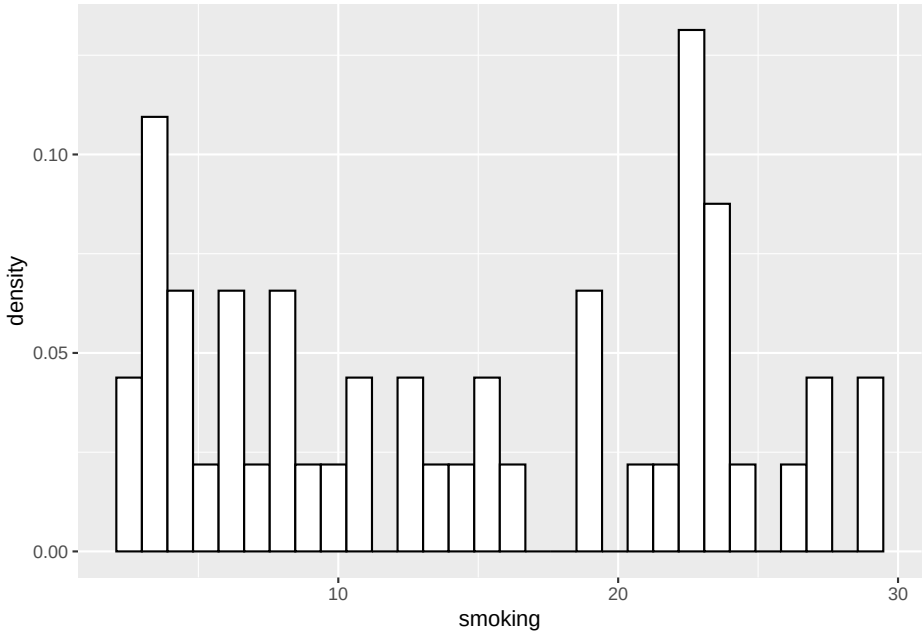



```
## # A tibble: 50 x 4
##     ...1 biking smoking heart.disease
##    <dbl>  <dbl>   <dbl>         <dbl>
##  1   146   71.5   21.6           3.67
```

```
##  2   207   67.1    5.82          2.55
##  3   493   21.6    7.60         12.4
##  4   458   68.7   22.4           5.36
##  5   190   32.1   12.5          11.5
##  6   196   70.7   22.7           2.71
##  7   493   21.6    7.60         12.4
##  8   140   16.3    3.46         13.6
##  9   158   54.1   20.4           8.05
## 10   213   15.5   18.8          14.9
## # i 40 more rows
```

```
## [1] 13.8539
```

```
## [1] 14.38825
```



Now, if we **return to the slides**, we can build a table using these mean and median values. Of course, the slide deck will have slightly different values, since it's based on a different run of the resampling process, but the principle is the same.

So, this is the basic idea of bootstrapping. We sample with replacement from our original sample, many many times. We did 3 here manually, but we generally use a program to do this many more times, such as a thousand or more.

7.1 Bootstrapping in the Context of Previous Examples

To further reinforce the point, let's now place ourselves in the position of three different researchers, each of varying levels of enthusiasm, and all three are researching the **same population of 498 people** that we have already explored in the last few examples.

Researcher 1 is a little like me as a Ph.D. student, and maybe more interested in 'experiencing life'. So, he has little time to actually collect data, and not much more enthusiasm for it. In the end, he manages to take a sample of 10 people from the population of 498.

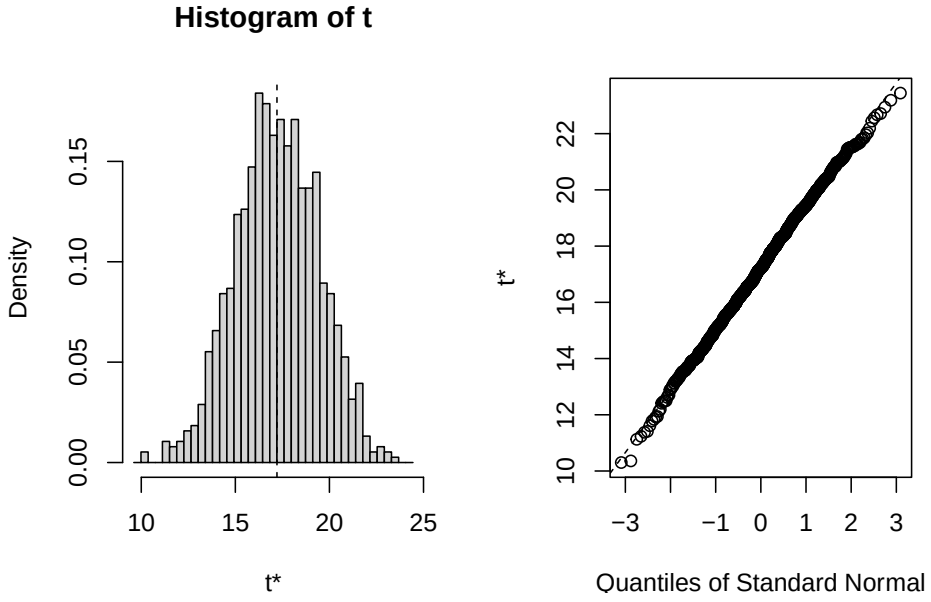
Researcher 2 is a bit more enthusiastic, and gets a sample of 50.

Researcher 3 is fairly conscientious, and takes a sample of 200 from the population of 498.

Now, what we can do, is run 1000 bootstrap replications of each of these varying-sized subsamples of the population, to see what might happen:

First, the 10:

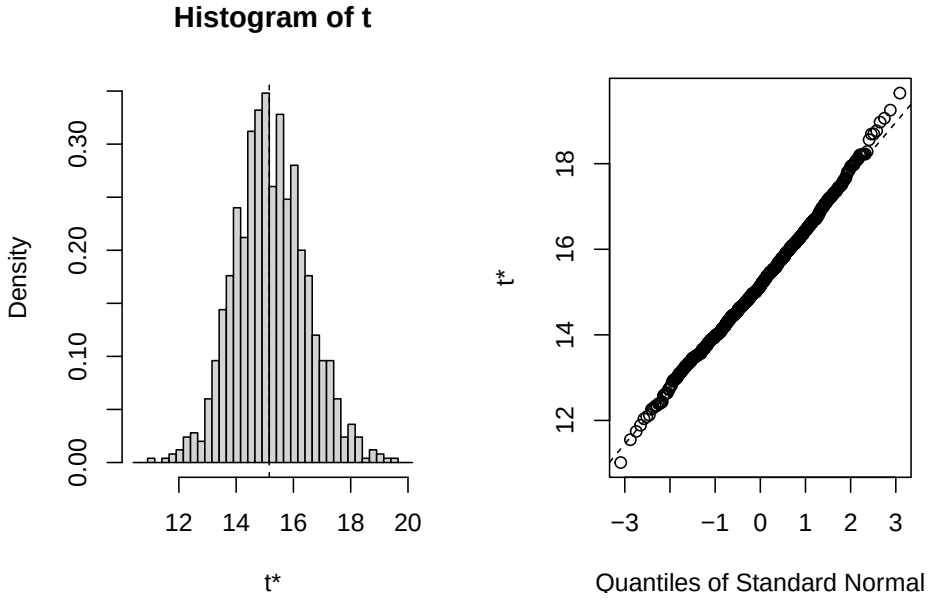
```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = sub.10, statistic = f1, R = 1000)
##
##
## Bootstrap Statistics :
##     original     bias    std. error
## t1* 17.21158 0.01989146    2.190944
```




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 1000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = results, type = "norm")
##
## Intervals :
## Level      Normal
## 95%   (12.90, 21.49 )
## Calculations and Intervals on Original Scale
```

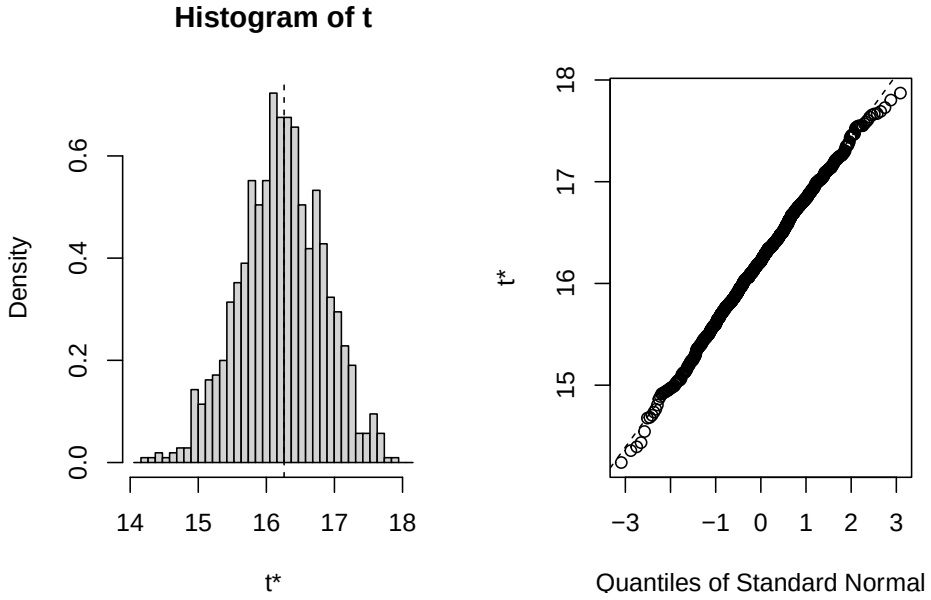
Now let's do it for the other two subsamples of n=50, and n=200

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = sub.50, statistic = f1, R = 1000)
##
##
## Bootstrap Statistics :
##     original     bias    std. error
## t1* 15.15627 0.05073014    1.252665
```




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 1000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = results, type = "norm")
##
## Intervals :
## Level      Normal
## 95%   (12.65, 17.56 )
## Calculations and Intervals on Original Scale
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = sub.200, statistic = f1, R = 1000)
##
##
## Bootstrap Statistics :
##     original     bias    std. error
## t1* 16.26091 -0.0406118   0.6097286
```




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 1000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = results, type = "norm")
##
## Intervals :
## Level      Normal
## 95%   (15.11, 17.50 )
## Calculations and Intervals on Original Scale
```

We'll now build a table with these values *back in the slide deck*. Again, remember the values in the slide deck will differ from these due to the randomness of the process.

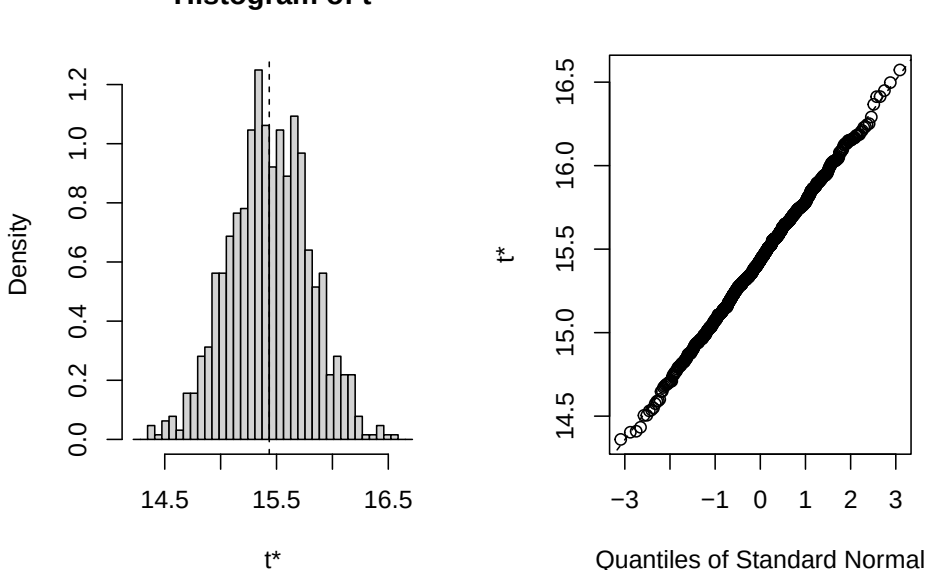
7.1.1 Bootstrapping the Original Sample

Now, let's shift our minds a bit, and consider that the data set of 498 actually represents **a sample of a larger population** (remember from the last chapter, it's simulated, but meant to represent a sample from the population).

So, let's bring in Researcher 4, the most conscientious of all. She is the one who manages to take a sample of 498 people from the population. And, *finally*, we can bootstrap the original full sample of 498:

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
```

```
##
## Call:
## boot(data = Heart, statistic = f1, R = 1000)
##
##
## Bootstrap Statistics :
##     original      bias    std. error
## t1* 15.43503 0.001437866    0.359051
```




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 1000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = results, type = "norm")
##
## Intervals :
## Level      Normal
## 95%   (14.73, 16.14 )
## Calculations and Intervals on Original Scale
```

This is a very nice set of results, which can tell us many interesting things. *So let's go back to the slides…..*

7.2 Bootstrapping Other Stuff...

We have so far only bootstrapped the mean. However, the basic principle can be applied to virtually any statistical estimate. So, we can revisit some of our

prior analyses, and use the bootstrap method to quantify the uncertainty in the estimates that we previously accepted without really thinking too hard about them.

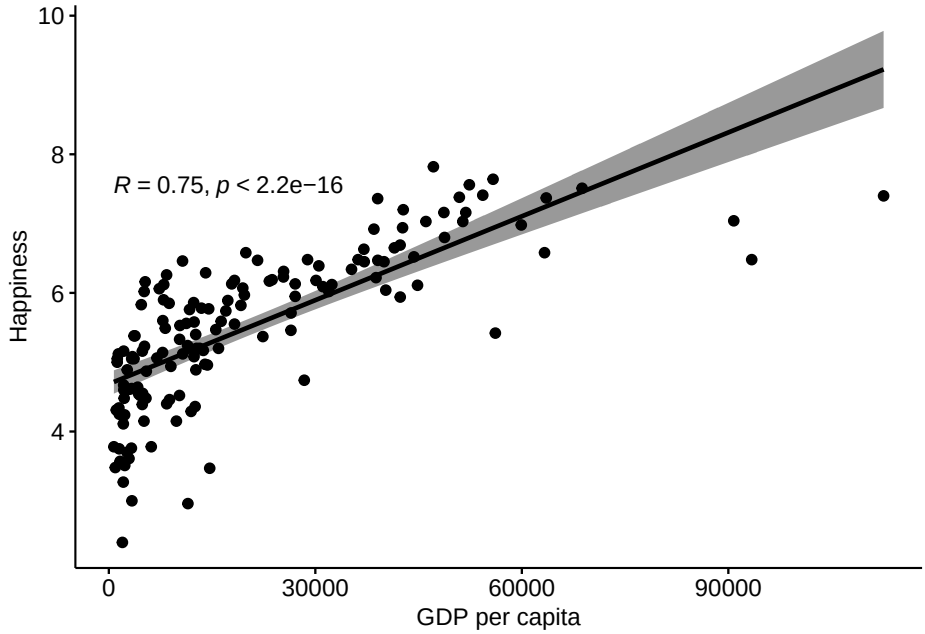
7.2.1 Correlations

First, let's revisit our recent correlation analysis of Happiness and GDP per capita.

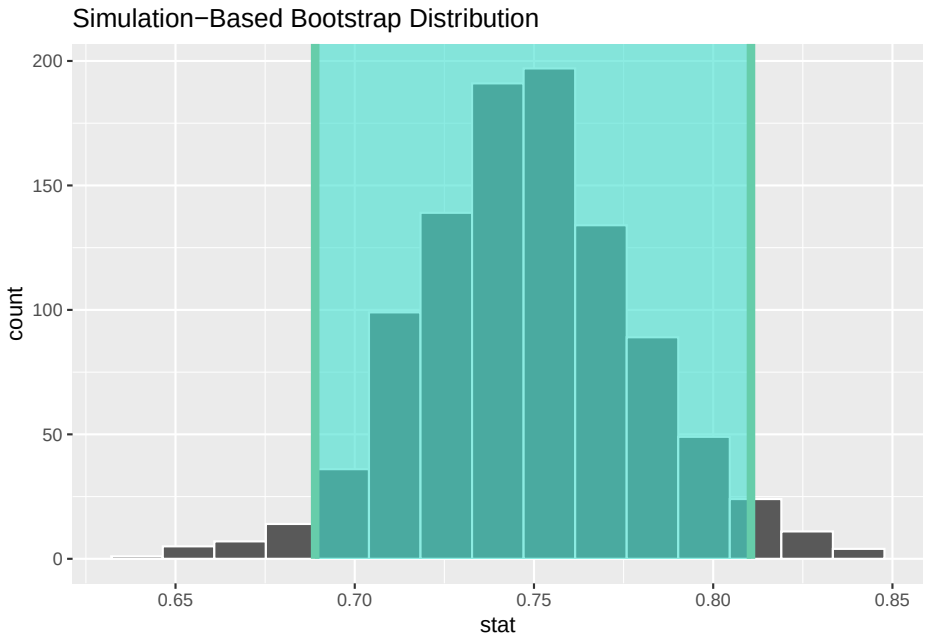
```
## # A tibble: 6 x 4
##   Country        Happiness GDPpc      Pop
##   <chr>              <dbl> <dbl>    <dbl>
## 1 Afghanistan         2.4   1971 38972236
## 2 Albania             5.2  13192  2866850
## 3 Algeria             5.12 10735 43451668
## 4 American Samoa     NA       NA    46216
## 5 Andorra            NA       NA    77723
## 6 Angola             NA     6110 33428490
```

```
##           vars   n        mean           sd     median     trimmed        mad
## Country*     1 249      125.00        72.02     125.00      125.00      91.92
## Happiness    2 153        5.49         1.12       5.53        5.52       1.16
## GDPpc        3 197    20463.88     20717.34   12655.00    17037.01   13338.95
## Pop          4 242 59178643.60 331869505.09 5596196.00 12318073.38 8185922.38
##              min          max        range  skew kurtosis          se
## Country*     1.0 2.490000e+02 2.480000e+02  0.00    -1.21        4.56
## Happiness    2.4 7.820000e+00 5.420000e+00 -0.26    -0.38        0.09
## GDPpc      731.0 1.125570e+05 1.118260e+05  1.58     2.55     1476.05
## Pop        809.0 4.663087e+09 4.663086e+09 11.65   152.44 21333379.77
```

If we run the same analysis as in Chapter 5, we'll get the same results: Correlation R = 0.75



Now, let's take uncertainty into account, by bootstrapping that correlation and creating some confidence intervals.




```
## # A tibble: 1 x 2
##   lower_ci upper_ci
```

```
##      <dbl>    <dbl>
## 1    0.689    0.811
```

```
## Response: Happiness (numeric)
## Explanatory: GDPpc (numeric)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1 0.745
```

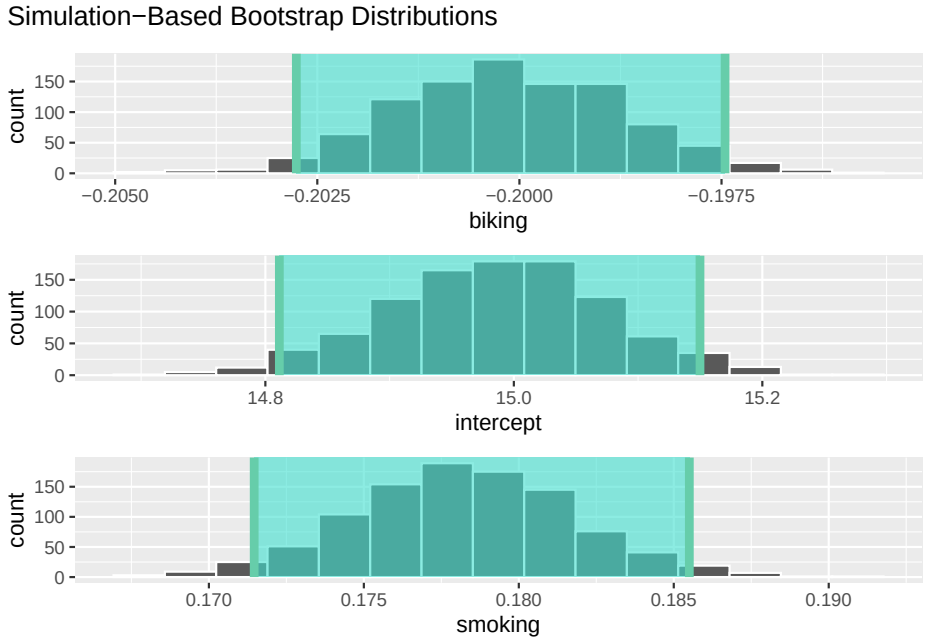
So, you can see the correlation is 0.75 with a 95% confidence interval of 0.69 - 0.81

Now, let's extend this to the multiple regression case we have previously used in Chapter 5, examining the relationships between smoking, biking, and heart disease.

```
## # A tibble: 6 x 4
##    ...1 biking smoking heart.disease
##   <dbl>  <dbl>   <dbl>         <dbl>
## 1     1  30.8    10.9          11.8
## 2     2  65.1     2.22          2.85
## 3     3   1.96   17.6          17.2
## 4     4  44.8     2.80          6.82
## 5     5  69.4    16.0           4.06
## 6     6  54.4    29.3           9.55
```

```
##               vars   n   mean     sd median trimmed    mad  min    max  range
## ...1             1 498 249.50 143.90 249.50  249.50 184.58 1.00 498.00 497.00
## biking           2 498  37.79  21.48  35.82   37.71  27.51 1.12  74.91  73.79
## smoking          3 498  15.44   8.29  15.81   15.47  10.86 0.53  29.95  29.42
## heart.disease    4 498  10.17   4.57  10.39   10.18   5.42 0.55  20.45  19.90
##                skew kurtosis   se
## ...1           0.00    -1.21 6.45
## biking         0.07    -1.22 0.96
## smoking       -0.04    -1.12 0.37
## heart.disease -0.03    -0.93 0.20
```

Here, we need to calculate multiple confidence intervals as we have multiple estimates.




```
## # A tibble: 3 x 2
##   term      estimate
##   <chr>        <dbl>
## 1 intercept   15.0
## 2 smoking      0.178
## 3 biking      -0.200
```

```
## # A tibble: 3 x 3
##   term      lower_ci upper_ci
##   <chr>        <dbl>    <dbl>
## 1 biking      -0.203   -0.197
## 2 intercept   14.8     15.1
## 3 smoking      0.171    0.186
```

It's worth reflecting on exactly what these confidence intervals mean, and to do so, we can move **back to the slides...**

7.3 Reading the literature

You'll find bootstrapping in management papers in a few places, such as confidence intervals for statistics that have no convenient formula, and robustness checks. But by far the most common is *mediation analysis*: testing whether X affects Y *through* some mediator M. If a paper claims that transformational leadership raises performance by increasing employees' engagement, or that a promotion boosts sales by changing how customers see the brand, it will almost

certainly report a bootstrapped confidence interval for an "indirect effect." This book doesn't teach mediation, but you can read those results with what this chapter has already given you.

7.3.1 Where you'll meet this

The indirect effect. A simple mediation model has two links: X to M (usually called a) and M to Y, holding X constant (b). The indirect effect is the product of the two, a $\times$ b. The trouble is that the sampling distribution of a product of two estimates is usually skewed rather than normal, so the older, formula-based tests of it are unreliable. The bootstrap gets round that exactly as this chapter did: resample the data thousands of times, calculate a $\times$ b in each resample, and take the middle 95% of those values as the confidence interval. So when a paper reports an indirect effect of 0.12 with a 95% bootstrap confidence interval of 0.04 to 0.21, based on 5,000 resamples, you know precisely what was done. The interval excludes zero, so the indirect effect is significant. Its width tells you how precisely it has been estimated, just as with any other interval.

What the bootstrap fixes, and what it doesn't. The bootstrap solves a statistical problem: getting an honest interval for a quantity with an awkward distribution. It does nothing about the much bigger problem, which is that a mediation claim is a causal claim about *two* links. Even if X was randomly assigned in an experiment, M wasn't, so the link from M to Y is exposed to every confounding problem in Chapter 4 (Bullock et al., 2010). And the bootstrap can't create information the sample doesn't contain. It resamples the data you have, so a small or unrepresentative sample stays small and unrepresentative however many times you resample it, as Researcher 1's sample of 10 showed.

"Full" and "partial" mediation. Older papers, and some newer ones, follow a set of steps from Baron and Kenny (Baron & Kenny, 1986), which among other things required a significant overall effect of X on Y before mediation could even be tested. That requirement is now widely rejected, and so is the habit of classifying results as "full" or "partial" mediation depending on whether the direct effect of X on Y remains significant (Zhao et al., 2010). Whether it does depends heavily on sample size, so "full mediation" is often just another way of saying "not enough data to detect the direct effect." One further point from the same authors connects straight to the red flag in Section 5.6. A mediator that isn't clearly distinct from X or from Y will appear to "fully mediate" almost automatically, so check its correlations in Table 1.

7.3.2 The connection back

A bootstrapped indirect effect is this chapter's procedure applied to one particular number. Everything the chapter showed about bootstrap intervals applies: they describe the uncertainty that comes from sampling, they get narrower as the sample grows, and they rely entirely on the original sample being a fair

picture of the population. What they can't tell you is whether the causal story connecting X, M and Y is true.

7.3.3 Questions worth asking

1. **What is the causal story, link by link?** What does the design do to support each link, and in particular the link from M to Y?
2. **Were X, M and Y measured at different times, or all at once?** If all at once, why should we believe the order the paper proposes?
3. **How was the indirect effect tested?** With a bootstrap confidence interval, or with an older test?
4. **How wide is the interval?** Does it include indirect effects too small to matter?
5. **Is "full mediation" doing any real work?** Or would a larger sample have found a direct effect too?

7.3.4 One red flag

Beware the single-survey mediation model. One questionnaire, completed by one person at one moment, measures three constructs, and the paper arranges them as X $\rightarrow$ M $\rightarrow$ Y with a bootstrapped indirect effect as its headline result. The statistics can't tell you which way round the arrows go: the same data are usually fitted just as well by rearranging the three variables into a different order, and the choice of order came from the theory, not the data. Add the common-method concerns from Chapter 5, and a significant indirect effect in a design like this tells you very little about mechanism. It may still be the best evidence available, but it should be presented as consistent with the proposed process, not as a demonstration of it.

7.3.5 If you want to read more

Xinshu Zhao, John Lynch and Qimei Chen's "Reconsidering Baron and Kenny" (Zhao et al., 2010) was written for consumer researchers as a non-technical guide to what mediation analysis can and can't show. It remains the most useful short account I know, and it will save you from several common mistakes if you ever run one yourself.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 8

T-Tests for Means

This Chapter supplements the material on t-tests in the second part of Lecture 6.

In this chapter, I will introduce the t-test, a classic analysis technique that is used to compare two groups together.

8.1 The Christmas Music Study

We can also use bootstrapping as an entry point to a new analysis situation, where we are comparing two groups. This could be for example in a classic experimental context; treatment and control.

Remember, t-tests can be done in any analysis setting, and it does not require bootstrapping at all. It just so happens that they are nicely explainable at this point.

So, here's the study I'll use for the rest of this chapter, and pick up again a couple of chapters from now. Participants come in and proofread a passage of text — a genuinely tedious task, by design — with background audio playing. I tell them the study is about concentration and background noise, which is true as far as it goes, but isn't the whole story. Half the participants get silence, half get Christmas music (which is, delightfully, one of the few things guaranteed to mildly irritate almost everybody, making it a gift for a researcher even if not for the participants). Before and after the task, I measure how irritated people feel, using four items — irritated / annoyed / tense / on edge, each rated 1-7 — and average them into a single composite score, `IRRIT`. The four items hold together well as one measure: Cronbach's alpha across them is a healthy 0.86 in this sample, so treating them as a single composite rather than four separate numbers is a reasonable thing to do.

Before looking at any numbers, there's a design question worth sitting with,

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because it's the reason this chapter runs two different-looking t-tests on what can feel like the same question.

Suppose the Christmas-music group's irritation goes up from before the task to after it. Is that enough, on its own, to conclude Christmas music is irritating? No — because the task itself is boring. Twenty minutes of proofreading a dull passage would probably wind most people up a bit regardless of what's playing in the background, including nothing at all. A rise from baseline to post-task doesn't tell you the *music* did anything, because the task alone might have. What actually answers that question is a comparison group who did the identical boring task in silence, so I can see whether the Christmas-music group ended up *more* irritated than boredom alone would predict. That's why this chapter needs both an independent-samples comparison (Christmas vs silence, after the task) and a paired comparison (the Christmas group's own before-and-after) — they answer two different questions, and only one of them can rule out "it was just the task."

8.2 Independent-Samples T-Test

The first thing to do is read in the data and describe it.

```
## # A tibble: 6 x 7
##      ID GROUP   IRRIT_IRRITATED IRRIT_ANNOYED IRRIT_TENSE IRRIT_ONEDGE IRRIT
##   <dbl> <chr>             <dbl>         <dbl>       <dbl>        <dbl> <dbl>
## 1     1 Silence               3             4           2            5  3.5
## 2     2 Silence               3             3           4            3  3.25
## 3     3 Silence               3             3           3            3  3
## 4     4 Silence               5             4           4            4  4.25
## 5     5 Silence               4             2           4            3  3.25
## 6     6 Silence               3             2           3            3  2.75
```

```
##                  vars  n  mean   sd median trimmed   mad min   max range  skew
## ID                  1 30 15.50 8.80  15.50   15.50 11.12 1.0 30.00 29.00  0.00
## GROUP*              2 30  1.50 0.51   1.50    1.50  0.74 1.0  2.00  1.00  0.00
## IRRIT_IRRITATED     3 30  4.10 1.40   4.00    4.21  1.48 1.0  6.00  5.00 -0.46
## IRRIT_ANNOYED       4 30  4.13 1.38   4.00    4.08  1.48 2.0  7.00  5.00  0.37
## IRRIT_TENSE         5 30  4.13 1.17   4.00    4.17  1.48 2.0  6.00  4.00 -0.25
## IRRIT_ONEDGE        6 30  3.93 1.20   4.00    3.92  1.48 1.0  7.00  6.00  0.12
## IRRIT               7 30  4.08 1.08   4.25    4.12  1.30 1.5  5.75  4.25 -0.37
##                  kurtosis   se
## ID                  -1.32 1.61
## GROUP*              -2.07 0.09
## IRRIT_IRRITATED     -0.52 0.26
## IRRIT_ANNOYED       -0.72 0.25
## IRRIT_TENSE         -0.95 0.21
## IRRIT_ONEDGE         0.37 0.22
```

Table 8.1: Data summary

Name	ED_IND
Number of rows	30
Number of columns	7
_____	
Column type frequency:	
factor	1
numeric	6
_____	
Group variables	None

```
## IRRIT              -0.82 0.20
```

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
GROUP	0	1	FALSE	2	Sil: 15, Chr: 15

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
ID	0	1	15.50	8.80	1.0	8.25	15.50	22.75	30.00	▇▇▇▇▇
IRRIT_IRRITATED	0	1	4.10	1.40	1.0	3.00	4.00	5.00	6.00	▂▇▇▇▅
IRRIT_ANNOYED	0	1	4.13	1.38	2.0	3.00	4.00	5.00	7.00	▆▆▅▂▂
IRRIT_TENSE	0	1	4.13	1.17	2.0	3.00	4.00	5.00	6.00	▂▅▇▅▂
IRRIT_ONEDGE	0	1	3.93	1.20	1.0	3.00	4.00	5.00	7.00	▂▅▇▅▂
IRRIT	0	1	4.08	1.08	1.5	3.25	4.25	4.94	5.75	▂▅▅▇▇

We can see that we have one FACTOR variable, which indicates which group each subject was in — Silence or Christmas.

So, let's run an *independent samples T-Test* with bootstrapped confidence interval, comparing the two groups' irritation scores at the end of the task (T2).

We use an independent samples test here because these are two different groups of people, who did the task under two different conditions. What we're trying to assess is whether the Christmas-music group ended up more irritated than the silence group.

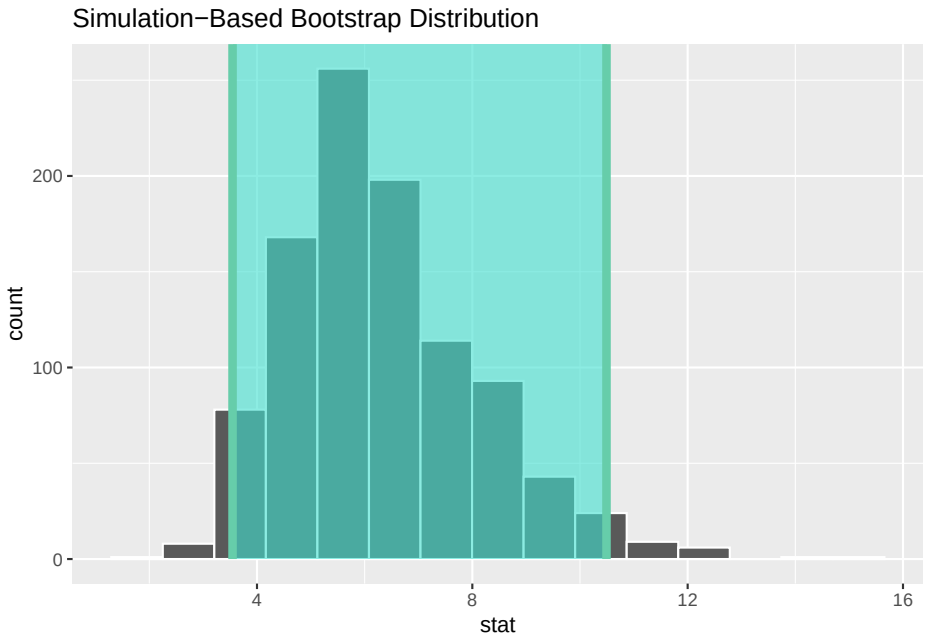
Before we get to the code, it's worth pausing on what we're actually bootstrapping, and why it isn't simply the difference in the two group means.

The raw mean difference — a bit over 1.5 points on our seven-point irritation scale — is perfectly interpretable in its own right: it's a real number in the actual units the scale was measured in. But its size alone tells us nothing about how impressed we should be by it, because that depends entirely on the scale we happened to use and how much natural variation there already is in irritation

scores. A difference of "1.5 points" is enormous on a scale that barely moves and unremarkable on one that swings wildly.

This is the same problem as the standardised versus unstandardised regression estimates back in Section 5. An unstandardised coefficient is interpretable in real-world units — cigarettes, miles cycled — but can't be compared across variables measured on different scales; standardising puts everything onto a common footing. The t-statistic does something similar here: it takes the mean difference and re-expresses it relative to how much variability and how much data we have, which is what lets us judge whether a given difference stands out from the noise. It isn't quite the same thing as standardising, though, and I'll come back to why at the end of this section.

```
## # A tibble: 2 x 2
##   GROUP      name
##   <fct>     <dbl>
## 1 Silence    3.28
## 2 Christmas  4.87
```




```
## Response: IRRIT (numeric)
## Explanatory: GROUP (factor)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1  5.88
```

```
## # A tibble: 1 x 2
```

```
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     3.55     10.5
```

Cool, so it seems that the Christmas-music group displayed more irritation. The confidence interval for the t-statistic does not contain 0, so it supports the idea that there is a real difference here. The interval is quite wide though - because of our small sample size. In a later Chapter we'll return to this issue of what confidence intervals actually mean.

I'll be honest: in a purely bootstrap context like this one, that standardisation step can feel like an unnecessary detour. We didn't need the t to reach our conclusion — the conclusion was always going to be "does the confidence interval contain zero or not", and that would have been exactly as true if we'd bootstrapped the raw mean difference directly instead. The t only really earns its keep once we start relying on known probability distributions rather than brute-force resampling, which is where we're headed in the next chapter. For now, take it as scaffolding for what's coming rather than something bootstrapping strictly requires.

There's one more way of putting that difference on a common footing, and it's the one you'll meet most often in published papers. The t-statistic divides the mean difference by its standard error, and the standard error shrinks as the sample grows. That makes t good at answering "does this difference stand out from the noise?" but not "how big is it?" Run exactly the same study with ten times as many people, get exactly the same difference, and the t would be roughly three times larger.

Cohen's d answers the second question. It divides the mean difference by the standard deviation of the irritation scores themselves, pooled across the two groups, so it tells you how many standard deviations apart the groups ended up. Unlike t, it doesn't grow just because the sample does. For our study, d is 2.15: the Christmas-music group finished about 2.1 standard deviations more irritated than the silence group. It's a direct cousin of the standardised regression coefficients in Chapter 5: the same idea in a different setting.

You'll often see d described using Jacob Cohen's labels, where 0.2 is "small," 0.5 "medium" and 0.8 "large." By those labels ours is large. But Cohen offered them as a convention for use when nothing better suggests itself (Cohen, 1988), and they've since hardened into exactly the kind of rule this book keeps warning about. Two cautions apply to our own number, too. First, with only 15 people per group, it's a noisy estimate, and as you'll see in Chapter 13, big-looking effects from small samples deserve particular scepticism. Second, like any standardised effect, d tells you nothing about whether the difference matters in practice. For that you have to go back to the original units, which here means a difference of 1.6 points on a seven-point scale.

8.3 Paired-Sample T-test

OK, so let's use a different design on the same study — the Christmas-music group's own scores, before the task (T1) and after it (T2) — using a paired samples t-test. Let me go *back to the slides…* to explain this difference.

The same basic process is needed, but with some modifications because of the type of comparison we are doing. And, as such, we have some new data.

```
##   ID IRRIT_T1_IRRITATED IRRIT_T1_ANNOYED IRRIT_T1_TENSE IRRIT_T1_ONEDGE
## 1  1                  3                1              1               3
## 2  2                  3                5              5               5
## 3  3                  4                4              5               3
## 4  4                  3                3              4               2
## 5  5                  2                4              2               2
## 6  6                  4                4              5               3
##   IRRIT_T2_IRRITATED IRRIT_T2_ANNOYED IRRIT_T2_TENSE IRRIT_T2_ONEDGE IRRIT_T1
## 1                  6                6              6               5      2.0
## 2                  6                5              6               5      4.5
## 3                  7                7              7               7      4.0
## 4                  4                4              6               3      3.0
## 5                  4                4              3               5      2.5
## 6                  4                5              4               4      4.0
##   IRRIT_T2
## 1     5.75
## 2     5.50
## 3     7.00
## 4     4.25
## 5     4.00
## 6     4.25
```

```
##                    vars  n mean   sd median trimmed  mad min   max range  skew
## ID                    1 15 8.00 4.47   8.00    8.00 5.93   1 15.00 14.00  0.00
## IRRIT_T1_IRRITATED    2 15 3.60 1.55   3.00    3.54 1.48   1  7.00  6.00  0.53
## IRRIT_T1_ANNOYED      3 15 3.20 1.57   3.00    3.15 1.48   1  6.00  5.00  0.00
## IRRIT_T1_TENSE        4 15 3.33 1.54   4.00    3.38 1.48   1  5.00  4.00 -0.20
## IRRIT_T1_ONEDGE       5 15 3.33 1.11   3.00    3.31 1.48   2  5.00  3.00  0.25
## IRRIT_T2_IRRITATED    6 15 5.27 1.03   5.00    5.23 1.48   4  7.00  3.00  0.23
## IRRIT_T2_ANNOYED      7 15 4.93 1.22   5.00    4.92 1.48   3  7.00  4.00  0.12
## IRRIT_T2_TENSE        8 15 5.13 1.06   5.00    5.15 1.48   3  7.00  4.00 -0.24
## IRRIT_T2_ONEDGE       9 15 4.93 0.96   5.00    4.92 0.00   3  7.00  4.00  0.12
## IRRIT_T1             10 15 3.37 1.15   3.25    3.29 1.11   2  5.75  3.75  0.50
## IRRIT_T2             11 15 5.07 0.87   4.75    5.00 0.74   4  7.00  3.00  0.68
##                    kurtosis   se
## ID                    -1.44 1.15
## IRRIT_T1_IRRITATED    -0.34 0.40
## IRRIT_T1_ANNOYED      -1.25 0.40
```

```
## IRRIT_T1_TENSE        -1.69 0.40
## IRRIT_T1_ONEDGE       -1.41 0.29
## IRRIT_T2_IRRITATED    -1.27 0.27
## IRRIT_T2_ANNOYED      -0.93 0.32
## IRRIT_T2_TENSE        -0.86 0.27
## IRRIT_T2_ONEDGE       -0.10 0.25
## IRRIT_T1              -0.97 0.30
## IRRIT_T2              -0.73 0.23
```

Now, for this bootstrap purpose we actually need to to calculate the **difference between the two measurements** (here, T1 and T2).

Then, we bootstrap a one-sample t-test with this difference variable. This essentially tests whether the difference is different from zero. If it is, that indicates some effect of the treatment.

Let us first calculate the new variable:

```
##   ID IRRIT_T1_IRRITATED IRRIT_T1_ANNOYED IRRIT_T1_TENSE IRRIT_T1_ONEDGE
## 1  1                  3                1              1               3
## 2  2                  3                5              5               5
## 3  3                  4                4              5               3
## 4  4                  3                3              4               2
## 5  5                  2                4              2               2
## 6  6                  4                4              5               3
##   IRRIT_T2_IRRITATED IRRIT_T2_ANNOYED IRRIT_T2_TENSE IRRIT_T2_ONEDGE IRRIT_T1
## 1                  6                6              6               5      2.0
## 2                  6                5              6               5      4.5
## 3                  7                7              7               7      4.0
## 4                  4                4              6               3      3.0
## 5                  4                4              3               5      2.5
## 6                  4                5              4               4      4.0
##   IRRIT_T2  DIF
## 1     5.75 3.75
## 2     5.50 1.00
## 3     7.00 3.00
## 4     4.25 1.25
## 5     4.00 1.50
## 6     4.25 0.25
```

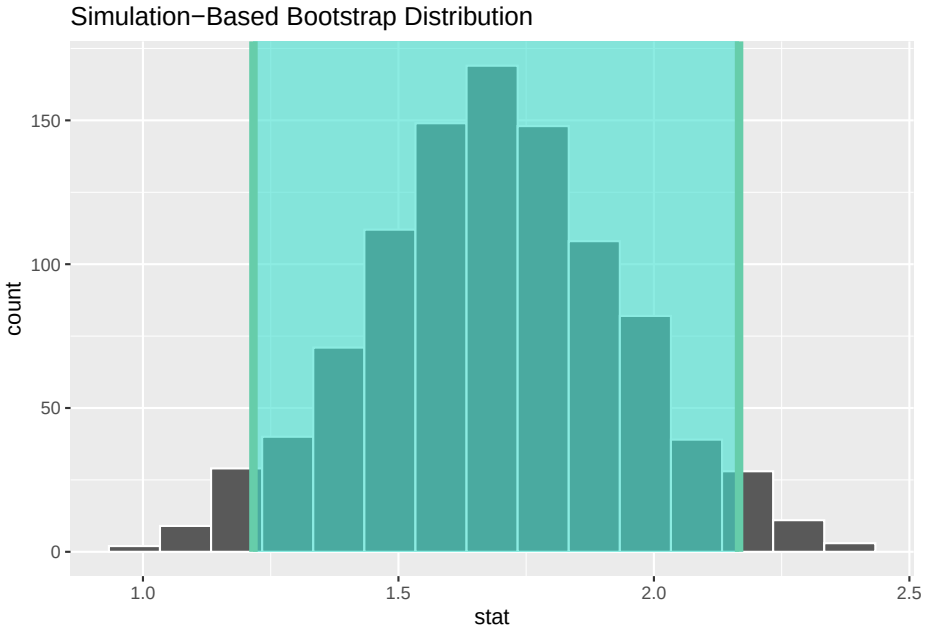
```
##                    vars  n mean   sd median trimmed  mad  min   max range  skew
## ID                    1 15 8.00 4.47   8.00    8.00 5.93 1.00 15.00 14.00  0.00
## IRRIT_T1_IRRITATED    2 15 3.60 1.55   3.00    3.54 1.48 1.00  7.00  6.00  0.53
## IRRIT_T1_ANNOYED      3 15 3.20 1.57   3.00    3.15 1.48 1.00  6.00  5.00  0.00
## IRRIT_T1_TENSE        4 15 3.33 1.54   4.00    3.38 1.48 1.00  5.00  4.00 -0.20
## IRRIT_T1_ONEDGE       5 15 3.33 1.11   3.00    3.31 1.48 2.00  5.00  3.00  0.25
## IRRIT_T2_IRRITATED    6 15 5.27 1.03   5.00    5.23 1.48 4.00  7.00  3.00  0.23
## IRRIT_T2_ANNOYED      7 15 4.93 1.22   5.00    4.92 1.48 3.00  7.00  4.00  0.12
```

```
## IRRIT_T2_TENSE        8 15 5.13 1.06   5.00    5.15 1.48 3.00  7.00  4.00 -0.24
## IRRIT_T2_ONEDGE       9 15 4.93 0.96   5.00    4.92 0.00 3.00  7.00  4.00  0.12
## IRRIT_T1             10 15 3.37 1.15   3.25    3.29 1.11 2.00  5.75  3.75  0.50
## IRRIT_T2             11 15 5.07 0.87   4.75    5.00 0.74 4.00  7.00  3.00  0.68
## DIF                  12 15 1.70 1.03   1.50    1.65 0.74 0.25  3.75  3.50  0.43
##                    kurtosis   se
## ID                    -1.44 1.15
## IRRIT_T1_IRRITATED    -0.34 0.40
## IRRIT_T1_ANNOYED      -1.25 0.40
## IRRIT_T1_TENSE        -1.69 0.40
## IRRIT_T1_ONEDGE       -1.41 0.29
## IRRIT_T2_IRRITATED    -1.27 0.27
## IRRIT_T2_ANNOYED      -0.93 0.32
## IRRIT_T2_TENSE        -0.86 0.27
## IRRIT_T2_ONEDGE       -0.10 0.25
## IRRIT_T1              -0.97 0.30
## IRRIT_T2              -0.73 0.23
## DIF                   -0.99 0.27
```

We can see there is a new 'DIF' variable here.

Notice we're bootstrapping the mean of this difference directly, not a t-statistic as we did for the independent-samples design above. It's the same underlying question — is this far from zero? — just reached without the standardisation step in between. Nothing is lost by skipping it: with paired data there's a single already-differenced variable rather than two separately-scaled groups to compare, so there's less need to standardise before asking the question.

Next, we bootstrap a confidence interval for the *mean of the difference variable*, to see whether it includes zero:



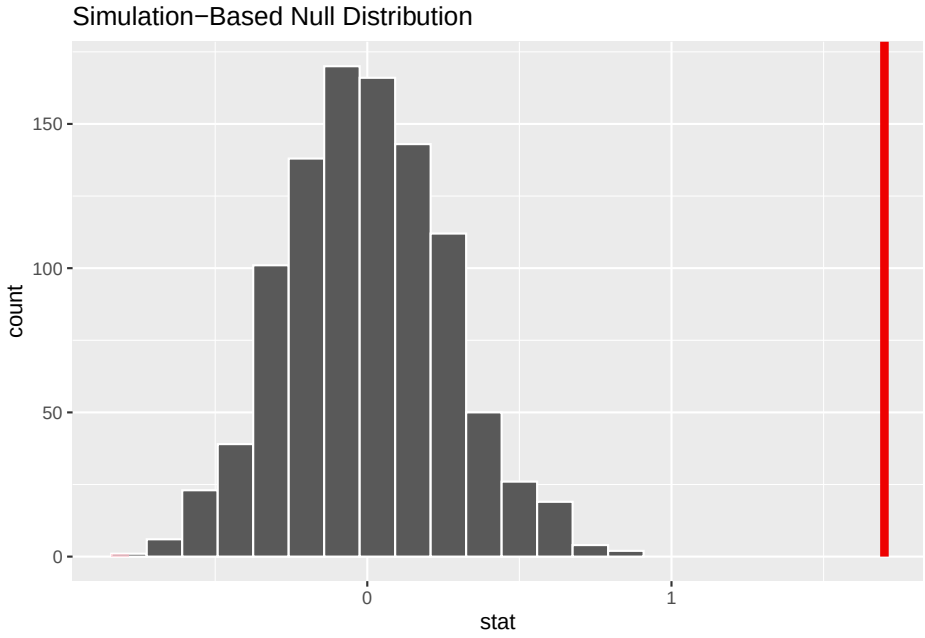

```
## Response: DIF (numeric)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1   1.7
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     1.22     2.17
```

```
## [1] 3.366667
```

```
## [1] 5.066667
```

We can do this in another way, but instead running a 1-sample t-test on the difference variable, testing whether it is different from zero:




```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

```
## Response: DIF (numeric)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1   1.7
```

```
## [1] 3.366667
```

```
## [1] 5.066667
```

Marvellous. We can see that the results from both variants of this test suggest that after the proofreading task with Christmas music playing, our sample on average reported more irritation than they did beforehand.

1: Because the 95% confidence interval does not include 0, I am confident in saying that there is some effect going on here.

2: Because the bootstrapped probability (as expressed by the p-value) of observing the mean difference in irritation between T1 and T2 of 1.7 is very low (0 in our results, but the true value will not be exactly zero), I am confident in saying there is an effect here.

Not surprised — but remember the design question from the start of the chapter: this rise, on its own, doesn't tell us the *music* did it, because the boring task

alone could have produced some of it. That's exactly what the independent-samples comparison above was for, and it agrees — the Christmas-music group ended up more irritated than the silence group who did the identical task.

But there's a limit to what even that comparison can tell us. All I've actually shown is that background music of *some* kind moves irritation more than no music at all — I haven't shown there's anything special about *Christmas* music specifically, as opposed to background music in general. The design as it stands can't separate "background music is irritating" from "Christmas music in particular is irritating," because silence is the only comparison group I have. That's a real limitation, not a minor caveat, and it's exactly the question Chapter 12 picks up, with a third group added to settle it. It's also exactly the confounding problem from Chapter 4: background music and Christmas music are tangled together in this design, so I can't yet credit one over the other.

8.4 Reading the literature

Comparing two groups is the simplest design there is, and it turns up in management papers far more often than the phrase "t-test" does. Sometimes you'll see a t-test reported as such, usually in an experiment comparing a treatment with a control. Much more often, the comparison is hiding inside a regression as a single coefficient. Either way, the questions from this chapter apply: what exactly is being compared with what, how big is the difference, and could something other than the treatment have produced it?

8.4.1 Where you'll meet this

Reported t-tests. In experimental papers you'll see something like $t(58) = 2.41$, $p = .019$. The number in brackets is the degrees of freedom, which for two independent groups is the total sample size minus two. If it's a decimal, such as $t(41.7)$, the authors have used *Welch's t-test*, which doesn't assume the two groups are equally variable. That's a good sign rather than an oddity: there's a strong case for using Welch's version by default (Delacre et al., 2017).

Group differences as regression coefficients. A classic t-test comparing two groups gives exactly the same answer as a regression with a single *dummy variable*, coded 1 for one group and 0 for the other. The coefficient on the dummy is the difference between the two group means, and its t-value is the t-test. So when a paper reports a coefficient for "female," "family firm" or "treated," it's telling you the difference between two group means, adjusted for whatever else is in the model. Two things follow. First, you need to know which group was coded 0, because the coefficient is always relative to that *reference category*: flip the coding and the sign flips. With more than two categories, such as industries or regions, every coefficient compares one category with the one left out, and the choice of which to leave out decides which comparisons you get to see. Second, "adjusted for" is doing a lot of work, for all the reasons in Chapter 4.

Cohen's d and its relatives. You'll also meet *Hedges' g*, which applies a small correction that matters in small samples. It's fine to read the two as essentially the same thing. The trap is in before-and-after designs like the paired test in this chapter. There, d can be calculated by dividing by the standard deviation of the *differences* rather than of the scores. When people's before and after scores are strongly related, as they usually are, that standard deviation is small, and the resulting d can be much larger than an independent-groups d for the same difference (Lakens, 2013). So a d from a within-person study isn't necessarily comparable with one from a between-groups study, and a paper comparing its effect with "typical" effects elsewhere should say which kind it calculated. And like every standardised effect, d needs translating back into natural units before you know whether it matters, as Section 2.6 argued.

8.4.2 The connection back

Every group comparison in a paper, whether it's labelled a t-test or hidden in a coefficient, comes down to the two questions this chapter asked about the Christmas music study. Is the difference bigger than chance alone would produce? And is the comparison group the right one, so that the difference can be put down to the thing the authors say caused it? Statistics can answer the first question. Only the design can answer the second.

8.4.3 Questions worth asking

1. **What exactly is being compared with what?** Which group is the reference category?
2. **What would the comparison group need to be to rule out the most obvious alternative explanation?** Is it?
3. **How big is the difference in the original units, as well as in standard deviations?**
4. **If it's a within-person design, how was the effect size calculated?** Is it being compared with effects calculated the same way?

8.4.4 One red flag

Beware the before-and-after study with no comparison group. A training programme, an incentive scheme or a culture initiative is introduced, scores are measured before and after, and they go up. The paired test is significant, and the paper concludes that the intervention worked. But as the Christmas music study showed, a rise from before to after can't tell you on its own what caused it. People change over time anyway, organisations change around them, and those who start with unusually low scores tend to score closer to average next time regardless. Without a group that went through the same period without the intervention, the rise describes what happened. It doesn't explain it.

8.4.5 If you want to read more

Daniël Lakens's primer on calculating and reporting effect sizes (Lakens, 2013) is free to read online, practical, and especially good on the different versions of d and when each one is appropriate. It's written for researchers rather than readers, but it's the clearest account I know of what the numbers in an effect-size table actually mean.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 9

Introduction to Probability

This Chapter provides examples and additional content for the first part of Lesson 7.

In this lesson, we will learn about the classical theory of probability, and how it can be used to construct measures of uncertainty of our estimates, and from there we can begin to draw inferences from the sample to the population.

9.1 Simulating Coin Flips

The first thing we'll do is simulate 5000 fair coin flips.

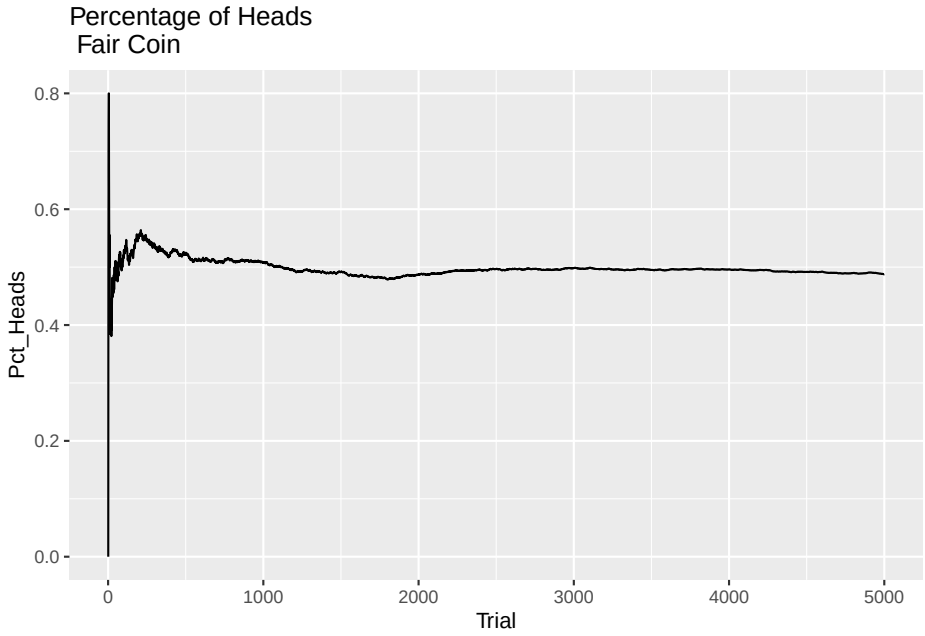
Remember, this is another random process, so results will differ each time slightly

The simulation is taken almost verbatim from a post by Philip Grosse on RPubs: https://rpubs.com/pgrosse/545948.

```
##   Heads Trial Cum_Heads Pct_Heads
## 1     0     1         0 0.0000000
## 2     1     2         1 0.5000000
## 3     1     3         2 0.6666667
## 4     1     4         3 0.7500000
## 5     1     5         4 0.8000000
## 6     0     6         4 0.6666667
```

Below, we'll plot the results of this simulation for ease of interpretation

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The thing to notice (in combination with the table of results above the plot) is that eventually the proportion of heads versus tails approaches 50%, *but it does not start out that way.*

In other words, for a fair coin toss sequence, the trials (tosses) are *independent*, which means that for any given toss of the coin, the chance of a head or tail is 50%. But, that's not the same as saying that you'll get equal numbers of heads or tails for every (or any) particular sequence of multiple tosses - it's not 'self-correcting' in any other way than sheer weight of numbers. So, **eventually** over enough trials, the sequence should start to look like 50/50 heads and tails. But, that is not the tosses 'self-correcting', simply that you are tossing a lot of coins. This is at the root of a number of fallacies about predicting random events, such as the Gambler's fallacy.

9.2 Premier League Goals and the Poisson Distribution

Note, much of the inspiration and code for this section comes from: https://bookdown.org/theqdata/honors_thesis/goal-scoring-and-the-poisson-process.html

The data are the EPL results for the 2021–22 season, from Football-Data.co.uk (n.d.).

```
## # A tibble: 6 x 107
##   Div   Date                Time                HomeTeam   AwayTeam   FTHG  FTAG
##   <chr> <dttm>              <dttm>              <chr>      <chr>     <dbl> <dbl>
```

```
## 1 E0    2021-08-13 00:00:00 1899-12-31 20:00:00 Brentford  Arsenal       2     0
## 2 E0    2021-08-14 00:00:00 1899-12-31 12:30:00 Man United Leeds         5     1
## 3 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Burnley    Brighton      1     2
## 4 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Chelsea    Crystal ~     3     0
## 5 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Everton    Southamp~     3     1
## 6 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Leicester  Wolves        1     0
## # i 100 more variables: TOTG <dbl>, FTR <chr>, HTHG <dbl>, HTAG <dbl>,
## #   HTR <chr>, Referee <chr>, HS <dbl>, AS <dbl>, HST <dbl>, AST <dbl>,
## #   HF <dbl>, AF <dbl>, HC <dbl>, AC <dbl>, HY <dbl>, AY <dbl>, HR <dbl>,
## #   AR <dbl>, B365H <dbl>, B365D <dbl>, B365A <dbl>, BWH <dbl>, BWD <dbl>,
## #   BWA <dbl>, IWH <dbl>, IWD <dbl>, IWA <dbl>, PSH <dbl>, PSD <dbl>,
## #   PSA <dbl>, WHH <dbl>, WHD <dbl>, WHA <dbl>, VCH <dbl>, VCD <dbl>,
## #   VCA <dbl>, MaxH <dbl>, MaxD <dbl>, MaxA <dbl>, AvgH <dbl>, AvgD <dbl>, ...
```

Next, let's see a few important bits of descriptive data. First, let's calculate the total number of games in a season:

```
##    Length     Class      Mode
##       380 character character
```

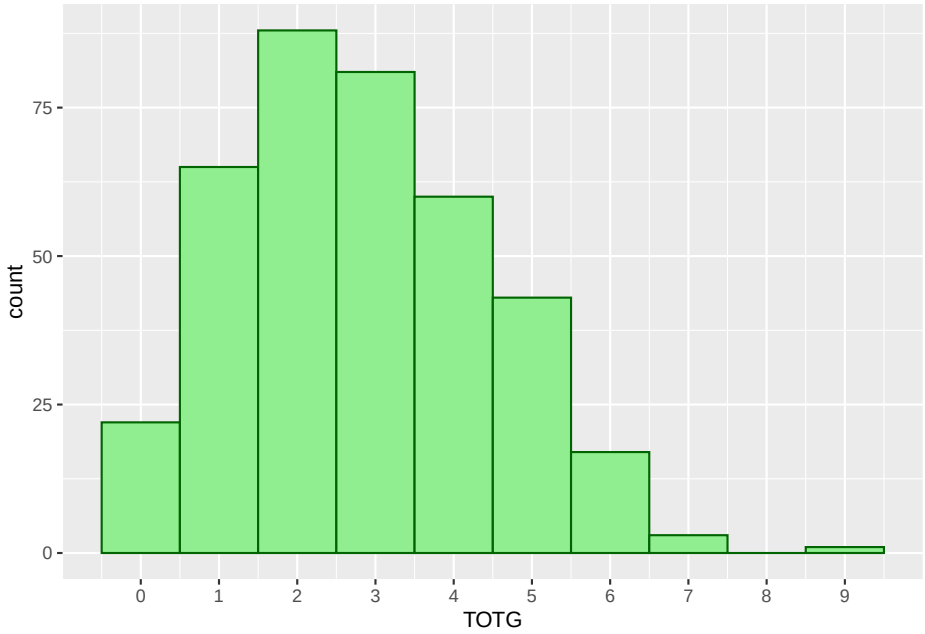
Next, how many goals in the season were there?

```
## [1] 1071
```

And, what was the mean number of goals per game?

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   2.000   3.000   2.818   4.000   9.000
```

OK, so now we have some idea of the goals per game, let's plot the distribution of goals.



If you've ever seen a Poisson distribution, you'll recognise this!

Let's check this out. First, let's create a table of all the matches with different numbers of goals.

```
## # A tibble: 9 x 2
##    TOTG ActualMatches
##   <dbl>         <int>
## 1     0            22
## 2     1            65
## 3     2            88
## 4     3            81
## 5     4            60
## 6     5            43
## 7     6            17
## 8     7             3
## 9     9             1
```

Let's pull some stats of the Total Goals variable which well help us in a second.

```
##  min Q1 median Q3 max     mean       sd   n missing
##    0  2      3  4   9 2.818421 1.626359 380       0
```

Importantly, because the Poisson distribution is described only by its mean, the first and most basic check we can do is whether the mean and the variance of the variable are the same (or at least very close). We already have the mean above (2.82), but what is the variance?

```
## [1] 2.645042
```

This is not exactly the same, but good enough to be going on with.

So, it seems to me at least that we can use the Poisson distribution to at least somewhat accurately describe the probability of occurrence of a given number of total goals scored in a premier league game, for this season at least.

The first thing I am going to do with that information is build a figure which compares the *actual* numbers of goals scored in games with the *predicted* numbers which would be scored if the goals scored were a perfect Poisson distribution.

Ideally, we would build the Poisson probabilities for 0-9 goals (which is the maximum number scored in 21-22). However, if you do this, you will find that the two tables (actual and predicted goals) will have different numbers of rows. This is because there were no games with 8 goals scored in 21-22, but one with 9.

This is a bit annoying, but not a big issue. What we do is build a Poisson probability distribution for 0-8 goals, and treat the final probability as that for '8 or more' goals, for the purposes of drawing our figure. This isn't strictly correct, because we have not collected up our actual goal data into that category, but we could do that if we wanted to.

Later, we'll break this down properly, but for our showy graph here we don't really need to do it.

```
##      PoisProb
## 1 0.059700132
## 2 0.168260108
## 3 0.237113915
## 4 0.222762284
## 5 0.156959477
## 6 0.088475579
## 7 0.041560239
## 8 0.016733465
## 9 0.005895244
```

So, this table of probabilities, based on the Poisson distribution, allows us to make some predictions of what we could expect.

Remembering again that there are 380 games in a season, we can see that there is a 0.16 chance (i.e. 16%) of seeing 4 goals in a game, which equates to 380 x 0.16 = 60.8 games we would expect to see in a season with 4 goals.

We can check this out by creating a new table comparing **actual** with **predicted** values…

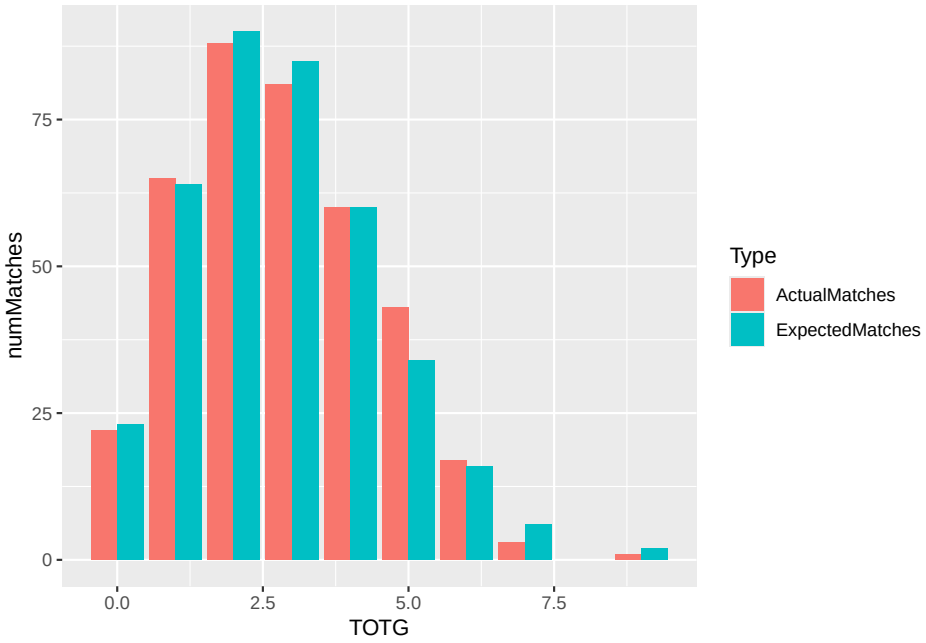
```
## # A tibble: 9 x 3
##    TOTG ActualMatches ExpectedMatches
##   <dbl>         <int>           <dbl>
```

```
## 1      0              22              23
## 2      1              65              64
## 3      2              88              90
## 4      3              81              85
## 5      4              60              60
## 6      5              43              34
## 7      6              17              16
## 8      7               3               6
## 9      9               1               2
```

Remember, as you can see above, the row for '8' is missing, and it goes straight to 9. So, to repeat, just take (for simplicity's sake here) the final row of Expected Matches as meaning '2 games with 8 or more goals'

Anyway, it's pretty scarily close! E.g. in 21-22 there were 60 matches with 4 goals, and that is identical to the predicted amount (the difference with the predicted 60.8 from the probability distribution is likely a rounding error).

Let's plot this for the big pay-off:



Wow, that is pretty accurate! Remember, the 'expected' figures are purely drawn from the *entirely theoretical* Poisson distribution with a mean of 2.82. They very closely match the actual data.

To be honest though, that graph is really just a bit of show, but what we want is the 'dough' (I had an old friend who was a golfer who always said 'drive for show, putt for dough…). In other words, what could we do with this information?

Well, there is a lot of money spent on sports betting. We could use this information to help us decide whether we should place a bet that **there will be a game in the 22-23 season with 10 goals?** - assuming that this is originally being written before the 22-23 season (which is true).

Remember, there were 380 games in 21-22, none of which contained 10 goals. So, for ease of thinking about this, let's make the assumption that the PL started in 2021, and there were no games before to count (in reality, we would go back to the last time 10 goals were scored and count from there, but let's go with this in the first instance, and we'll expand later on, promise).

So, first, we need to calculate a new set of probabilities out to 10 goals.

```
##        PoisProb
## 1  0.0597001317
## 2  0.1682601080
## 3  0.2371139154
## 4  0.2227622837
## 5  0.1569594775
## 6  0.0884755792
## 7  0.0415602392
## 8  0.0167334647
## 9  0.0058952437
## 10 0.0018461421
## 11 0.0005203206
## 12 0.0001333166
```

So, the number we want is the second-last probability: 0.00052.

This is because the table starts from the probability of 0 goals.

This means we can expect a game with 10 goals to happen every 1923 games

Calculate this by dividing 1 by the probability.

So, given there are 380 games per season, you would expect a game with 10 goals to happen once every 5 seasons.

So, if I started counting from the 21-22 season, the answer is NO, I would not bet on there being a game with 10 goals in the 22-23 Premier League season.

But, let's add some (pretty obvious) further context, as of November 2022, there have been 5 games in the history of the Premier League where 10 goals have been scored (and 1 with 11).

The Premier League has been going since 1992, and so far in Nov 22 there have been…30 seasons. So, we are probably due one.

AND, the last game with 10 goals was in 2013 (Man Utd 5, West Brom 5)

So, I reckon we are *definitely-maybe-probably* due one.

By the way, as of the start of the 22-23 season, there had been 21 matches with 9 goals. We would expect given our Poisson distribution that a game with 9 goals should happen every 1.4 seasons, meaning over 30 seasons we would expect…. 21.

9.2.1 2023 Update

So, the above example is *totally unchanged* (other than some typo corrections and nicer wording) from when I originally coded it in 2022. The million dollar question: *was there a game with 10 goals in the 22-23 season…?*

The answer is no, there wasn't. But there were two with 9! Luckily I am not a betting man. I do think we are due one, so this season (2023-2024) is probably a good bet, but as of January 20th 2024, we still haven't had one… Unless of course you think something has fundamentally changed about the premier league in recent years, changing the probability of goals (and thus suggesting recent seasons are drawn from a *Poisson distribution with different characteristics* than past ones), which is in fact an interesting question in its own right…

9.2.2 2024 Update

Did we get one in 2023-2024…again, no. The highest in fact this season was 8 - and there were 3 games with 8 goals in.

This is interesting. At this point (30 September, 2025) I find it tempting to put on a bet for a 10 goal game. I might do that, even though I am not much of a gambler. Here's why:

1. The probabilities suggest we are well overdue one, unless there has been something fundamentally changing about the distribution of premier league goal.
2. The sort of change would logically need to be a reduction of average goals scored per game, which would change the mean of the Poisson distribution, reducing the probabilities of a 9-goal game.
3. However, the average goals per game is actually ***increasing***, according to the Premier League's own figures (2024).
4. So, one could also draw the conclusion that the Poisson distribution is no longer a good fit - but this could be checked, and I would guess this isn't the case (you can check yourself).

Interesting.

Chapter 10

Introduction to Statistics

This Chapter builds on the material in the second part of Lesson 7. Here, we begin to apply our probability concepts to the task of statistical inference in the 'classical' way.

The theory explained and demonstrated here is the foundation of the (very large) majority of the quantitative business and management research you will read and use. Almost any time someone refers to a finding being **(statistically) significant** or to a **p-value**, they are using this idea. However, a major weakness in many social science fields - business and management being one - is that a rather large proportion of researchers don't actually understand the core principles of statistical inference, and are instead often just copying what other people have written, or told them to do.

So, if you can get your head around the next 3-4 chapters, you'll be in a strong position to work with what's already out there, and also to do your own solid research, if you wish.

10.1 The Distribution of Sample Means, and the Central Limit Theorem

The first thing to cover is an extension of the ideas in the last chapter about sampling distributions. It's called the *Central Limit Theorem*, and is often the source of much angst in statistics students (including myself). However, here I am actually going to demonstrate how it works, and show how is actually quite amazing.

To do so, I will use a synthetic data set of monthly incomes from 20 people. Let us treat this data set as the **population**.

```
## # A tibble: 6 x 2
```

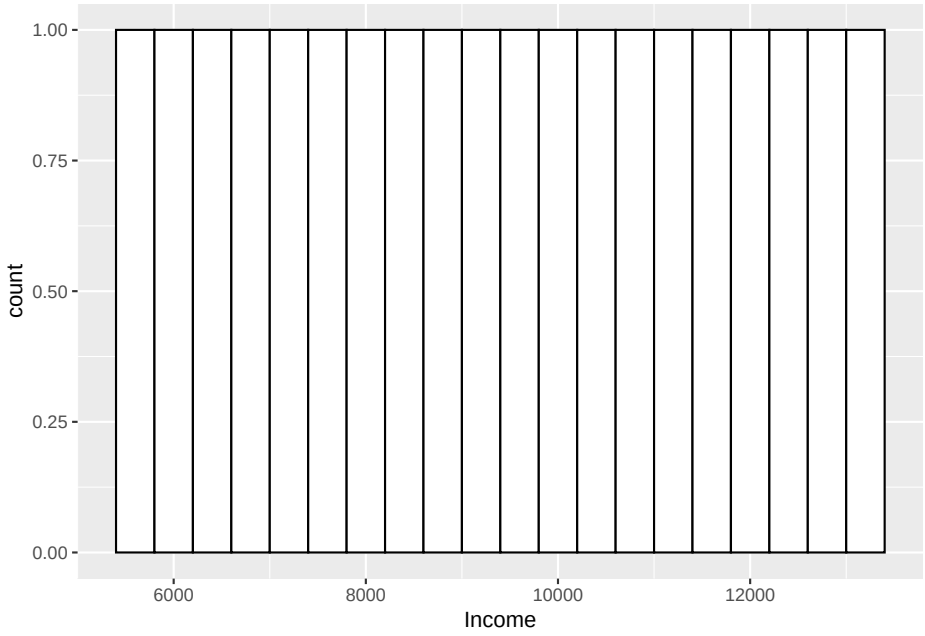

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```
##   Person Income
##   <chr>   <dbl>
## 1 A        5600
## 2 B        6000
## 3 C        6400
## 4 D        6800
## 5 E        7200
## 6 F        7600
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5600    7500    9400    9400   11300   13200
```

```
## [1] 2366.432
```

Above, you'll see the important descriptive statistics, including the standard deviation of 2366.432. Let's plot the frequency distribution, and once we do so we will find that it is essentially uniform - every value occurs once and only once in the population.



Now, what I am going to do is take a *single random sample of 2* from that population. This sample is taken 'without replacement', if you are wondering, and so it is in essence exactly as if a researcher went into the field and took a sample from this population.

```
## [1] "One Random Combination of 2 of the 20 Income Values"
```

```
## [1]  5600 10000
```

Let's take the mean of the sample:

```
##    res1
## 1  5600
## 2 10000
```

```
## [1] 7800
```

The mean of the random sample of 2 will of course be different each time the sample is taken (e.g. when writing this text, the mean was 10600, but this may not be the same in the version you are reading!)

Just for reference, the mean of the population (remember, the 20 people) was 9400

So, let's now do that for **every possible combination of 2 values** from this population.

Given there are n=20 values in the population, there are k=190 possible combinations of two values.

Note: This is not the same as bootstrapping - we are *not* sampling with replacement here. We are instead taking combinations. The thing to think about is that this is the equivalent of taking every single possible sample of 2 that you could take from this population.

Remember the ping-pong balls I used in class to demonstrate!

Let's do it - the calculation below is adapted from a GeeksforGeeks tutorial: https://www.geeksforgeeks.org/calculate-combinations-and-permutations-in-r/

```
## [1] "Every Possible Combination of the 20 Income Values"
```

```
##          [,1]  [,2]
##   [1,]  5600  6000
##   [2,]  5600  6400
##   [3,]  5600  6800
##   [4,]  5600  7200
##   [5,]  5600  7600
##   [6,]  5600  8000
##   [7,]  5600  8400
##   [8,]  5600  8800
##   [9,]  5600  9200
##  [10,]  5600  9600
##  [11,]  5600 10000
##  [12,]  5600 10400
##  [13,]  5600 10800
##  [14,]  5600 11200
##  [15,]  5600 11600
##  [16,]  5600 12000
##  [17,]  5600 12400
##  [18,]  5600 12800
```

```
##  [19,]  5600 13200
##  [20,]  6000  6400
##  [21,]  6000  6800
##  [22,]  6000  7200
##  [23,]  6000  7600
##  [24,]  6000  8000
##  [25,]  6000  8400
##  [26,]  6000  8800
##  [27,]  6000  9200
##  [28,]  6000  9600
##  [29,]  6000 10000
##  [30,]  6000 10400
##  [31,]  6000 10800
##  [32,]  6000 11200
##  [33,]  6000 11600
##  [34,]  6000 12000
##  [35,]  6000 12400
##  [36,]  6000 12800
##  [37,]  6000 13200
##  [38,]  6400  6800
##  [39,]  6400  7200
##  [40,]  6400  7600
##  [41,]  6400  8000
##  [42,]  6400  8400
##  [43,]  6400  8800
##  [44,]  6400  9200
##  [45,]  6400  9600
##  [46,]  6400 10000
##  [47,]  6400 10400
##  [48,]  6400 10800
##  [49,]  6400 11200
##  [50,]  6400 11600
##  [51,]  6400 12000
##  [52,]  6400 12400
##  [53,]  6400 12800
##  [54,]  6400 13200
##  [55,]  6800  7200
##  [56,]  6800  7600
##  [57,]  6800  8000
##  [58,]  6800  8400
##  [59,]  6800  8800
##  [60,]  6800  9200
##  [61,]  6800  9600
##  [62,]  6800 10000
##  [63,]  6800 10400
##  [64,]  6800 10800
```

```
##  [65,]  6800 11200
##  [66,]  6800 11600
##  [67,]  6800 12000
##  [68,]  6800 12400
##  [69,]  6800 12800
##  [70,]  6800 13200
##  [71,]  7200  7600
##  [72,]  7200  8000
##  [73,]  7200  8400
##  [74,]  7200  8800
##  [75,]  7200  9200
##  [76,]  7200  9600
##  [77,]  7200 10000
##  [78,]  7200 10400
##  [79,]  7200 10800
##  [80,]  7200 11200
##  [81,]  7200 11600
##  [82,]  7200 12000
##  [83,]  7200 12400
##  [84,]  7200 12800
##  [85,]  7200 13200
##  [86,]  7600  8000
##  [87,]  7600  8400
##  [88,]  7600  8800
##  [89,]  7600  9200
##  [90,]  7600  9600
##  [91,]  7600 10000
##  [92,]  7600 10400
##  [93,]  7600 10800
##  [94,]  7600 11200
##  [95,]  7600 11600
##  [96,]  7600 12000
##  [97,]  7600 12400
##  [98,]  7600 12800
##  [99,]  7600 13200
## [100,]  8000  8400
## [101,]  8000  8800
## [102,]  8000  9200
## [103,]  8000  9600
## [104,]  8000 10000
## [105,]  8000 10400
## [106,]  8000 10800
## [107,]  8000 11200
## [108,]  8000 11600
## [109,]  8000 12000
## [110,]  8000 12400
```

```
## [111,]  8000 12800
## [112,]  8000 13200
## [113,]  8400  8800
## [114,]  8400  9200
## [115,]  8400  9600
## [116,]  8400 10000
## [117,]  8400 10400
## [118,]  8400 10800
## [119,]  8400 11200
## [120,]  8400 11600
## [121,]  8400 12000
## [122,]  8400 12400
## [123,]  8400 12800
## [124,]  8400 13200
## [125,]  8800  9200
## [126,]  8800  9600
## [127,]  8800 10000
## [128,]  8800 10400
## [129,]  8800 10800
## [130,]  8800 11200
## [131,]  8800 11600
## [132,]  8800 12000
## [133,]  8800 12400
## [134,]  8800 12800
## [135,]  8800 13200
## [136,]  9200  9600
## [137,]  9200 10000
## [138,]  9200 10400
## [139,]  9200 10800
## [140,]  9200 11200
## [141,]  9200 11600
## [142,]  9200 12000
## [143,]  9200 12400
## [144,]  9200 12800
## [145,]  9200 13200
## [146,]  9600 10000
## [147,]  9600 10400
## [148,]  9600 10800
## [149,]  9600 11200
## [150,]  9600 11600
## [151,]  9600 12000
## [152,]  9600 12400
## [153,]  9600 12800
## [154,]  9600 13200
## [155,] 10000 10400
## [156,] 10000 10800
```

```
## [157,] 10000 11200
## [158,] 10000 11600
## [159,] 10000 12000
## [160,] 10000 12400
## [161,] 10000 12800
## [162,] 10000 13200
## [163,] 10400 10800
## [164,] 10400 11200
## [165,] 10400 11600
## [166,] 10400 12000
## [167,] 10400 12400
## [168,] 10400 12800
## [169,] 10400 13200
## [170,] 10800 11200
## [171,] 10800 11600
## [172,] 10800 12000
## [173,] 10800 12400
## [174,] 10800 12800
## [175,] 10800 13200
## [176,] 11200 11600
## [177,] 11200 12000
## [178,] 11200 12400
## [179,] 11200 12800
## [180,] 11200 13200
## [181,] 11600 12000
## [182,] 11600 12400
## [183,] 11600 12800
## [184,] 11600 13200
## [185,] 12000 12400
## [186,] 12000 12800
## [187,] 12000 13200
## [188,] 12400 12800
## [189,] 12400 13200
## [190,] 12800 13200
```

```
## [1] "Number of combinations without repetition"
```

```
## [1] 190
```

That was a rather long exhibit up there, but I wanted to show you that this was actually really done properly, and we have 190 combinations.

Next, let us create the means of each of the 190 combinations

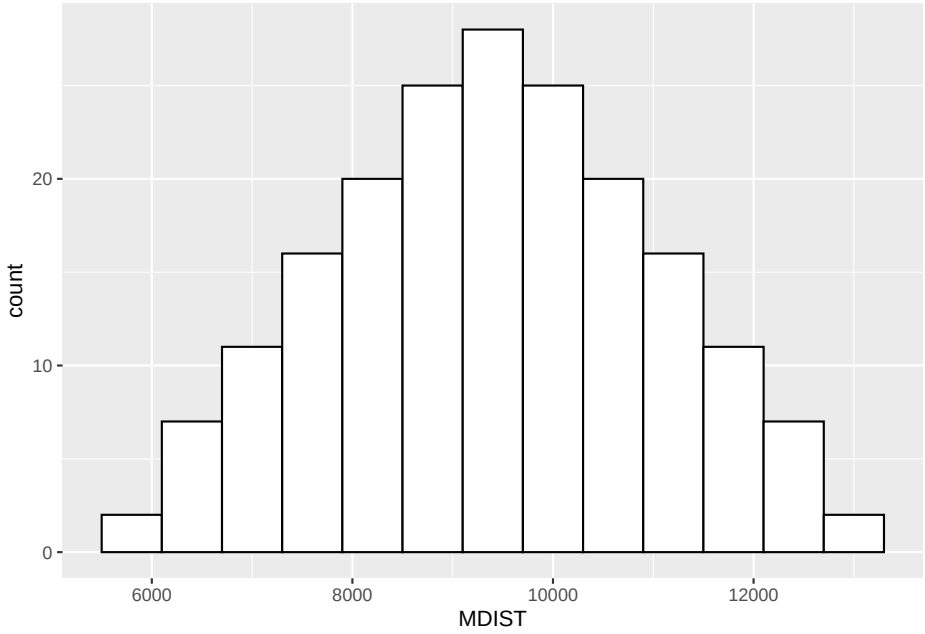
```
##     X1   X2 MDIST
## 1 5600 6000  5800
## 2 5600 6400  6000
## 3 5600 6800  6200
```

```
## 4 5600 7200  6400
## 5 5600 7600  6600
## 6 5600 8000  6800
```

That's just the first 6 (the head of the data set), but you can rest assured that there are 190 means created here.

OK, so here is the kicker. Let's plot a histogram of these 190 means.

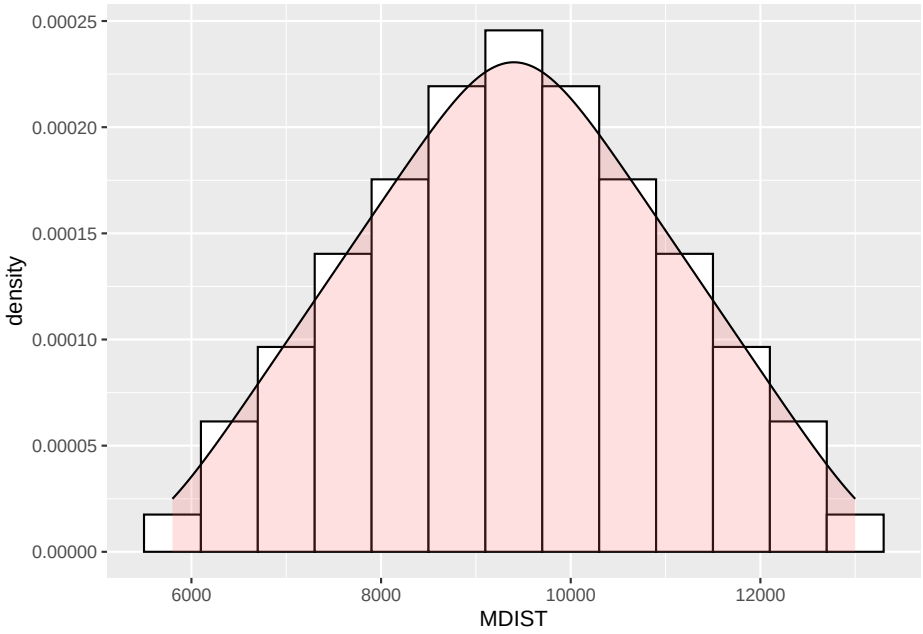
Remember, a histogram is a frequency distribution:



Well well well!

Even though the original population was a completely uniform distribution with a mean of 9400, the distribution of all of the sample means looks quite a lot like a normal distribution / bell curve!

Let's overlay one on it as well to make the point…



Further, let's take the mean of those means...

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5800    8200    9400    9400   10600   13000
```

Hello! It turns out, the mean of those means is the *population mean!!!*

This, in a demonstration, is *the central limit theorem.*

Let's **move back to the slides** to talk about this some more.

10.2 Demonstrating Confidence Intervals

Below I will demonstrate the idea of confidence intervals as a measure of precision, and at the same time help to really lock in the correct interpretation of what a confidence interval really means.

To do so, I will continue the in-class example of WBS Ph.D. graduate salaries. However, here, I will **simulate** the population distribution that we have sampled from. Let's recap:

Sample size (N) = 42

Sample mean = 170,000

Sample SD = 46617.4

I will simulate a population distribution that this sample could quite conceivably have come from. It will have the following features:

Population mean = 165,000

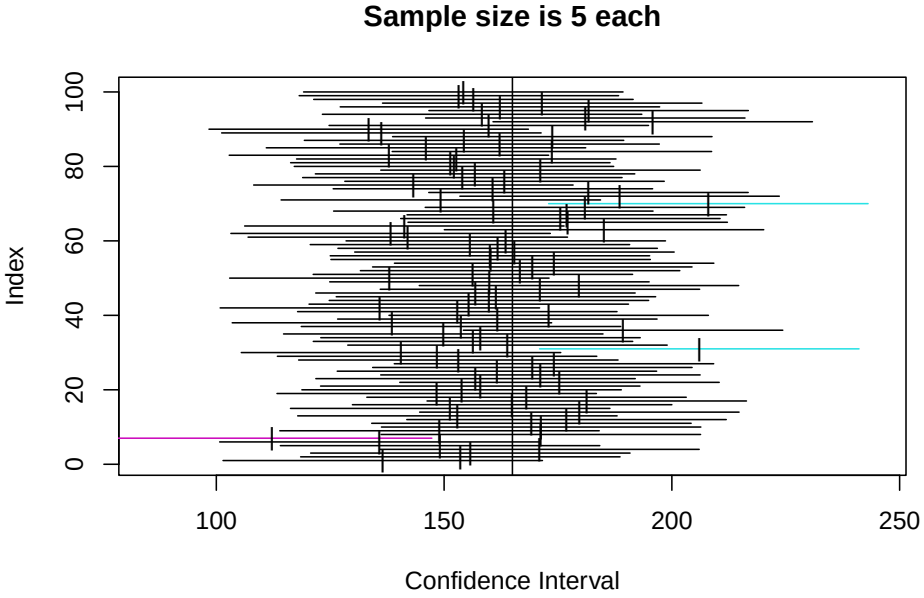
Population SD = 40000

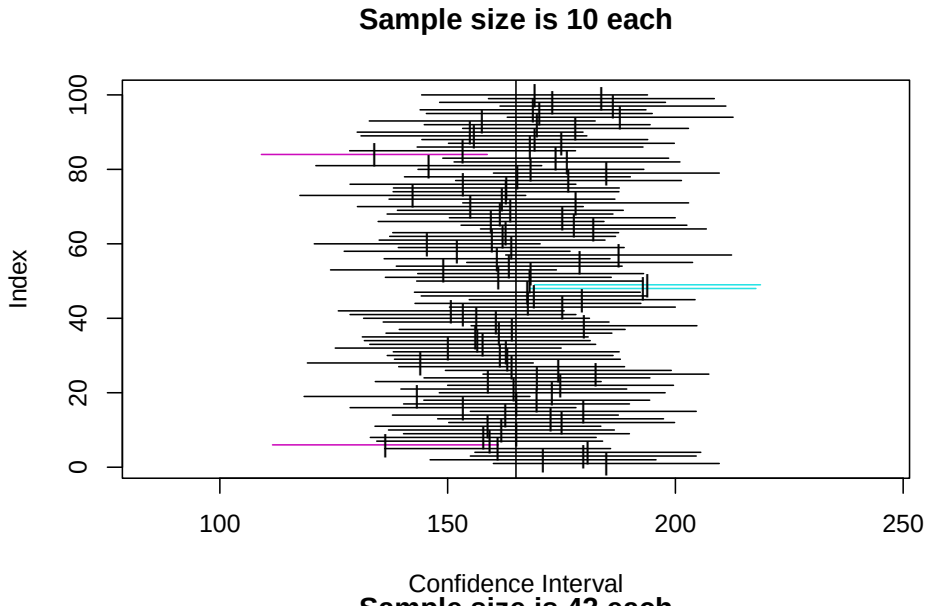
I will also assume that the population distribution is *normal*. That is, the distribution of all business and management Ph.D. starting salaries is shaped like a bell curve - the normal distribution (remember birth weights from the last chapter). This is a reasonable assumption in my view, given the features of the phenomena we are investigating.

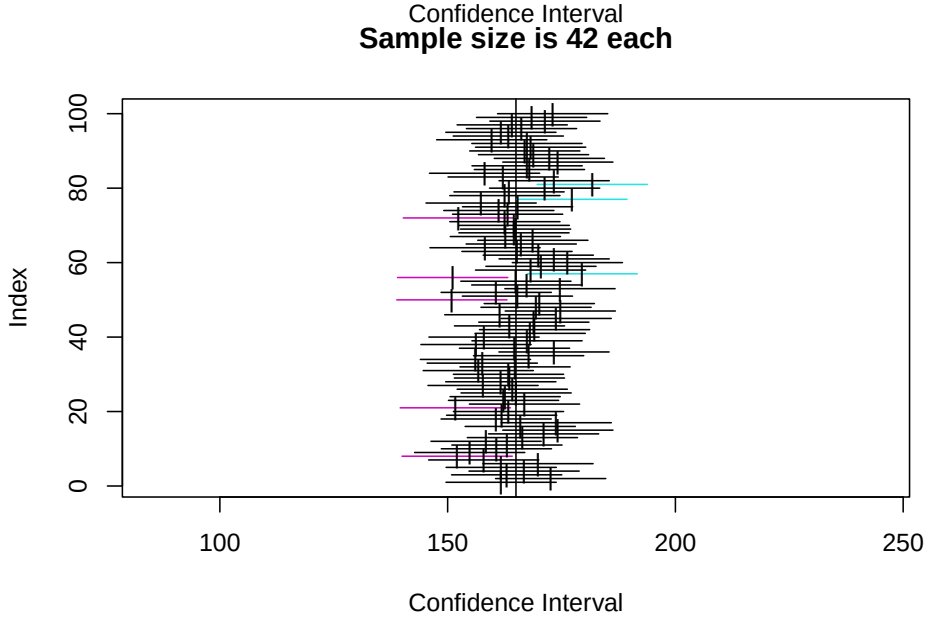
The important thing to remember is not really the population distribution of salaries (which is just used here to simulate the population), but the *sampling distribution of the means*. Remember, the central limit theorem shows us that **whatever the shape of the population distribution we are sampling from**, the sampling distribution of the means will be centred on the population mean, and will get closer and closer to a normal distribution as the sample size grows. With small samples from a very lopsided population the approximation can be rough, but it often kicks in surprisingly quickly, as the 20-person uniform population above showed with samples of just 2.

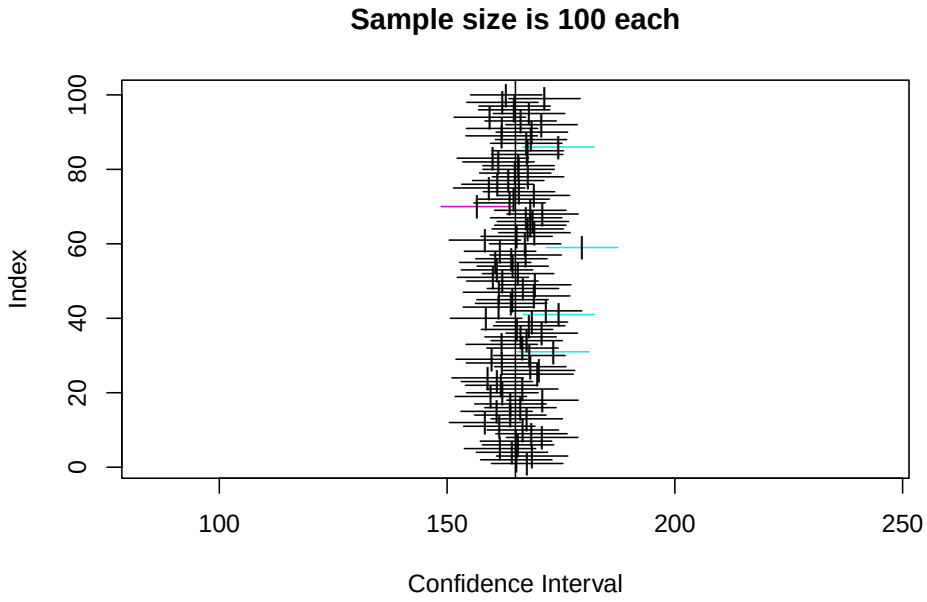
So, below, I will simulate the population, and then calculate 100 confidence intervals for 4 different sample sizes (5, 10, 42, 100), to demonstrate two things:

a) How confidence intervals get more precise as sample size increases, and
b) How to interpret your specific confidence interval as simply one possibility of many.




Sample size is 10 each
Index
0
20
40
60
80
100
100
150
200
250
Confidence Interval


Sample size is 42 each
Index
0
20
40
60
80
100
100
150
200
250
Confidence Interval



So, you can see that the confidence intervals for each of the 100 samples at a given sample size all cluster around the mean, but are wider as sample size is lower. That makes sense intuitively.

Looking specifically at our example N of 42, we can estimate which one of the 100 that might be, and see how the CI for sample means of about that size does include the real population mean of 165. You can also see there are a number of coloured CIs which do **not** include the population mean. This shows how to interpret a CI.

Specifically, **your actual sample** is one of a very large amount of possible samples from the population (here we took 100 but we could have taken many more). The CI calculation essentially creates a CI with the width so that, out of 100 random samples, approximately 95 will actually contain the *true population mean*.

As such, we can see how it is clearly a measure of the **precision** of the estimate. For a compelling visualisation of this, look how wide the CIs for N=5 are.

One thing to remember though is the 'law of large numbers' here. You might notice that not all of the CI plots contain exactly 5 intervals that do not contain the mean. This is analogous to the idea of coin flipping - it's just probability over very large numbers of trials, not saying that 'out of every 100, exactly 5 will not include the population mean'. But, over a very large number of trials, we can expect the amount of intervals that do not include the mean to tend towards 5%, in the same way that over a large number of coin flips, the proportion of heads tends towards 50%.

10.3 Rate of Change in Football Goals per Season

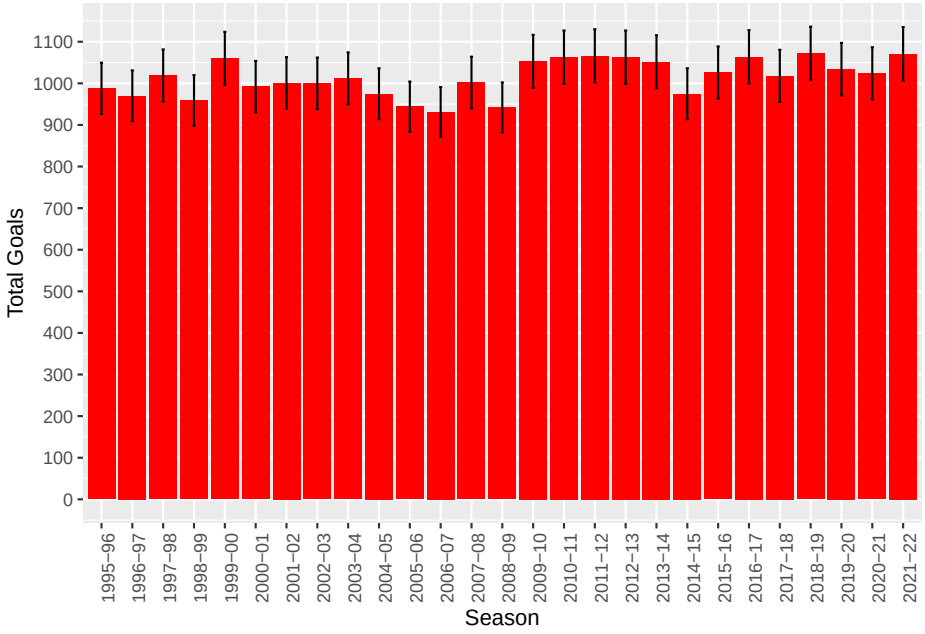
In this example, I'll build on the idea of confidence intervals, to show how we can test hypotheses about the world. I'll continue on with my football goals example from Chapter 9. Here, I'm using data from footballhistory.org (n.d.).

I hand-entered this into a spreadsheet, and calculated the additional stuff.

```
## # A tibble: 6 x 7
##   Season  Games Goals   GPG    SE Lower95CI Upper95CI
##   <chr>   <dbl> <dbl> <dbl> <dbl>     <dbl>     <dbl>
## 1 1995-96   380   988  2.6   31.4      926.      1050.
## 2 1996-97   380   970  2.55  31.1      909.      1031.
## 3 1997-98   380  1019  2.68  31.9      956.      1082.
## 4 1998-99   380   959  2.52  31.0      898.      1020.
## 5 1999-00   380  1060  2.79  32.6      996.      1124.
## 6 2000-01   380   992  2.61  31.5      930.      1054.
```

You can see here I have calculated the standard errors from the yearly goal totals (which represent that year's underlying rate of goal occurrence), then used that to calculate the 95% Confidence Interval limits

We can use these to create a nifty chart with the error bars…drawing from the code used by Spiegelhalter in his book for Figure 9.4, available on his GitHub: https://github.com/dspiegel29/ArtofStatistics/blob/master/09-4-homicide-rates-E%2BW/09-4-homicide-trends-x.Rmd



From this chart, and looking at the data itself, we can see that the 95% Intervals overlap, so it is hard to conclude that the underlying rate of goals has changed significantly year on year. Yes, even in the pandemic.

The closest we get in fact is between the 2008-2009 and 2009-10 season. Interestingly, this corresponds to when Man City were bought, and it is evident that the top teams scored a lot more that year.

In fact, you could add data for the next few seasons, to directly test the conclusion that the Premier League drew in the article I referenced in Chapter 9 (2024).

This is a stringent test however, and the ONS suggest that you can also test the change by using a Z-test (Office for National Statistics, 2017), which *directly tests the hypothesis* that the change is zero, using the assumption that the events are Poisson distributed (we agree) and also that when the number of events are large (generally over 20), we can use an approximation to the normal distribution.

The z-test is simply explained in a short BMJ article (Pocock, 2006).

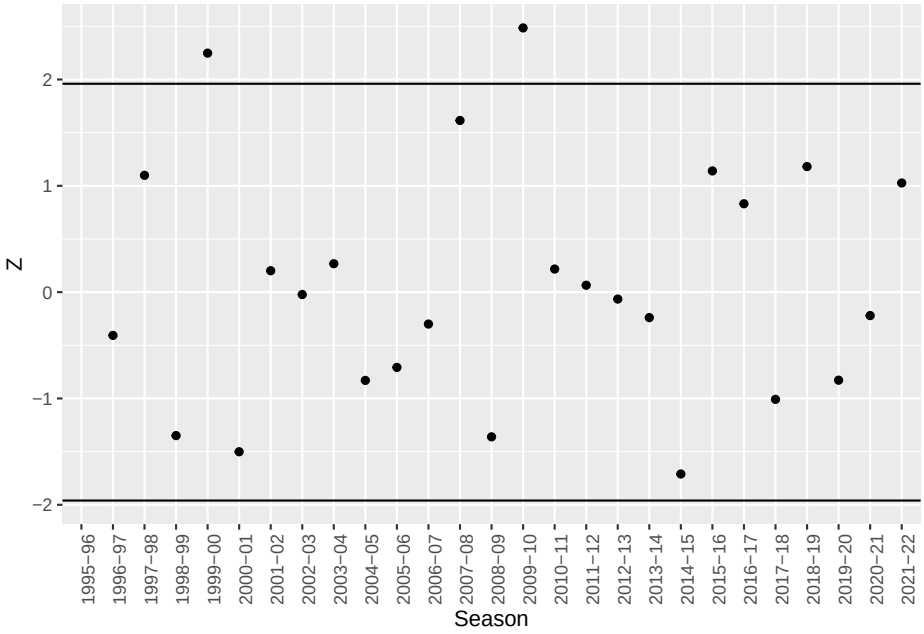
It links us nicely to the next lesson, because it is aiming to test a *specific hypothesis* that the difference is zero…

If you open the data file below, you can see I have calculated the z-test results for the difference between each season, year-on-year.

```
## # A tibble: 6 x 9
##   Season  Games Goals   GPG    SE Lower95CI Upper95CI Change      Z
##   <chr>   <dbl> <dbl> <dbl> <dbl>     <dbl>     <dbl>  <dbl>  <dbl>
## 1 1995-96   380   988  2.6   31.4      926.     1050.     NA NA
## 2 1996-97   380   970  2.55  31.1      909.     1031.    -18 -0.407
## 3 1997-98   380  1019  2.68  31.9      956.     1082.     49  1.10
## 4 1998-99   380   959  2.52  31.0      898.     1020.    -60 -1.35
## 5 1999-00   380  1060  2.79  32.6      996.     1124.    101  2.25
## 6 2000-01   380   992  2.61  31.5      930.     1054.    -68 -1.50
```

The ‘Z’ column is what we are interested in, and we are looking for a z-value greater than + or - 1.96 for a 95% test (analogous to the 95% intervals we’ve been dealing with so far).

A simple way to visualise this is to plot the z-values for each season, and include ‘control lines’ which represent the + or -1.96 z value, beyond which we consider there to be a significant difference



So, we can see that the 1999-2000 season, and the 2009-10 seasons exceed our z values, making them significantly different from the seasons before.

Of course, we could do z-tests for any combination of two seasons, if we had a good reason. You can see on the ONS website they do this for different years' murder rates to make a point.

10.4 Reading the literature

Every estimate in a paper comes with a measure of how uncertain it is, even if it's tucked away in brackets or buried in a footnote. This chapter, and Chapter 6 before it, showed where that uncertainty comes from: the sample you happen to have is one of a huge number you could have drawn, and each would have given a slightly different answer. The standard error is a measure of how much estimates would vary across those possible samples, and a confidence interval is built from it. Once you can read those two numbers properly, a surprising amount of a results section opens up.

10.4.1 Where you'll meet this

Numbers in brackets. In most regression tables each estimate has a second number underneath it in brackets. It's usually the standard error, but it can be the t-value, and occasionally it's a confidence interval, so check the table note. If it's a standard error, you can build a rough 95% confidence interval yourself: the estimate plus or minus about two standard errors. That takes a few seconds,

and it tells you much more than a row of stars does. An estimate of 0.30 with a standard error of 0.14 is "significant," but its interval runs from roughly 0.02 to 0.58, so the data are consistent with an effect that's barely there and with one that's nearly twice as large as the estimate.

Confidence intervals on estimates. More and more papers report intervals directly. Read them the way the simulation in Section 10.2 taught you: as the output of a procedure that captures the true value in about 95 of every 100 samples, not as a range that the true value is 95% certain to be in. In practice the useful questions are simple. How wide is it? Does it include values that would be too small to matter? Does it include values that would be implausibly large? A wide interval that only just excludes zero is a much weaker finding than its p-value suggests.

Error bars in figures. Bar charts and line plots often carry error bars, and they're one of the most misread things in any paper. First, find out what the bars are. They could be standard deviations, standard errors or confidence intervals, and a surprising number of figures don't say. Standard-error bars are only about half the length of 95% confidence intervals, so the same data can look far more or far less convincing depending on which was chosen. Second, be careful with overlap. Two 95% intervals that overlap can still come from a difference that's statistically significant. In fact, for two independent groups the intervals can overlap by up to about half of one arm and the difference will still have p of around .05 (Cumming & Finch, 2005). You saw this in this chapter's football example: the intervals for consecutive seasons overlapped, yet the z-tests found two significant changes.

Clustered and robust standard errors. Every standard error you've met in this book assumes that each observation is an independent draw, like the ping-pong balls in the lecture. Much management data isn't like that. Employees are nested in teams and firms, firms appear year after year in a panel, and observations within the same cluster tend to resemble each other. When that happens, the data contain less independent information than the sample size suggests, and ordinary standard errors come out too small. *Clustered standard errors* correct for this. They change the uncertainty, not the estimates themselves. As a reader, ask what the observations were clustered by, and whether that matches how the data were actually collected. Also ask how many clusters there were. With only a handful, such as a few industries or a dozen countries, even clustered standard errors can be too small (Cameron & Miller, 2015). You'll also see *robust* (or "heteroskedasticity-robust") standard errors, which deal with a different problem: spread in the data that isn't constant. They're not a substitute for clustering.

What a standard error doesn't cover. A standard error measures one thing: how much an estimate would bounce around across random samples from the same population. It says nothing about whether the sample was random in the first place, whether the people who answered differ from those who didn't, or whether the measures capture what they claim to. Those are the "inference gaps"

from Chapter 6, and many of them are the selection problems from Chapter 4. That doesn't make intervals from non-random samples meaningless. It means they describe the precision of the estimate under an assumption, and the gap between that assumption and reality isn't in the interval.

10.4.2 The connection back

Every standard error in every paper is a compressed version of the simulations in this chapter. Imagine all the samples you could have drawn; the standard error is how spread out their estimates would be, and a confidence interval is a way of turning that spread into a range. Both depend on the samples being random and independent, which is exactly what clustering and non-random sampling call into question.

10.4.3 Questions worth asking

1. **What's in the brackets?** Standard errors, t-values, or something else?
2. **What range of values is the estimate consistent with?** Does that range include effects too small to matter?
3. **Are the observations really independent?** If not, were the standard errors clustered, at what level, and how many clusters were there?
4. **In the figures, what are the error bars?** Is any claim about differences based on whether the bars overlap?
5. **What sources of error does the interval leave out?**

10.4.4 One red flag

"The confidence intervals overlapped, so there was no difference." This conclusion turns up in text and in figures alike, often with error bars that aren't even labelled. Overlap isn't the test. Two 95% intervals can overlap substantially and the difference between them can still be significant, and a figure with standard-error bars can make a trivial difference look convincing. If a paper's argument depends on how two sets of error bars look, the authors should have tested the difference itself, as Chapter 11 will argue more generally.

10.4.5 If you want to read more

Geoff Cumming and Sue Finch's "Inference by eye" (Cumming & Finch, 2005) is a short paper about reading confidence intervals and error bars in figures. It's full of pictures, and it will change the way you look at every bar chart you see from now on.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 11

Classical Statistical Hypothesis Testing

This Chapter relates to the first part of Lecture 8, and contains some (re)worked examples from earlier chapters / lessons, focusing specifically on the hypothesis testing aspect of them.

11.1 Testing a Real Hypothesis: Are WBS PhD Salaries Different?

Back in Section 10.2 we used the WBS PhD graduate salary data — 42 graduates, mean starting salary $170,000, standard deviation $46,617.4 — to build a confidence interval. That was useful, but it didn't actually answer the question I cared about when I started looking at this data: can I claim, in the PhD programme prospectus, that WBS graduates earn more than graduates elsewhere?

A confidence interval alone doesn't let me say that. It tells me something about precision, not about a comparison. To make a comparison I need a number to compare against, and I need to state, in advance, how different my sample would have to be before I'm willing to call it a real difference rather than the ordinary noise you'd expect from any sample of 42 people.

That comparison number is my null hypothesis, and I didn't invent it out of thin air. The American Marketing Association runs a doctoral special interest group, and every year they survey new marketing PhD graduates about starting salary and send me the results — mostly, I suspect, to depress me about what I earned in my first job. The most recent figure was US$145,000. It's not a perfect number (a proper job would involve going round every PhD director in the business school and averaging across fields), but it's a real, defensible estimate

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of "what graduates elsewhere earn", which is exactly what a null hypothesis needs to be.

- **Null hypothesis:** the average starting salary of WBS PhD graduates is not different from US$145,000.
- **Sample:** 42 WBS graduates, sample mean $170,000, sample SD $46,617.4 — the same sample as Section 10.2.

Before doing any calculation, there's a decision to make, and it's the decision that usually gets made by accident, halfway through the arithmetic, once everyone's attention is already used up. Did I hypothesize a direction? I could have framed the alternative as directional — WBS graduates earn *more* than $145,000 — or non-directional — WBS graduates earn a *different* amount, which could be more or less. A directional hypothesis is the stronger claim, and it has to be committed to in advance, not read off the result after the fact.

Here's the honest version: privately, of course I was expecting WBS graduates to earn more. But my null hypothesis, as written, was "not different from $145,000", which commits me to the non-directional alternative whatever I was quietly hoping. I didn't encode the direction I actually believed in, so I don't get to use it. That's not me being pedantic about my own case for the sake of the example — it's the actual mechanism. A lot of people hypothesize a direction in their head and then run a two-tailed test anyway, mostly because no one ever made them sit down and decide properly. Here, the test is two-tailed.

With that settled, the arithmetic is almost an anticlimax. What I want to know is: if the population mean really is $145,000, how far away — in standardised units — is my observed sample mean of $170,000?

```
## [1] 3.475495
```

Same standardisation logic as everywhere else in this book: take what I observed, subtract what I'd expect under the null, divide by how much variation sampling alone would produce. The effect is to redraw the whole problem on a scale where the null sits at zero, and my sample mean lands wherever it lands. A z of around 3.5 means my sample mean is roughly three and a half standard errors out from where the null says it should be — well beyond the 1.96 we've used as our 95% cut-off elsewhere.

```
## [1] 0.0005099124
```

That p-value — somewhere around 0.0005, about 1 in 2,000 — means: *if* the null is true, the probability of drawing a sample of 42 with a mean this far out, in either direction, is about 1 in 2,000. It does not mean there's a 1-in-2,000 chance the null is true. A p-value says nothing about the probability of any hypothesis. It's a statement about how surprising the data would be under an assumption, not a verdict on the assumption.

So, do I reject the null? This is a judgement, not a law of nature. Nothing about this p-value compels a conclusion — I could, in principle, decide this is

just an unusually extreme sample from a population that really does average \$145,000, and nothing in the maths would stop me. What I actually do, like almost everyone else, is apply a cut-off — conventionally $p < 0.05$ — and treat anything below it as evidence against the null. That cut-off exists because people like rules, not because 0.05 has any special status in nature. Here the data are miles past any conventional threshold, so the call is easy. It won't always be, and when it isn't, the arithmetic still won't make the decision for you.

Directional vs non-directional, side by side:

	One-tailed (directional)	Two-tailed (non-directional)
Alternative hypothesis	States a direction — e.g. "bigger than \$145,000"	States only "different from \$145,000"
Must be decided	Before looking at the data	Before looking at the data
Relationship between the p-values	Half the two-tailed p, for the same z	Double the one-tailed p, for the same z
This example	Not what I actually committed to	What I actually committed to

11.2 Premier League Goals and the Poisson Distribution

In this section, we are not really worried about the *actual* goals or anything like that. What we are concerned about is formally testing the hypothesis that Premier League goals per game actually do follow a Poisson distribution. To do this, we will use what is called a **chi-square** test, which uses the chi-square statistic.

Chi-square is a fantastically versatile statistic, which is often used to compare tables of observations. Commonly, we compare a table of observed events with a table of events that we would expect if the null hypothesis were true in the population. The chi-square value is the measure of the difference between observed and expected counts in the table. We can use it here to compare the amount of goals we actually did observe in the 21-22 season, with those that we would *expect* to observe if the goals per game really do follow a Poisson distribution.

Again, much of the idea and code for this section comes from: https://bookdown.org/theqdata/honors_thesis/goal-scoring-and-the-poisson-process.html

The data are the EPL results for the 2021–22 season, from Football-Data.co.uk (n.d.), with a new column of Total Goals (TOTG) created by me in Excel.

```
## # A tibble: 6 x 107
##   Div   Date                Time                HomeTeam   AwayTeam   FTHG  FTAG
##   <chr> <dttm>              <dttm>              <chr>      <chr>     <dbl> <dbl>
## 1 E0    2021-08-13 00:00:00 1899-12-31 20:00:00 Brentford  Arsenal       2     0
## 2 E0    2021-08-14 00:00:00 1899-12-31 12:30:00 Man United Leeds         5     1
## 3 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Burnley    Brighton      1     2
## 4 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Chelsea    Crystal ~     3     0
## 5 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Everton    Southamp~     3     1
## 6 E0    2021-08-14 00:00:00 1899-12-31 15:00:00 Leicester  Wolves        1     0
## # i 100 more variables: TOTG <dbl>, FTR <chr>, HTHG <dbl>, HTAG <dbl>,
## #   HTR <chr>, Referee <chr>, HS <dbl>, AS <dbl>, HST <dbl>, AST <dbl>,
## #   HF <dbl>, AF <dbl>, HC <dbl>, AC <dbl>, HY <dbl>, AY <dbl>, HR <dbl>,
## #   AR <dbl>, B365H <dbl>, B365D <dbl>, B365A <dbl>, BWH <dbl>, BWD <dbl>,
## #   BWA <dbl>, IWH <dbl>, IWD <dbl>, IWA <dbl>, PSH <dbl>, PSD <dbl>,
## #   PSA <dbl>, WHH <dbl>, WHD <dbl>, WHA <dbl>, VCH <dbl>, VCD <dbl>,
## #   VCA <dbl>, MaxH <dbl>, MaxD <dbl>, MaxA <dbl>, AvgH <dbl>, AvgD <dbl>, ...
```

```
##    Length     Class      Mode
##       380 character character
```

The above is a quick way of calculating / checking the total number of games if you don't know already.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   2.000   3.000   2.818   4.000   9.000
```

```
## [1] 1071
```

First, let's create a table of all the matches with different numbers of goals, just as we did in Chapter 9.

```
## # A tibble: 9 x 2
##    TOTG ActualMatches
##   <dbl>         <int>
## 1     0            22
## 2     1            65
## 3     2            88
## 4     3            81
## 5     4            60
## 6     5            43
## 7     6            17
## 8     7             3
## 9     9             1
```

Here, we have a problem. The chi-square test has a condition that the expected values in every column will be greater than 5, and as we go down the Poisson probabilities (i.e. for higher total goals), it is clear that this will start to become a problem for us.

The solution is simple, we combine the numbers for larger numbers of goals into a single 'x and above' variable. Here, let's do it for 6 goals and above.

To do so, we first need to create a new table, where the values for 6 and above are combined into a new row.

```
## # A tibble: 7 x 2
##   TOTG      ActualMatches
##   <chr>             <int>
## 1 0                    22
## 2 1                    65
## 3 2                    88
## 4 3                    81
## 5 4                    60
## 6 5                    43
## 7 6 or more            21
```

Now we can create the Poisson distribution for the relevant mean, which means we first need to create the stats…

```
##  min Q1 median Q3 max     mean       sd   n missing
##    0  2      3  4   9 2.818421 1.626359 380       0
```

Below is some code which will help us to create the distributions and also check them.

```
## [1] 2.818421
```

```
## [1] 2.645042
```

As we saw in Chapter 9, while for a Poisson distribution these should be exactly the same, we decided that they were good enough to be going on with.

So, let's actually check this formally with the Chi-square test.

We do so by first creating a table of the Poisson probabilities for x or fewer goals, for the Poisson with a mean of 2.818 (i.e. the variable MeanGoals)

```
##     PoisProb
## 1 0.05970013
## 2 0.16826011
## 3 0.23711392
## 4 0.22276228
## 5 0.15695948
## 6 0.08847558
## 7 0.04156024
```

This allows us to make some predictions of what we could expect.

Remembering again that there are 380 games in a season, we can see that there is a 0.16 chance of seeing 4 goals in a game, which equates to 380 x 0.16 = 60.8 games we would expect to see in a season with 4 goals.

We can check this out by creating a new table comparing actual with predicted values…

```
##        TOTG ActualMatches   PoisProb ExpectedMatches
## 1         0            22 0.05970013              23
## 2         1            65 0.16826011              64
## 3         2            88 0.23711392              90
## 4         3            81 0.22276228              85
## 5         4            60 0.15695948              60
## 6         5            43 0.08847558              34
## 7 6 or more            21 0.04156024              16
```

This looks quite frighteningly close up until you get to the 5 row. It will be interesting to see how we go with a formal chi-square test.

```
##
##  Chi-squared test for given probabilities
##
## data:  NewGoalsTable$ActualMatches
## X-squared = 4.2244, df = 6, p-value = 0.6463
```

The P-value here is actually 0.64. This suggests that there is not a significant difference between the distribution of the actual goals, and that which would be expected if they did follow a Poisson distribution.

Notice what that sentence does and doesn't say. It says the goals data are consistent with a Poisson distribution. It doesn't say they *are* Poisson. A p-value of 0.64 means that data like these would be unsurprising if goals followed a Poisson distribution, but plenty of other distributions we didn't test might fit them just as well. Failing to reject a null is not the same as showing it's true. Absence of evidence is not evidence of absence. That might sound like pedantry, but it matters enormously when you read papers, because one of the most common moves in published research is to run a test, find nothing, and conclude that nothing is there: no bias, no difference between the groups, no problem. With a small sample, a test can easily fail to detect even a large effect. If you want to claim that something is absent, you need a test designed to show that, and I'll come back to what that looks like at the end of this chapter.

11.3 Correlation Significance Tests

Here, we revisit our correlation between GDP per capita, and Happiness metrics, which I pulled from Our Word in Data:

```
##    Country           Happiness         GDPpc             Pop
##  Length:249         Min.   :2.400   Min.   :   731   Min.   :8.090e+02
##  Class :character   1st Qu.:4.670   1st Qu.:  4917   1st Qu.:4.153e+05
##  Mode  :character   Median :5.530   Median : 12655   Median :5.596e+06
##                     Mean   :5.492   Mean   : 20464   Mean   :5.918e+07
```

```
##                     3rd Qu.:6.260   3rd Qu.: 30100   3rd Qu.:2.421e+07
##                     Max.   :7.820   Max.   :112557   Max.   :4.663e+09
##                     NA's   :96      NA's   :52       NA's   :7
```

```
## # A tibble: 6 x 4
##   Country        Happiness GDPpc      Pop
##   <chr>              <dbl> <dbl>    <dbl>
## 1 Afghanistan         2.4   1971 38972236
## 2 Albania             5.2  13192  2866850
## 3 Algeria             5.12 10735 43451668
## 4 American Samoa     NA       NA    46216
## 5 Andorra            NA       NA    77723
## 6 Angola             NA     6110 33428490
```

Let's not worry about plotting the data, and go straight to the correlation:

```
##
##  Pearson's product-moment correlation
##
## data:  x and y
## t = 13.502, df = 146, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.6636288 0.8092331
## sample estimates:
##      cor
## 0.745184
```

Here are our results. The estimate is a correlation, and we test that using the **t statistic**. The *t-value* is simply the estimate divided by the standard error (which we can't see in this output), and is interpreted essentially as 'how far from 0 is the estimate, in standard errors'.

The *p*-value for t is very very small, and obviously less than 0.05.

Conclusion - reject null hypothesis, accept alternative hypothesis (as always, pending better evidence).

Importantly, this does not mean that the **true correlation in the population** is 0.745, simply that it is *very unlikely to be zero*.

We can then look at our estimate of 0.745, and - even better - our confidence interval (see Section 10.2 for information on how to interpret confidence intervals), to gain some indication of the likely true correlation in the population.

11.4 Regression Significance Tests

The process to asses the significance of regression estimates is very very similar to that for correlations. Let's revisit the heart disease data set we used earlier,

in Chapter 5.

```
##       ...1           biking          smoking        heart.disease
##  Min.   :  1.0   Min.   : 1.119   Min.   : 0.5259   Min.   : 0.5519
##  1st Qu.:125.2   1st Qu.:20.205   1st Qu.: 8.2798   1st Qu.: 6.5137
##  Median :249.5   Median :35.824   Median :15.8146   Median :10.3853
##  Mean   :249.5   Mean   :37.788   Mean   :15.4350   Mean   :10.1745
##  3rd Qu.:373.8   3rd Qu.:57.853   3rd Qu.:22.5689   3rd Qu.:13.7240
##  Max.   :498.0   Max.   :74.907   Max.   :29.9467   Max.   :20.4535
```

```
## # A tibble: 6 x 4
##    ...1 biking smoking heart.disease
##   <dbl>  <dbl>   <dbl>         <dbl>
## 1     1  30.8    10.9          11.8
## 2     2  65.1     2.22          2.85
## 3     3   1.96   17.6          17.2
## 4     4  44.8     2.80          6.82
## 5     5  69.4    16.0           4.06
## 6     6  54.4    29.3           9.55
```

Let's go straight to the multiple regression model.

```
##
## Call:
## lm(formula = heart.disease ~ biking + smoking, data = Heart)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1789 -0.4463  0.0362  0.4422  1.9331
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 14.984658   0.080137  186.99   <2e-16 ***
## biking      -0.200133   0.001366 -146.53   <2e-16 ***
## smoking      0.178334   0.003539   50.39   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.654 on 495 degrees of freedom
## Multiple R-squared:  0.9796, Adjusted R-squared:  0.9795
## F-statistic: 1.19e+04 on 2 and 495 DF,  p-value: < 2.2e-16
```

We interpret these just as we did the correlation significance tests.

The t-value is large, and the p-value (two-tailed) is small.

Interestingly, here we are given 'stars' for the different levels of significance, so to some extent the software is doing some decision making for you. To be honest, I always caution against relying solely on looking for 'stars' (it's actually a bit

of a running joke amongst the people who have known me a long time that I once told an entire class in the 1990s to 'just look for the stars'). That's because the actual significance or not decision is based on the critical value and one- or two-tailed decision. The software often makes an assumption of 0.05 critical value for p, two-tailed, and calculates the 'stars' based on that. Sometimes that can conflict with the decision you have made yourself about what should be significant or not. That can trip you up if you didn't know to change these values in the software package.

Further, it also sort of entrenches the idea that things can be 'more' or 'less' statistically significant. Take a look at the results above, you'll see 3 stars represents a significance of '0', and 2 stars represents '0.001', boring old '0.05' only gets a single star, and '0.1' gets a dot. I'm not a fan here because this encourages the analyst to post-hoc make decisions about 'marginally' significant, or 'very' significant. These concepts do not actually exist in real statistical theory. Instead, you decide your critical value ***a priori***, and you either pass or fail it. As Yoda may have said if he was a statistician, "Pass or do not pass. There is no marginally significant".

Am I perfect? No. Specifically, do any of my papers use the language of 'marginal' significance? Sure, I bet you could find them. I am never a fan though, and I can promise you I argued about it at the time!

11.5 Reading the literature

Hypothesis tests are the basic currency of quantitative management research. Almost every number that matters in a results section arrives with a test attached, and you've now done one of those tests by hand. That's the good news: the machinery in a published paper is the same machinery as in the salary example. The less good news is that papers use it in ways this chapter wouldn't recommend, and the conventions of reporting make those ways look completely normal. (What happens when a paper runs lots of tests at once is a problem of its own, which Chapter 13 takes on.)

11.5.1 Where you'll meet this

The coefficient table. Each row usually gives an estimate and a standard error, often in parentheses underneath, along with some combination of a t-value, a p-value and stars. The t-value is exactly the salary calculation: the estimate, minus what the null says it should be, divided by its standard error. In a regression table the null is almost always that the coefficient is zero, and it's almost never written down. That's worth noticing, because zero isn't always the comparison that matters. If a theory predicts that one effect is bigger than another, testing each of them against zero doesn't test the theory.

Stars and daggers. Everything I said earlier in this chapter about stars applies to papers, with one addition: the thresholds aren't standard, so always read the

note under the table. Some journals use a dagger (†) or a plus sign for $p < .10$, which quietly adds a fourth grade of significance. Some have gone the other way. In 2016 the editors of the *Strategic Management Journal* announced that they would no longer accept papers reporting cut-off levels of significance. They asked instead for standard errors or exact p-values without asterisks, and for an explicit discussion of effect sizes (Bettis et al., 2016). So a table with exact p-values and no stars isn't a less rigorous paper. It may well be a more rigorous one.

"Marginally significant" and its relatives. "Approached significance," "trended towards significance," "directional support." You know my view of these. The threshold was supposed to be chosen before the data arrived, and this language exists to rescue a result that failed it.

A difference that was never tested. This is one of the most common errors in published research, and one of the hardest to spot. A paper finds that an effect is significant in one group, say manufacturing firms, but not in another, say service firms, and concludes that the effect differs between them. That doesn't follow. A coefficient with $p = .03$ and one with $p = .20$ can be almost exactly the same size; one of them just has a bigger standard error. To claim that two effects differ, you need a test of the difference itself, usually an interaction term. The title of Gelman and Stern's paper on this says it all: the difference between "significant" and "not significant" is not itself statistically significant (Gelman & Stern, 2006).

"No effect." This is the other half of the same problem. As the Premier League example earlier in this chapter showed, a non-significant result means the data weren't surprising under the null. It doesn't mean the null is true. Yet papers routinely treat non-significance as proof of absence: the groups didn't differ, the control variable didn't matter, non-response bias wasn't a concern. Where a paper genuinely needs to show that something is absent or negligible, there are tests designed for exactly that. The most common are *equivalence tests*, which ask whether an effect is smaller than a pre-specified amount that would matter in practice (Lakens, 2017). You'll rarely see them in management journals, which is precisely why a "no difference" claim is worth questioning.

Very large samples. Many management papers use firm-level or employee-level databases with tens of thousands of observations. Standard errors shrink as samples grow, so in data that size even tiny effects come out as significant. The p-value then tells you very little, and the size of the estimate tells you nearly everything. The question from Section 2.6 applies with full force: what does this mean in natural units?

One-tailed tests. Occasionally a paper reports one-tailed tests. That's legitimate if, as in the salary example, the direction was committed to in advance. Be more wary when one-tailed tests turn up only for results whose two-tailed p-value would have fallen between .05 and .10. Halving the p-value is the entire effect of the switch.

Counts. You'll often meet outcomes that are counts: patents filed, acquisitions made, alliances formed. These are usually modelled with *Poisson regression* or its more flexible cousin, *negative binomial regression*. Poisson regression assumes the counts behave like the Premier League goals in this chapter, and in particular that their variance equals their mean. Business counts are usually more spread out than that (the term is *overdispersed*), which is why negative binomial models are so common: they relax that assumption. Their coefficients are often reported as *rate ratios*, and like every ratio in Section 2.6, a rate ratio needs a base rate before it means anything.

11.5.2 The connection back

Every p-value in every table is the salary calculation. There's an estimate, a null that usually goes unstated, a standard error, and a threshold that should have been chosen before anyone looked. Stars, daggers and adjectives are decoration on top of that. Strip them away, and the questions are always the same: compared with what, how big, and how sure?

11.5.3 Questions worth asking

1. **What's the null for this estimate?** Is zero really the comparison that matters for the theory?
2. **How big is the effect, in units someone would care about?** This matters most when the sample is large.
3. **When the authors claim two effects differ, did they test the difference?** Or did they compare one significant result with one non-significant result?
4. **When they claim something doesn't matter, what's the evidence?** Is it anything more than a non-significant p-value?
5. **Were the thresholds set in advance, and applied the same way throughout?** Check the table notes, and check whether any tests were one-tailed.
6. **If you covered up the stars, would you tell the same story?** Read only the estimates and standard errors, and see whether your conclusions change.

11.5.4 One red flag

Watch for a double standard. The same paper that calls p = .08 "marginal support" for a hypothesis will often call p = .08 on a comparison of groups "no significant difference." The second case usually arises when the authors want the groups to be the same, for example in a check that treatment and control groups were comparable, or that early and late respondents didn't differ. The threshold has moved to wherever it helps the argument. I doubt it's usually deliberate; it's what wanting a result does to anyone's judgement, mine included. But once you've spotted it, you know which way the whole paper leans.

11.5.5 If you want to read more

Two short papers this time. The American Statistical Association's statement on p-values (Wasserstein & Lazar, 2016) is the closest thing statistics has to an official position on what a p-value can and can't tell you. It's free to read online and only a few pages long. For something closer to home, the *Strategic Management Journal* editorial that dropped the stars (Bettis et al., 2016) makes the same case for management research. It includes a thought experiment about three junior scholars studying the same hypothesis that I suspect will feel uncomfortably familiar.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 12

ANOVA

This Chapter provides analysis and examples related to the second part of Lecture 8.

Let's pick up exactly where Chapter 8 left off. That chapter showed that people who did a boring proofreading task with Christmas music playing ended up more irritated than people who did the identical task in silence — but it couldn't tell us whether that was really about *Christmas* music, or just about having background music of any kind playing while you're bored. Silence was the only comparison available, so the design couldn't separate the two explanations.

Study 2 fixes that by adding a third group: people who did the same task with background music that wasn't Christmas music, but was matched to it on tempo and familiarity, so it isn't a softer comparison to beat just because it's blander. Now there are three groups — Silence, Other (music), and Christmas — which finally lets us ask the question Chapter 8 left hanging: is it Christmas music specifically that's the problem, or just having something playing in the background at all? Randomly assigning people to one of the three groups is exactly the confound-neutralising move from Chapter 4 — it's what lets this design pull the two explanations apart at all.

With three groups instead of two, running three separate t-tests would walk straight into the multiple comparisons problem (more on that in the next chapter). ANOVA instead tests whether there's a difference among all three groups with a single significance test, which is where we'll start.

```
##        ID           GROUP    IRRIT_IRRITATED IRRIT_ANNOYED    IRRIT_TENSE
##  Min.   : 1   Silence  :15   Min.   :1.000   Min.   :2.000   Min.   :2.000
##  1st Qu.:12   Other    :15   1st Qu.:3.000   1st Qu.:3.000   1st Qu.:3.000
##  Median :23   Christmas:15   Median :4.000   Median :4.000   Median :4.000
##  Mean   :23                  Mean   :4.133   Mean   :3.978   Mean   :4.111
##  3rd Qu.:34                  3rd Qu.:5.000   3rd Qu.:5.000   3rd Qu.:5.000
```

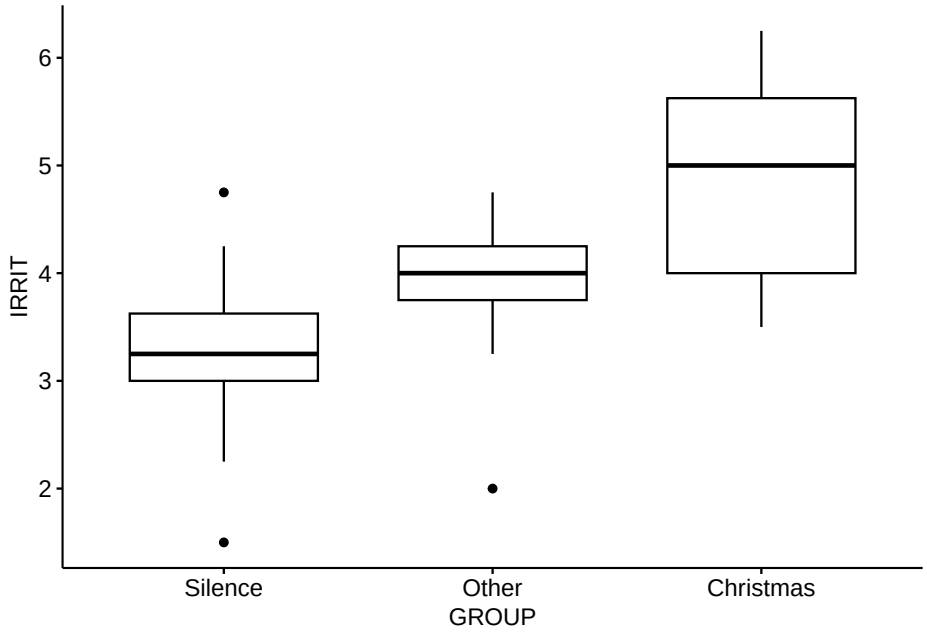

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```
##  Max.   :45                 Max.   :7.000   Max.   :7.000   Max.   :7.000
##   IRRIT_ONEDGE       IRRIT
##  Min.   :1.000   Min.   :1.500
##  1st Qu.:3.000   1st Qu.:3.500
##  Median :4.000   Median :4.000
##  Mean   :3.756   Mean   :3.994
##  3rd Qu.:5.000   3rd Qu.:4.750
##  Max.   :6.000   Max.   :6.250
```

```
## # A tibble: 6 x 7
##      ID GROUP   IRRIT_IRRITATED IRRIT_ANNOYED IRRIT_TENSE IRRIT_ONEDGE IRRIT
##   <dbl> <fct>             <dbl>         <dbl>       <dbl>        <dbl> <dbl>
## 1     1 Silence               3             4           2            5  3.5
## 2     2 Silence               3             3           4            3  3.25
## 3     3 Silence               3             3           3            3  3
## 4     4 Silence               5             4           4            4  4.25
## 5     5 Silence               4             2           4            3  3.25
## 6     6 Silence               3             2           3            3  2.75
```

Let's create a quick table of the group means, and visualise it with a simple boxplot of the groups.

```
## # A tibble: 3 x 5
##   GROUP     variable     n  mean    sd
##   <fct>     <fct>    <dbl> <dbl> <dbl>
## 1 Silence   IRRIT       15  3.28 0.807
## 2 Other     IRRIT       15  3.87 0.661
## 3 Christmas IRRIT       15  4.83 0.88
```



Interesting picture. Silence is our baseline, Other is the matched background music without any Christmas content, and Christmas is, well, Christmas music. This is what we might call 'model-free' analysis. We are just looking at the data to see if intuitively it looks like we expect it to.

Let's run an ANOVA now, which places a statistical model on it, and tests whether or not the data supports the model.

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd      F        p p<.05   ges
## 1  GROUP   2  42 14.822 1.35e-05     * 0.414
```

Results here suggest there is a significant effect (p-value is very small). We also have an *effect size* measure ('ges', or generalised eta-squared) which is very useful to us, and suggests the effect is quite large. This can be interpreted in a similar way to R^2 in a regression: it's the proportion of the variance in the dependent variable (irritation) that is explained by group membership.

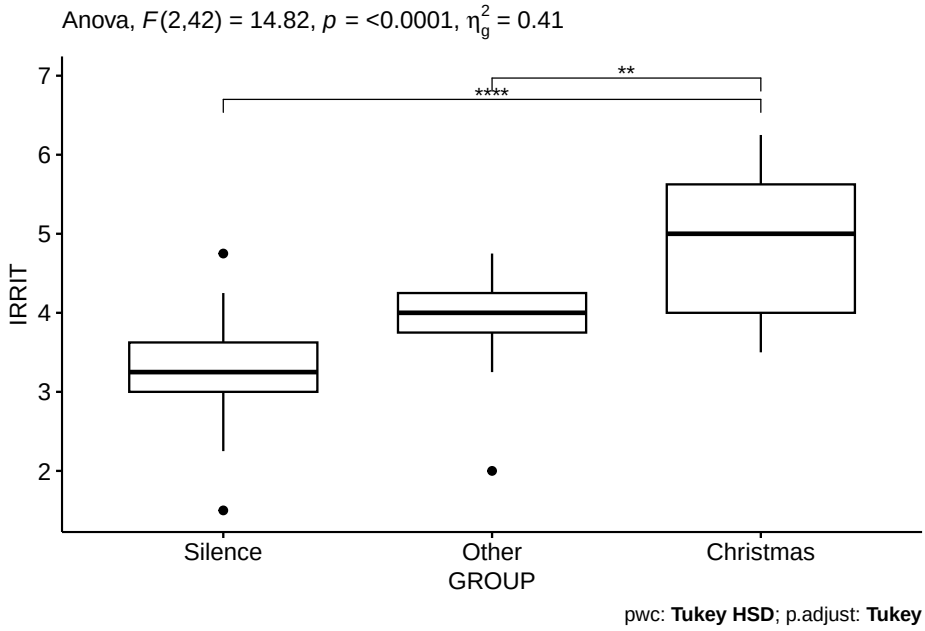
However, ANOVA only tests the *general effect* of the treatment / group. We don't know whether this is because of the difference between **all of the groups**, or only **some**. If it's background music *in general* that's irritating, we'd expect Silence to differ from both Other and Christmas, but Other and Christmas to look much the same as each other. If it's *Christmas music specifically*, we'd expect the opposite pattern: Christmas differs from both Silence and Other, but Silence and Other look much the same as each other. Only a post-hoc comparison of every pair can tell us which of these it actually is.

12.1 Post-Hoc Testing

We can investigate this using **post-hoc tests**, which compare the difference between each pair of groups. There are many different types of post-hoc test for ANOVA, but the most typical one to use is the Tukey's test, which is what we will do here.

```
## # A tibble: 3 x 9
##   term  group1  group2    null.value estimate conf.low conf.high      p.adj
## * <chr> <chr>   <chr>          <dbl>    <dbl>    <dbl>     <dbl>      <dbl>
## 1 GROUP Silence Other              0    0.583   -0.115      1.28 0.118
## 2 GROUP Silence Christmas          0    1.55     0.851      2.25 0.00000878
## 3 GROUP Other   Christmas          0    0.967    0.268      1.67 0.00464
## # i 1 more variable: p.adj.signif <chr>
```

We can actually plot these results in a really effective way:



This is the *Christmas-specifically* pattern from a moment ago: Christmas differs clearly from both Silence and Other, and Silence and Other don't differ significantly from each other (p = .118). Taken at face value, that looks like a clean answer — it really is something about Christmas music, not background music in general.

But look at the Silence-vs-Other comparison a bit more carefully, because this is exactly the case that a bare significant/not-significant reading undersells. Its 95% confidence interval is -0.12 to 1.28. That interval *only just* crosses zero — the lower bound is a hair's breadth into negative territory. Reading "not

significant" as "proven: no difference" throws away exactly the information the interval is offering, which is that the data are entirely compatible with a real, positive difference too; there just isn't enough precision at n = 15 per group to rule zero out. A p-value collapses that whole interval down to one bit of information, pass or fail, and right at a boundary this close, that's precisely where the collapse costs you the most. I'd want a bigger sample before I'd rule out *any* effect of background music in general — this result argues for "Christmas specifically, at least as far as we can tell from here," not "definitely, only Christmas, case closed."

However, this type of post-hoc analysis has the potential for what is known as a *multiple comparisons problem*, which we might need to deal with.

Let's take a look **back in the slide deck...**

Remember, there are many other things that if we were doing ANOVA that we would also look to deal with - such as the various assumptions required of ANOVA, and so forth. But, they are beyond our scope in this class. Suffice to say that this has only scratched the surface of ANOVA so far.

12.2 Reading the literature

Experiments are common in some areas of management research, particularly in marketing, organisational behaviour and behavioural strategy, and ANOVA is how most of them are analysed. The simplest ones look just like the music study in this chapter: a few groups, one outcome, an F-test and some pairwise comparisons. But most published experiments go a step further and cross two manipulations in a *factorial design*, because the hypothesis is usually not "X affects Y" but "X affects Y, but only when Z." That's an *interaction*, and interactions are where most of the reading difficulty lies, whether they turn up in an ANOVA or a regression.

12.2.1 Where you'll meet this

The F-test. Results are usually reported as something like $F(2, 42) = 15.31$, $p < .001$. The first number in brackets is the degrees of freedom for the groups (the number of groups minus one) and the second is for the error, which is roughly the total sample size minus the number of groups. So you can work backwards from the brackets to the sample size, which is worth doing when a paper is vague about how many people were in each condition. As in this chapter, a significant F only tells you that the groups aren't all the same. It doesn't tell you which ones differ, which is what the follow-up comparisons are for.

Eta-squared and its relatives. The effect size in this chapter was generalised eta-squared, the proportion of the variation in the outcome explained by group membership. With one factor, that's the same idea as R^2 in a regression. Papers also report plain eta-squared (η^2) and, very often, *partial* eta-squared (η^2p). In

a one-factor design they're all the same number. In a factorial design they're not. Partial eta-squared leaves the other factors' effects out of the denominator, so it's usually larger, and partial eta-squareds from different designs can't be compared directly with each other (Lakens, 2013). You'll also see .01, .06 and .14 described as small, medium and large, and those labels deserve the same scepticism as Cohen's labels for d in Chapter 8.

Factorial designs and interactions. In a 2×2 design, say a price promotion (yes or no) crossed with brand familiarity (known or unknown brand), the analysis gives two *main effects* and one *interaction*. The main effect of the promotion is its average effect across both kinds of brand. The interaction asks whether the promotion's effect is different for known and unknown brands. When there's a real interaction, the main effects can be badly misleading: a promotion that helps unknown brands and does nothing for known ones has a positive "main effect" that describes neither. So when a paper reports an interaction, the main effects deserve much less attention than they usually get, and the *simple effects* (the effect of one factor at each level of the other) deserve much more. It also matters what shape the interaction takes. An effect that reverses direction is a very different finding from one that merely shrinks, and shrinking interactions in particular need far larger samples to detect reliably than most experiments have.

Interactions in regression. The same idea appears in survey papers as *moderation*: a regression with X, Z and their product $X \times Z$. Two things commonly go wrong in reading these. First, once the product term is in the model, the coefficients on X and Z aren't main effects any more. The coefficient on X is the effect of X when Z equals zero, which may be a meaningless value, such as a firm with zero employees. That's one reason papers *centre* their variables (subtract the mean), which makes zero mean "average." Second, papers often claim that centring was done to reduce multicollinearity. It doesn't: centring changes how the lower-order coefficients are interpreted, but it leaves the precision of the estimates and the test of the interaction exactly as they were (Echambadi & Hess, 2007). Look for a plot or table of *simple slopes*, which shows the effect of X at a few chosen values of Z. Without one, it's very hard to tell what an interaction actually means.

12.2.2 The connection back

An ANOVA is a regression with the groups coded as dummy variables, just as the t-test in Chapter 8 was. The F-test asks whether the group means are further apart than the variation within the groups would lead you to expect, and eta-squared is R^2 by another name. What the factorial design adds is the question of whether one effect depends on another. That's usually the most interesting question in a paper, and the easiest one to misread.

12.2.3 Questions worth asking

1. **Which effect size is reported?** Is it being compared with others calculated the same way?
2. **Is there an interaction?** If so, are the main effects being interpreted as if there weren't?
3. **What shape is the interaction?** Does an effect reverse, or just get smaller? Is there a plot of the simple effects or simple slopes?
4. **In a regression with a product term, what does zero mean for each variable?** What, therefore, does each coefficient actually describe?
5. **How many people were in each cell?** You can work it out from the degrees of freedom if the paper doesn't say.

12.2.4 One red flag

A subtle interaction from a small experiment. A typical example is a 2×2 design with twenty or thirty people per cell, reporting that an effect is present under one condition and weaker under another. That's precisely the kind of finding that needs a large sample, and precisely the kind of design that doesn't have one. Put it next to what Chapter 13 says about power and about how many ways there are to analyse an experiment, and ask how much weight the finding can really bear. Then look at whether anyone has replicated it.

12.2.5 If you want to read more

Julie Irwin and Gary McClelland's "Misleading heuristics and moderated multiple regression models" (Irwin & McClelland, 2001) is written for marketing researchers and goes through the most common mistakes in reading and reporting interactions, including the ones above. It's a little more technical than most of the readings I've suggested, but it's short, and it's the paper I'd give anyone about to test a moderation hypothesis.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Chapter 13

Issues with Significance Testing

This chapter provides examples and analysis for the third part of Lecture 8, and also parts of Lecture 9.

In this chapter, I'll explore examples of two common 'issues' or 'debates' regarding statistical significance testing, applied to two of our prior examples.

But first, let's unpack more about the Christmas music study, and the multiple comparisons problem.

13.1 More on the Multiple Comparisons Problem

In this section, I'll provide some visualisations which can help us to understand the problem with multiple comparisons.

However, it's worth noting that the results depend on a number of assumptions about the research and what the real world actually looks like. The latter for sure we can never know. So, do bear that in mind.

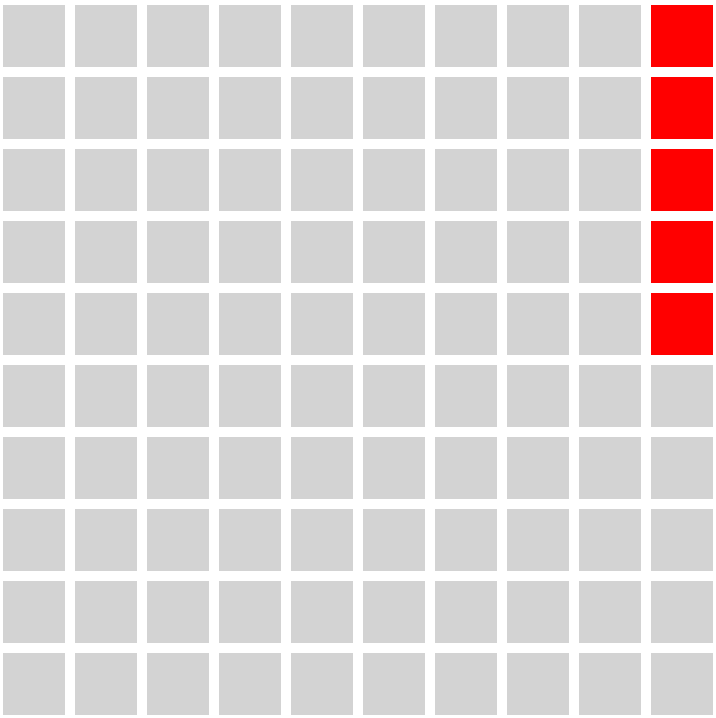
First, let's visualise the situation I just discussed in the slides. We test 100 hypotheses, and in every case, the null is true in the population (remember, we cannot know the true state of the population in reality). Below, each grey box represents a hypothesis we test. These grid figures are adapted from an answer on Cross Validated: https://stats.stackexchange.com/q/143424.

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Now, remember, **even if all null hypotheses are true**, our significance level, or $p = 0.05$, so we have to expect that *on average* we would expect **5** of these hypotheses tests to return p-values < 0.05, and thus we would reject the null. This is known as a **false positive**.

You might wonder how we could reduce the chances of false positive to zero. You could do this by setting your required p to be exactly zero. Do you see the problem here? You would certainly solve the false positive problem. But, you would also accept *all* null hypotheses. You would never actually find a statistically significant result in your research. Unless the null hypothesis really *was* true in the population in every single case, you would be inevitably missing out on making some discoveries.

Moving on, let's visualise the situation in hand here. Below, the same 100 hypothesis tests (remember, the null is true in all cases), but this time I have marked in red the expected level of false positives.



Now, that is just the *expected* level for every 100 tests. If you remember from the discussion in Section 10.2, it may not be you get *exactly* 5 false positives in 100 tests, just that you would expect the long run average number of false positives to be 5%, in an all-null world.

So, we can look at it through a different lens. What is the probability that you would get **at least one false positive** in any given batch of multiple tests.

That is known as the *familywise error rate*, and it applies whenever you can define a cluster or 'family' of tests.

It's the same principle as rolling a 6-sided die, and needing to score a '6'. Think of the '6' as the 'false positive' here.

In a single die role, the probability of getting a '6' is $\frac{1}{6}$, and of course the equivalent probability of *not* getting a '6' is $1 - \frac{1}{6} = \frac{5}{6}$.

But, what if you get to roll the die 3 times? It's fairly easy to intuit that, even though the probability of getting a '6' on any single roll stays the same, the probability of getting a single '6' in 3 rolls is higher. In fact, we can calculate it.

To do so, we use the probability of *not* getting a 6, as follows:

$P(\text{no } 6) = (\frac{5}{6})^3 = 0.579$

Recalling the laws of probability then…

P(at least one 6) = 1 - P(no 6) = 1 - .579 = .421

So, *now* we have a 42% probability of at least one 6 in 3 rolls, even though the probability of getting a 6 in any single roll is unchanged.

We can calculate the familywise error rate for any set of multiple comparisons in exactly the same way.

One of the key advantages of ANOVA is that it is a single test, and thus does not fall prey to the multiple comparisons problem. However, in the case of *post-hoc* testing for the Christmas music study (Study 2), we are doing 3 pairwise comparisons (i.e. 3 tests) across the 3 groups, with a stated significance level of 0.05.

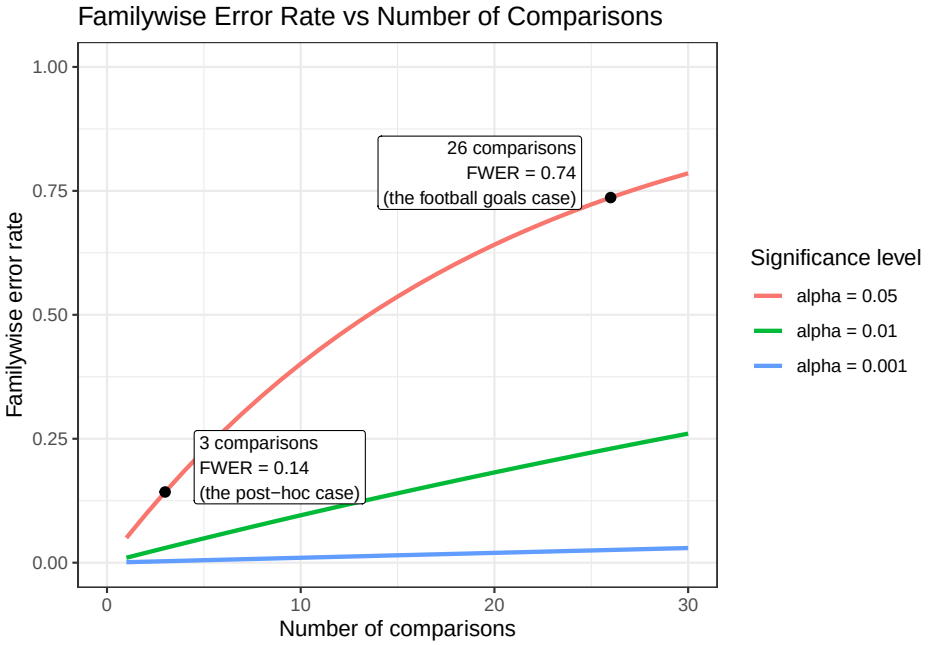
For 1 test, the false positive rate is $1 - 0.95 = .05$

But, for 3 tests, the false positive rate is $1 - 0.95^3 = 0.14$

We are 'rolling the dice' multiple times.

In fact, the familywise error rate is easy to see for yourself, once you know the formula: it's simply $1-(1-\alpha)^n$, where α is your chosen significance level and n is the number of tests you're running. The figure below plots that out for a few different significance levels, so you can see how quickly the error rate climbs as the number of comparisons grows.

I first saw this demonstrated properly using a lovely interactive calculator built by Dr Daniel Roelfs, during his time at the Norwegian Centre for Mental Disorders Research (NORMENT) in Oslo, and he kindly gave permission for me to reproduce the idea here. Check out his website: https://danielroelfs.com/about/. A book page can't rely on an external app staying up forever for every reader, so the version below reproduces the same idea as a static figure instead.



Watching that curve climb feels kind of scary, right? There are many ways to deal with the problem, but they pretty much all amount to making some correction to the required significance level, making it more stringent, in order to reduce the false positives.

For example, in the Christmas-music post-hoc tests in Chapter 12, we used Tukey's test, which uses a correction for multiple comparisons. Another method dealt with below is the *Bonferroni Correction*, which is generally the most well-known of them.

13.2 The Bonferroni Correction: Rate of Change in Football Goals per Season

Here, I will expand on the football goals rate-of-change example from Chapter 10, to incorporate corrections for the multiple comparisons problem. Again, I'm using data from footballhistory.org (n.d.).

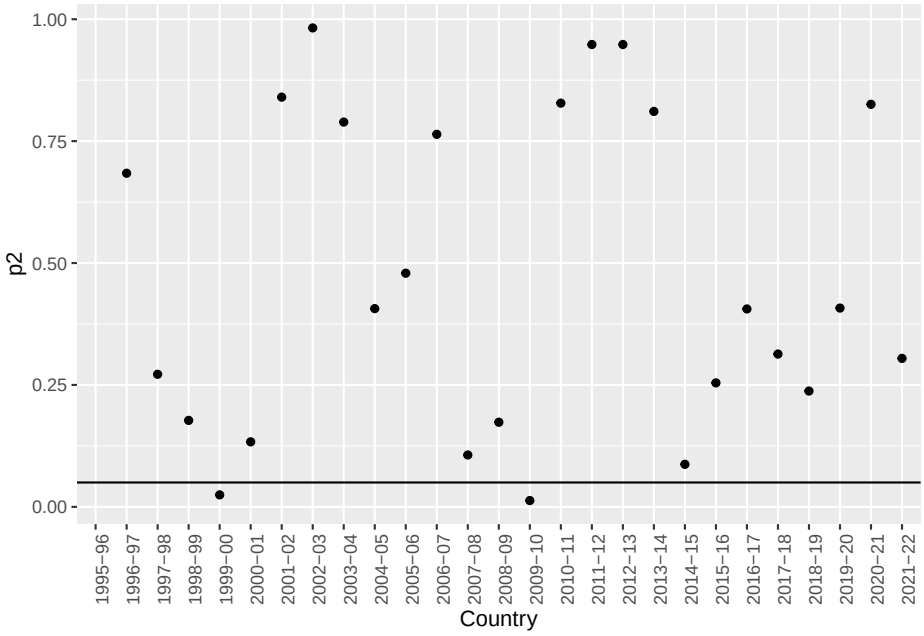
I'm building on the z-test table from the end of Chapter 10. But here, using the z-scores in the data file, I have calculated the p-value (2 tailed as we do not hypothesize a direction for the difference) for the z-scores for the difference between each season, year-on-year.

```
##     Season              Games          Goals              GPG
##  Length:27          Min.   :380   Min.   : 931.0   Min.   :2.450
##  Class :character   1st Qu.:380   1st Qu.: 981.5   1st Qu.:2.583
```

```
##  Mode  :character   Median :380   Median :1018.0   Median :2.679
##                     Mean   :380   Mean   :1013.9   Mean   :2.668
##                     3rd Qu.:380   3rd Qu.:1056.5   3rd Qu.:2.780
##                     Max.   :380   Max.   :1072.0   Max.   :2.821
##
##        SE          Lower95CI        Upper95CI          Change
##  Min.   :30.51   Min.   : 871.2   Min.   : 990.8   Min.   :-77.000
##  1st Qu.:31.33   1st Qu.: 920.1   1st Qu.:1042.9   1st Qu.:-35.500
##  Median :31.91   Median : 955.5   Median :1080.5   Median : -2.000
##  Mean   :31.84   Mean   : 951.5   Mean   :1076.3   Mean   :  3.192
##  3rd Qu.:32.50   3rd Qu.: 992.8   3rd Qu.:1120.2   3rd Qu.: 44.750
##  Max.   :32.74   Max.   :1007.8   Max.   :1136.2   Max.   :111.000
##                                                    NA's   :1
##        Z                negged               p2
##  Min.   :-1.71027   Min.   :-2.48514   Min.   :0.01295
##  1st Qu.:-0.79795   1st Qu.:-1.30711   1st Qu.:0.19236
##  Median :-0.04369   Median :-0.83063   Median :0.40618
##  Mean   : 0.07032   Mean   :-0.88179   Mean   :0.46986
##  3rd Qu.: 0.97794   3rd Qu.:-0.24627   3rd Qu.:0.80549
##  Max.   : 2.48514   Max.   :-0.02236   Max.   :0.98216
##  NA's   :1          NA's   :1          NA's   :1
```

```
## # A tibble: 6 x 11
##   Season  Games Goals   GPG    SE Lower95CI Upper95CI Change      Z negged
##   <chr>   <dbl> <dbl> <dbl> <dbl>     <dbl>     <dbl>  <dbl>  <dbl>  <dbl>
## 1 1995-96   380   988  2.6   31.4      926.      1050.     NA NA     NA
## 2 1996-97   380   970  2.55  31.1      909.      1031.    -18 -0.407 -0.407
## 3 1997-98   380  1019  2.68  31.9      956.      1082.     49  1.10  -1.10
## 4 1998-99   380   959  2.52  31.0      898.      1020.    -60 -1.35  -1.35
## 5 1999-00   380  1060  2.79  32.6      996.      1124.    101  2.25  -2.25
## 6 2000-01   380   992  2.61  31.5      930.      1054.    -68 -1.50  -1.50
## # i 1 more variable: p2 <dbl>
```

Here, we can plot the p-values (2-tailed), and again we see that two seasons seem to have significant differences. In other words, the p-values are less than 0.05 for the test as to whether the number of goals scored differs from the season before



We can see that the 1999-2000 season, and the 2009-10 seasons have p values less than 0.05

The question is **are we suffering from the multiple comparisons problem**? Should we correct for it?

It's hard to say actually. Of course, we are indeed running multiple tests, 26 in fact. So, the chance of a false positive is high. The Bonferroni correction would immediately reduce the false positive chances, but at what cost?

Let's see how this would work. In order to calculate a Bonferroni correction, you can either adjust the p-value directly that you calculate for each test, or instead simply adjust the 'cut-off' value for p, known as the *critical p*, to lower it from 0.05 and make the significance test 'harder' to pass. The formula to adjust the cut-off value is simply:

$\alpha_b = (\frac{\alpha}{n})$

Where

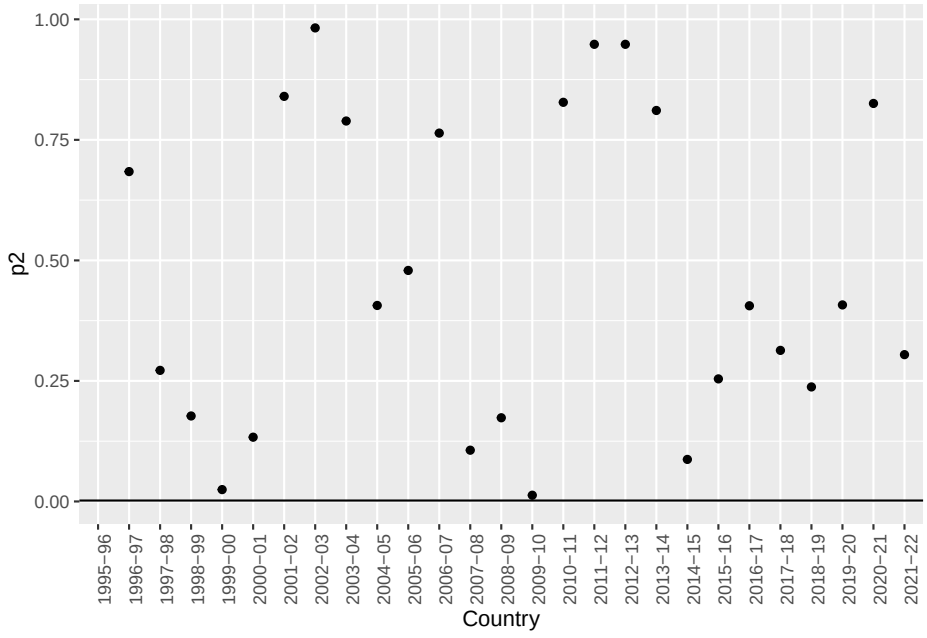
α_b = Bonferroni-adjusted critical p value

α = original critical p value (here this is 0.05)

n = number of comparisons (here this is 26)

So, the formula gives us a new Bonferroni-adjusted critical p value of 0.0019

Let's see what happens with this new critical p value:



We can see that none of our tests now rejects the null. The Bonferroni correction is known as a highly conservative test. That is, it is based on the idea that the *null hypothesis is true* in each case in the population. We can thus consider it as the most stringent and conservative way to correct for the chance of false positives when doing multiple comparisons.

But, as I suggested above, that might not always be the best idea. Indeed, if you want to totally avoid any chance of a false positive, why not simply make the required alpha 0? Then, you would never get a false positive. Of course, you would never detect a **true positive** either.

What if it is the alternative hypothesis (that is, the H of an effect existing) that is true in all cases? In such cases, there can of course be *no false positives*. Therefore, in such a situation you would be *increasing the chances of a false negative* by reducing the chances of a false positive. So, what are the potential costs of each of these mistakes?

For example, Thomas Perneger's (1998) paper in the BMJ is scathing about the Bonferroni adjustment.

Mind you, I am not saying that's the final word, just that there are multiple perspectives on the issues!

It's never as simple as it seems when making statistical decisions, is it?

13.3 Statistical Power: Brief Demonstration

Different types of analysis and research design require different types of power calculation, so it is hard to give a uniform example. But, for simplicity's sake, let's calculate the required sample size for the Christmas music study (Study 2) we conducted earlier, in Chapter 12.

Remember, really, we should have done this **before collecting data**.

To calculate power, all we need the parameters of the experiment and analysis design. As such, it is easily possible to do this before collecting data, and to design your studies around it. Really, we should do this a lot more in business and management - it's routine in fields like medicine.

So, we had 3 groups, and used ANOVA

Let's set a significance of 0.05, a required power of 0.8, and assume the effect size is moderate (say 0.25)

```
##
##      Balanced one-way analysis of variance power calculation
##
##               k = 3
##               n = 52.3966
##               f = 0.25
##       sig.level = 0.05
##           power = 0.8
##
## NOTE: n is number in each group
```

So, we really wanted to have around 50 in each group to have an 80% chance of detecting a moderate effect presuming the null was true.

You can see that my study (with only 15 in each group) was rather underpowered — assuming a moderate effect. But now here's a slightly awkward wrinkle. If instead I take the effect size *this actual experiment* produced — a generalised eta-squared of 0.41 back in Chapter 12, which converts to a Cohen's f of about 0.84 — and plug that into the same calculation:

```
##
##      Balanced one-way analysis of variance power calculation
##
##               k = 3
##               n = 5.679485
##               f = 0.84
##       sig.level = 0.05
##           power = 0.8
##
## NOTE: n is number in each group
```

That says I'd only have needed about 6 people per group for 80% power. That's *fewer* than the 15 I actually used, not closer to it from above — a different, and slightly uncomfortable, lesson from the one I set out to teach.

Here's the honest reading. What I've just done — taking the effect size my own small sample happened to produce, and treating it as the 'true' population effect size for a power calculation — is a well-known bad practice, sometimes called *observed* or *post-hoc* power. Small samples give you very noisy estimates of the true effect size: a difference that looks huge in a sample of 15 per group could easily be a real, more modest effect combined with a slice of good luck in exactly this data. So this isn't really evidence that a study of 15 would reliably detect an effect this size in general — it's evidence that *this specific sample* turned up a strikingly large difference. The honest way to plan a study is to commit to a plausible effect size **before** you collect any data (which is what the moderate f = .25 assumption above was trying to do, arbitrary as it was), not to reverse-engineer one from a result you already have. If anything, this is itself a small demonstration of why "just check whether the effect size you found was big enough" isn't a fix for a badly-powered study after the fact — the number you'd get depends entirely on which random sample you happened to draw, small samples especially.

13.4 How to find an effect that isn't there

"p-hacking" is one of those terms that everyone has heard and nobody quite agrees on. If you walked around a business school asking what it means, you'd get a surprisingly wide range of answers. At one end it means deliberate cheating: running analyses until something works, and reporting only that. I think it's important to separate that from something much more common and much more interesting, which is researchers who are doing their honest best and still end up somewhere they shouldn't. This section is mostly about the second kind, because it's the kind you're most at risk of yourself.

Let's build it with a simulation. We'll create a world in which the null hypothesis is *true*: two groups, A and B, that are genuinely no different on anything we measure. Then we'll run 2,000 separate studies in that world.

It's worth being very clear about what that means, because the whole logic depends on it. These are not 2,000 tests run on one dataset. They are 2,000 completely independent studies, each with its own freshly collected data. Think of it as 2,000 different researchers around the world, each setting out to study the same question without knowing about any of the others. Each study has 20 people per group to start with, two outcome measures that are moderately related to each other (the way two measures of similar things usually are), and a yes/no background variable about each person.

Let's start with researchers who behave exactly as Chapter 11 says they should. Each of them decides in advance to test outcome 1 with a single t-test, runs

their study, and does exactly that: one study, one test. Because there is no real effect, any significant result is a false positive, and each researcher has a 5 percent chance of getting one. Across all 2,000 studies, that's what we see: 5 percent of them "find" an effect. This is what $p < .05$ actually promises. It can't tell any individual researcher whether they are one of the unlucky ones. What it tells you is how often the procedure goes wrong across many independent studies like theirs.

Now let's give every researcher some flexibility, one choice at a time. Each of them still runs just one study, with one dataset. What changes is how many ways they're allowed to analyse it. None of these choices is outrageous. Each one is the kind of thing you'll find in published papers, often without comment.

- **Two outcomes.** They measured two things, so why not check both, and their average? If any of the three is significant, that's the finding.
- **A few more people.** If the result is "nearly there," it seems reasonable to collect another 10 per group and look again.
- **A control variable.** Perhaps that background variable should be controlled for. Try it with and without.
- **Outliers.** Some values look a bit extreme. Try the analysis with and without anyone more than two standard deviations from their group's mean.

After each new choice is added, we count how many of the 2,000 researchers can now find *some* path through their own data to $p < .05$. Figure 13.1 shows what happens.

By the time all four choices are available, 32 percent of researchers studying an effect that doesn't exist end up with a "significant" result. That's roughly one study in three, from a procedure that is supposed to be wrong one time in 20.

I want to draw three lessons from this.

First, no single step looks like cheating. Each choice, taken alone, is defensible, and some of them appear in textbooks as perfectly good practice. That's precisely the problem. The rate climbs because the choices *combine*: each new fork in the road gives the data another chance to throw up something by luck. Notice, too, that these extra analyses are not new studies. Our 2,000 researchers are independent of one another, but each researcher's forks are all different looks at the *same* data. That's part of what makes them so easy to walk down: each one feels like the same analysis done slightly differently, not like a fresh roll of the dice. But every fork is another chance for luck to produce $p < .05$, and those chances add up.

Second, the forks aren't equal. Adding a control variable barely moved the rate here, while outlier removal roughly doubled it. Outliers are a genuinely hard case. I grew up in a survey tradition where variables ran from 1 to 5, so if you found a 55 it was a typo and you removed it without a second thought. But deciding what counts as an outlier *after* you've seen whether it helps your

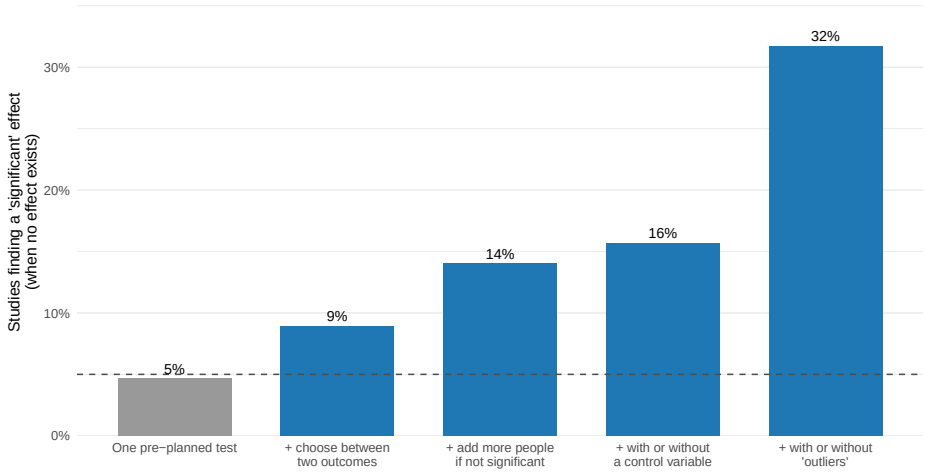


Figure 13.1: How researcher flexibility inflates false positives. Each bar shows the percentage of independent simulated studies, all of an effect that does not exist, in which the researcher could find at least one analysis with $p < .05$. Each bar adds one more analytic choice to those before it. The dashed line marks the 5 percent that $p < .05$ is supposed to deliver.

result is a very different thing. The honest question is always how the value came to exist, not whether the analysis looks better without it.

Third, and most important, you can't see any of this from outside. The published paper from our last researcher would report one clean t-test with $p = .03$, and it would look exactly like a paper from one of our honest single-test researchers. The p-value itself isn't wrong. The arithmetic is correct. But it answers the question "how surprising would this result be if it were the only test I ran?", and it wasn't the only test.

This idea goes further than you might think. The statistician Andrew Gelman calls it the "garden of forking paths," and it's probably my favourite idea in this whole area (Gelman & Loken, 2014). His point is that a researcher doesn't need to *try* lots of analyses for the problem to arise. It's enough that they *would* have made different choices if the data had come out differently. Looking back from the end, every choice seems like the only logical one: this measure, this method, these outliers removed. But someone else starting from the same place might have walked a different path and found something else.

Our simulation is also on the gentle side. The paper that made this problem famous used a slightly different set of four choices and found that combining them pushed the false-positive rate to around 61 percent (Simmons et al., 2011). Real studies usually offer far more than four forks.

The most effective remedy is to decide the path in advance and write it down.

That's what pre-registration does, and I'll say more about how to read pre-registered work below.

13.5 What happens next: the power pose story

The simulation shows how a result can assemble itself out of noise. What it can't show is what happens to that result once it leaves the lab. For that, I want to tell you a real story. I've taught it for years, and I feel rather differently about it now than when I started.

13.5.1 The finding

In 2010, Dana Carney, Amy Cuddy and Andy Yap published a study in *Psychological Science*, one of psychology's top journals (Carney et al., 2010). Forty-two people were randomly assigned to hold either two expansive, "high-power" poses, such as leaning back with your hands behind your head, or two contractive, "low-power" poses, for one minute each. Compared with the low-power group, the high-power posers showed higher testosterone and lower cortisol, took more risks in a gambling task, and said they felt more powerful. Two minutes of posture, in other words, apparently changed your hormones and your behaviour.

You can see why that caught on, especially in business schools. If standing like a superhero before a job interview or a big presentation made you more confident and more willing to take risks, every manager, and every MBA student, would want to know about it. Cuddy's 2012 TED talk on the research became one of the most-watched TED talks ever. I actually noticed the craze before I knew where it came from. At the 2015 Conservative Party conference, a line of senior politicians were photographed in the same odd, wide-legged stance, and I remember thinking that nobody stands like that unless someone has told them to.

13.5.2 Why it was always going to be checked

Now look at the study through the eyes of someone who has just read the rest of this chapter. Forty-two people split across two conditions is almost exactly the size of the studies in our simulation. The claim was surprising: that a minute or two of posture changes your hormones. It measured several different outcomes. And it appeared in a top journal and became spectacularly famous.

If you set out to design the study most likely to be a false positive, and then the study most likely to be checked, you'd struggle to beat it. Fame matters here. Scrutiny follows something like a power law: the more attention a finding gets, the more people go looking at how it was produced. Power posing became famous at exactly the moment researchers started doing that looking.

13.5.3 What happened next

In 2015, a team led by Eva Ranehill ran the study again with 200 participants, nearly five times the original sample. Their experimenter didn't know which condition each participant was in (Ranehill et al., 2015). People who held the high-power poses still said they felt more powerful. But there was no effect on testosterone, on cortisol, or on any of three behavioural tasks. The original authors replied that the new study differed in important ways, such as holding the poses for longer. That's a fair point to make about any replication that isn't exact.

Then, in 2016, something almost unheard of happened. Dana Carney, the paper's lead author, posted a statement on her website saying she no longer believed power pose effects were real (Carney, 2016). She was clear that the data were real, and nobody has suggested otherwise. But she set out, point by point, how the analysis had been done. If you've just read the simulation, the list will look very familiar:

- Participants were run in batches, and the effect was checked along the way before deciding whether to run more. At the time, she wrote, this felt less like p-hacking than like saving money.
- A few participants were excluded, and an outlier was removed from the hormone analyses but not from the others.
- Several questions about feeling powerful were asked, and the ones that "worked" were reported.
- For risk-taking, two versions of the same test gave p = .052 and p = .05, and the smaller one was reported.

Those are our simulation's "a few more people", "outliers" and "two outcomes" forks, plus one we didn't simulate: choosing between two tests of the same thing after seeing both results. None of them would have looked like cheating at the time. She also noted that the experimenters knew the hypothesis. And she pointed out that people who won the gambling task received extra money, while winning itself is known to raise testosterone, so the hormone result may have had nothing to do with posture.

The argument didn't end there. A p-curve analysis of 33 power posing studies, a method I'll describe in the reading section below, found no evidence of real effects on outcomes beyond feelings of power (Simmons & Simonsohn, 2017). Cuddy and colleagues replied with a larger set of studies that included those feelings, and reached the opposite conclusion (Cuddy et al., 2018). In 2017 a journal devoted a special issue to pre-registered replications. Taken together, they found essentially no effect on behaviour or hormones, but a consistent effect on how powerful people *say* they feel (Jonas et al., 2017). That's roughly where things stand: posture may well change how powerful you feel, but the original claims about hormones and behaviour haven't survived.

13.5.4 Why I tell this story differently now

When I first taught this case, I thought it was funny. Over the years I've come to feel increasingly sorry for Amy Cuddy in particular. She became the public face of the finding and took the brunt of what followed, and some of it was far from civil. Yet nothing in the original study was unusual for its time. Running participants until the result looks good, trying a few measures, picking the more favourable test: these were ordinary practice, and hardly anyone thought of them as problems. I'm quite sure that in their position, at that time, I'd have done the same.

What should have happened is simple. A surprising result from 42 people should have been a reason to run the study again, bigger, before telling the world. But almost nobody did that in 2010. Journals didn't ask for it, reviewers didn't ask for it, and every incentive pointed the other way. Would a top journal publish that study today without asking for a replication first? Probably not, and that's a fair measure of how much has changed.

Power posing wasn't an isolated case. In 2015 a large collaboration tried to repeat 100 published psychology studies, and only about a third produced a significant result the second time (Open Science Collaboration, 2015). Power posing is simply the clearest example of the pattern, and the one with the most dramatic arc.

13.5.5 What this means when you read

The lesson for you as a reader isn't that psychology is rubbish, or that small studies are worthless. It's that a single surprising result from a small sample is an interesting hypothesis, not a finding, however good the journal.

There's also a simple way to see how fragile such a result is. Remember the confidence-interval simulation in Section 10.2, where around 5 in every 100 intervals missed the true value purely by chance? (One of the sample sizes there was 42, as it happens.) A lucky small-sample study is one of those. Its interval sits clear of zero, but only just, and it's wide enough to include effects too small to matter to anyone. A p-value just under .05 hides all of that. The interval shows it.

13.5.6 The forks nobody sees

There's one more part of this I want to raise, because I think it's the next frontier and almost nobody talks about it. Pre-registration protects you from the forking paths you might walk yourself. It does very little about the ones reviewers send you down.

It's common to receive a revise-and-resubmit asking you to rethink your theory, reanalyse your data, or test something new. You then face a real dilemma. You can report the new analysis honestly, as something added at a reviewer's

request. Or you can rewrite the paper as if you'd planned it all along. You know you shouldn't, but it often produces a cleaner paper that's more likely to be accepted. Reviewers also frequently tell you what they expect you to find, and people who know what they're supposed to find are more likely to find it, even without meaning to. The result is that many clean-looking published papers may be some way away from the papers that were originally submitted, and a reader has no way of knowing. One study tracked 142 management PhD theses into print, and found the ratio of supported to unsupported hypotheses more than doubled along the way (O'Boyle et al., 2017). How much of that was reviewers rather than authors, nobody can say, which is rather the point.

Some journals are trying to fix this. When we had a paper under review at *The Leadership Quarterly*, we were offered the chance to submit a plan for the revision, with publication guaranteed whatever the results. We took it. We still got rejected. But I'd love to see that become normal.

13.6 Reading the literature

The simulation earlier in this chapter showed that you can't see the forking paths from outside. As a reader, that's the frustrating part. You're handed the one path the authors walked, and it looks like the only one there was. But the paths leave traces, and once you know what to look for, you'll start noticing them.

13.6.1 Where you'll meet this

Lots of tests in one place. A typical results table in a management paper might test a dozen coefficients across four models. Count the tests. If there are 40 chances to find $p < .05$ and the paper reports three significant results, that's roughly what luck alone would produce. You met the formal version of this, the *familywise error rate*, earlier in this chapter, where running many comparisons raised the chance of at least one false positive. Forking paths are the same problem. The only difference is that the extra tests are hidden rather than printed in a table.

Sample sizes nobody explains. If a paper doesn't say how its sample size was decided, you can't rule out the "collect a few more" fork from our simulation. Good papers justify their sample size in advance, usually with a power analysis (earlier in this chapter). Odd numbers, or samples described as having been "extended," are worth a second look.

Exclusions. Look for phrases like "after removing n cases," "failed attention checks," or "outliers were excluded." None of these is wrong in itself. What matters is whether the rule was set before the results were known, and whether the main finding survives with the excluded cases put back in. Many good papers report both versions. When a paper reports only one, the question is worth asking.

Control variables that move around. If the set of controls changes from one model or table to the next without explanation, each change is a fork. In Chapter 4 you saw that controls should be chosen for causal reasons. Controls that look chosen for their effect on the p-value are a different thing.

"Robustness checks." These are often valuable, and some of the best work goes much further: a *specification curve* or *multiverse analysis* runs the analysis every reasonable way at once and shows how often the result holds (Simonsohn et al., 2020; Steegen et al., 2016). But ordinary robustness checks are also chosen by the authors, and you only see the ones that were reported. A robustness section in which every single check comes out the same way deserves a little gentle scepticism.

Hypotheses that fit a bit too perfectly. Sometimes the hypotheses in the introduction match the results a little too neatly, including an unexpected interaction or a subgroup effect presented as if it had been predicted all along. This has a name, *HARKing*: hypothesising after the results are known (Kerr, 1998). You usually can't prove it. But a highly specific prediction that nobody would have thought to make in advance is a clue.

Pre-registration and registered reports. A pre-registration is a time-stamped plan, usually on a site such as the Open Science Framework, setting out the hypotheses, sample size and analysis before any data are collected. A *registered report* goes one step further: the journal reviews the plan and agrees to publish whatever the results turn out to be. Both are among the most important developments in research practice in the last 15 years. As a reader, don't just note that a study was pre-registered; go and look at the plan. Does the paper match it? Are any deviations disclosed and explained? A pre-registration that nobody checks is just a link. Also bear in mind that pre-registration fits experiments far more easily than it fits analyses of secondary data, so its absence isn't a red flag on its own in many management fields.

13.6.2 The connection back

Every fork gives chance another opportunity, and every p-value quietly assumes there was only one. That's the whole of it. It's the same logic as the multiple comparisons problem earlier in this chapter, and the same logic as the simulation above. What changes from paper to paper is only how visible the forks are.

13.6.3 Questions worth asking

1. **How many tests could they have run?** Not just how many they report: think about the outcomes, subgroups, controls and exclusions they could have tried.
2. **Was the sample size set in advance?** Is there a power analysis, and was it done before data collection?

3. **Are the exclusion rules justified, and does the result survive without them?**
4. **If it was pre-registered, does the paper match the plan?**
5. **Where do the key p-values sit?** A single p = .04 is unremarkable. A paper whose every central result falls between .01 and .05 is a pattern worth noticing. Across many studies, genuine effects tend to produce a pile of very small p-values rather than a cluster just under the line (Simonsohn et al., 2014).
6. **Would the story survive if the weakest result were removed?**

13.6.4 One red flag

The too-perfect paper: every hypothesis supported, every key result just under .05, a sample size nobody explains, and a robustness section that agrees with everything. No single feature proves anything, and I'd hate you to treat honest researchers who got lucky as suspects. But put together, that profile is exactly what a walk through the garden of forking paths produces. Real data is messier than that, and papers that admit to some mess are, in my experience, often the ones you can trust most.

13.6.5 If you want to read more

Gelman and Loken's short article "The statistical crisis in science" (Gelman & Loken, 2014) is the clearest statement of the forking-paths idea and is free to read online. For a book-length account of how these problems play out across science, and what's being done about them, Stuart Ritchie's *Science Fictions* (Ritchie, 2020) is accessible, fair and genuinely hard to put down.

A condensed version of these questions, combined with those from the other chapters and arranged in the order you'd meet them when reading a paper, is in Appendix A.

Appendix A

A Checklist for Reading Quantitative Papers

Every chapter in this book ends with a section on reading the literature, and each of those ends with a handful of questions. This appendix gathers them into one checklist, arranged in the order you'd meet things when reading a paper rather than the order the book teaches them. Where several chapters asked versions of the same question, I've merged them. Each question points back to the sections where the ideas behind it are explained.

Two warnings before you use it. First, not every question applies to every paper: an experiment raises different issues from an archival study, and you should skip whatever doesn't fit. Second, this isn't a scoring sheet. A paper can fail several of these questions and still be the best available evidence on something important. The point is to know what a paper has actually shown, not to hunt for reasons to dismiss it.

A.1 Design and sample

1. **What is the causal claim, link by link?** For each link, what does the design do to support it? (Sections 4.5, 7.3)
2. **What is being compared with what?** Is the comparison group the one needed to rule out the most obvious alternative explanation, or is it a before-and-after with no comparison at all? (Section 8.4)
3. **What's the most obvious third variable?** If you can think of one in thirty seconds, is it dealt with? (Section 4.5)
4. **Who is in the sample, and how did they get there?** Is this selection into treatment or selection into the sample? What would the missing cases look like? (Sections 4.5, 10.4)

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5. **Was the sample size set in advance?** Is there a power analysis done before data collection, and how many cases are in each group or cell? (Sections 13.6, 12.2)
6. **If the study was pre-registered, does the paper match the plan?** (Section 13.6)
7. **When, and from whom, was each variable measured?** If the predictor, mediator and outcome come from the same respondent at the same moment, what did the design do about common method variance, and why should we believe the proposed order? (Sections 5.6, 7.3)
8. **Are the observations really independent?** If not, were the standard errors clustered, at what level, and across how many clusters? (Section 10.4)

A.2 Measures

9. **What do we know about each measure, apart from its alpha?** (Section 5.6)
10. **Are the constructs the paper treats as different actually different?** Look for correlations in Table 1 that approach the measures' reliabilities, and for mediators that are barely distinct from X or Y. (Sections 5.6, 7.3)
11. **Which variables were logged or otherwise transformed?** How does that change what their coefficients mean, and what was done about zeros? (Section 3.3)

A.3 Descriptives and correlations

12. **Is the mean a sensible summary of each variable, and is there enough variation to work with?** Compare the mean with the median, the minimum and maximum, and the scale's limits. Be wary of a standard deviation bigger than the mean on a variable that can't be negative. If normality is discussed, is it the right thing (usually the residuals) that's being checked? (Section 3.3)
13. **Do the key variables actually correlate in Table 1?** If not, how do the authors explain a significant effect in the regression? (Section 5.6)
14. **Which predictors are strongly correlated with each other?** Do the authors claim to have separated their effects, and how convincingly? (Section 5.6)

A.4 Main results

15. **What does each number in the table represent?** What's in the brackets? Are the coefficients standardised? What's the null, which is

usually zero and usually unstated, and which group is the reference category? (Sections 10.4, 11.5, 8.4)

16. **How big is the effect, in units someone would care about?** Is it relative or absolute? Is it in percent or percentage points? Is it an odds ratio for a common outcome? This matters most when the sample is large. (Sections 2.6, 8.4, 11.5)
17. **What range of values is the estimate consistent with?** Does the interval include effects too small to matter? (Sections 10.4, 7.3)
18. **Which effect size is reported, and how was it calculated?** Is it compared only with effects calculated the same way? (Sections 8.4, 12.2)
19. **Were the significance thresholds set in advance, and applied the same way throughout?** Check the table notes and look for one-tailed tests. (Section 11.5)
20. **If you covered up the stars, would you tell the same story?** And where do the key p-values sit? A paper whose every central result falls between .01 and .05 is a pattern worth noticing. (Sections 11.5, 13.6)
21. **Is there an interaction, and is it read correctly?** Are main effects, or lower-order coefficients in a regression, interpreted as if the interaction weren't there? Does an effect reverse, or just get smaller? Is there a plot of simple effects or simple slopes? (Section 12.2)
22. **In a hierarchical regression, what went in first?** Would the story change if the order changed? (Section 5.6)
23. **For mediation, how was the indirect effect tested?** Was it a bootstrap interval or an older test? Is "full mediation" doing any real work? (Section 7.3)

A.5 Robustness

24. **How many tests could the authors have run?** Count the outcomes, subgroups, controls and exclusions they could have tried, not just the ones they report. (Section 13.6)
25. **Are the exclusion rules justified?** Does the result survive without them? (Section 13.6)
26. **If endogeneity was corrected for, what does the correction assume, and did it change anything?** How strong would an unmeasured confounder need to be to explain the result away? (Section 4.5)
27. **Would the story survive if the weakest result were removed?** (Section 13.6)

A.6 Discussion and claims

28. **What verb are the authors using?** Did the design earn the drift from "is associated with" to "drives"? (Section 4.5)
29. **When the authors claim that two effects differ, or that something**

doesn't matter, did they test it? A significant result beside a non-significant one isn't a tested difference. Overlapping error bars aren't a test either. And a non-significant p-value isn't evidence of absence. (Sections 11.5, 10.4)

30. **Whose effect is this?** An average can hide groups where the effect is much larger, and groups where it's absent. A small effect on each person can still add up across a population, or the other way round. (Section 2.6)

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